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BACHELOR THESIS

Physiological and geodata analyses of methane reduction potential in ruminants

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ABSTRACT

Methane is the second most abundant greenhouse gas in the atmosphere (1935 ppb) after CO₂ (423 ppm) and is responsible for around 30 % (gross) of anthropogenic global temperature rise. Because of its high global warming potential, especially on a short time period, methane is an excellent target for mitigation. Right now, methane is still accumulating globally in the atmosphere through the emission of mainly three sectors: energy, waste and agriculture. While the emission from the energy and the waste sector could be reduced with established technologies easily, the agriculture sector is classified as "hard-to-abate". This bachelor thesis investigates one possible mitigation strategy to solve this: the "immunization of cattle". For this purpose, a more holistic approach was chosen, combining a biochemistry and a geography part.

In the biochemistry part, I performed a microbiological investigation of *Methanobrevibacter* as the dominant rumen archaea, which are the biological entities that generate methane in the digestive tract of ruminants. The goal was to achieve DNA transfer in *Methanobrevibacter*, which has not been demonstrated before. Thereby, the focus was primarily on two species (*Methanobrevibacter thaueri* CW^T and *Methanobrevibacter boviskoreani* AbM4). It was shown that the species can be cultivated in pure culture in liquid and solid media and are sensitive to puromycin and DMSO. Furthermore, promising results were achieved to transform these species for the first time via interdomain conjugation with *Escherichia coli*, but verification via whole-genome sequencing is still pending. In the future, the established genetic tool can help to identify a specific and potent vaccine that targets only the anaerobic, methanogenic archaea in the rumen of cattle. The long-time goal is to shift the microbiome towards more non-methanogenic microbes, such as acetogenic bacteria, so that the digestive efficiency of cattle is not compromised.

In the geographical part, I tried to evaluate the potential role of the investigated mitigation strategy to slow down climate change. For this purpose, the changed methane emission from enteric fermentation processes in cattle were calculated for different success scenarios of microbiome shift for three countries (Germany, New Zealand, USA) and its impact on national CH₄ emission. The calculation is based on the 2006 published guidelines by the IPCC. The geodata analyses showed for a shift of at least 30 % a relevant impact to achieve international emission targets. Given the current state of vaccine development, this method cannot play a significant role in reducing emissions over the next decade. However, in the long term, the technology could offer a solution that allows for a sustainable cattle population size that is socially and politically acceptable, even though a reduction will likely still be necessary.

Future research should further develop genetic tools to intensify the investigation of methanogenic archaea. Additionally, it should be assessed whether immunization of cattle could affect emissions beyond enteric methane, either amplifying or offsetting the overall emission reduction potential.

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LIST OF ABBREVIATIONS

CO ₂ -eq.	CO ₂ equivalent.
<i>hpt</i>	hypoxanthine phosphoribosyltransferase.
8-Aza	8-Azahypoxanthin.
AA	amino acids.
AB	antibiotic.
bp	base pairs.
CAP	chloramphenicol.
CFU	colony forming unit.
CI	confidence intervall.
CoM	coenzyme M.
CRT	common reporting table.
Cyc-Se.	cycloserin.
DAP	diaminopimelic acid.
DC	dairy cattle.
DE	digestibility of food.
DEU	Germany.
DMSO	dimethylsulfoxid.
DTT	dithiothreitol.
EF	emission factor.
EnterFe	enteric fermentation.
eq.	equivalent.
<i>E. coli</i>	<i>Escherichia coli</i> .
<i>E. coli</i> DH5 α	<i>Escherichia coli</i> DH5 α .
<i>E. coli</i> WM6026	<i>Escherichia coli</i> WM6026.
<i>E. coli</i> WM6029	<i>Escherichia coli</i> WM6029.
EtOH	ethanol.

gDNA	genomic DNA.
GDP	gross domestic product.
GE	gross energy intake.
GHG	greenhouse gas.
GMP	global methane pledge.
GWP	global warming potential.
IPCC	intergovernmental panel on climate change.
LULUCF	land use and land use change and forestry.
<i>M. boviskoreani</i>	<i>Methanobrevibacter boviskoreani</i> AbM4.
<i>Methanob.</i>	<i>Methanobrevibacter</i> .
MIC	minimum inhibitory concentration.
<i>M. millerae</i>	<i>Methanobrevibacter millerae</i> SM9.
<i>M. thaueri</i>	<i>Methanobrevibacter thaueri</i> CW ^T .
<i>M. wolinii</i>	<i>Methanobrevibacter wolinii</i> SH ^T .
NDC	non-dairy cattle.
NZL	New Zealand.
OD600	optical density at 600 nm.
OECD	organisation for economic co-operation and development.
PB	planetary boundaries.
Puro	puromycin.
rRNA	ribosomal RNA.
rRNA	ribosomal RNA.
Sqr.	square.
ss-DNA	singlestranded-DNA.
<i>S. epidermidis</i>	<i>Staphylococcus epidermidis</i> .

UNFCCC	united nations framework convention on climate change.
USA	United States of America.
Ym	methane conversion factor.

1 INTRODUCTION

The pressing situation of the planets "health" is a scientific consensus (IPCC, 2022b; Planetary Boundaries Science (PBScience), 2025). We can already see the negative impact in our day to day live that the taken projectories over the last decades are to be proven true. So a temperature rise in Europe of 2.5 °C towards preindustrial times can be measured as well as a higher frequency of compound events (C3S/ECMWF & WMO, 2026; Ionita & Nagavciuc, 2026). This led to an extreme heat wave this summer (2026) in Europe, which, in Germany alone, is estimated to have resulted in more than 14 000 deaths (RKI, 2026). Therefore, it is highly necessary to develop strategies for adaption to this changing environment, but more so for mitigation to stabilize the climate and to keep at least the 2 °C goal of the Paris Agreement (UNFCCC, 2015).

One of the main drivers for this development is the anthropogenic-induced climate change. Climate is one of nine earth systems processes - Climate Change, Biosphere Integrity, Land System Change, Freshwater Change, Biogeochemical Flows, Ocean Acidification, Atmospheric Aerosols Loading, Stratospheric Ozone, Novel Entities - that are relevant for containing a stable system on earth, to grant condition, in which mankind can knowingly thrive (Planetary Boundaries Science (PBScience), 2025). Like six other of them, climate change exceeded its planetary boundaries (PB) already (ibid.). Just Atmospheric Aerosols Loading, Stratospheric Ozone are within their PB.

The relevant value for climate change is the anthropogenic radiative forcing, which not only exceeded the PB ($+1 \text{ W/m}^2$) 2025 but the boundary for the zone of an increasing risk ($+1.5 \text{ W/m}^2$) as well with a value of 2.97 W/m^2 (Irfan et al., 2024; Planetary Boundaries Science (PBScience), 2025, p.13). Figure 1 shows the impact of this on global temperature since the industrial revolution, with an increase of just under 1.7 °C. It also shows how much of this increase is caused by each of the three most important anthropogenic greenhouse gas (GHG) -CO₂, CH₄, and N₂O - broken down by emission source ("Fossil Fuels and Industry" or "Agriculture and Land"). Nitrous dioxide has a minor impact and is responsible for a total of

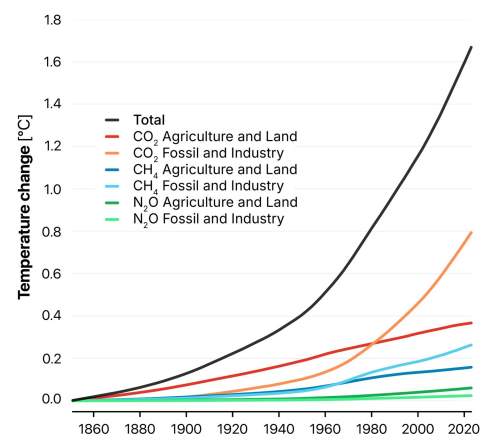


Figure 1: The share of the three main GHG (CO₂, CH₄, N₂O) on the anthropogenic temperature rise since 1850. Each GHG is subdivided in two groups depending on the emission source ("Agriculture and Land" or "Fossil and Industry"). Source: (Planetary Boundaries Science (PBScience), 2025), licensed under CC BY 4.0

approximately 0.1 °C. CO₂, on the other hand, has the biggest responsibility for the temperature rise (more than 50 %, Figure 1) due to its high abundance in the atmosphere (423 ppm in 2025) (GML, 2025; Planetary Boundaries Science (PBScience), 2025).

The second most abundant GHG is methane, with the biggest relative increase in atmospheric concentration of around 265 % from 720 ppb 1750 to 1935 ppb in 2026 (Appelhans et al., 2025; Planetary Boundaries Science (PBScience), 2025; Thoning et al., 2022). Methane is also responsible (Figure 1) for around 30 % (gross) of the anthropogenic temperature rise (Appelhans et al., 2025, p. 9). If compared to CO₂ some features makes it highly interesting as a potential target for mitigation. The stronger impact of methane on radiative forcing and its relatively short lifetime in the atmosphere of 12.4 years (15-40 % of CO₂ is still in the atmosphere after 1000 years) (Appelhans et al., 2025) lead to global warming potential (GWP) values that strongly decline over longer time horizons (GWP20: 79.7 and GWP100: 27 for non-fossil CH₄ in IPCC (2023a, Table 7.15)).

This means that, as a mitigation strategy addressing CH₄, reducing methane emissions is sufficient to achieve a notable impact within two decades, whereas mitigation strategies for CO₂ must also focus on the removal of CO₂ from the atmosphere. Because of that, international initiatives aim at reducing methane emissions such as the global methane pledge (GMP), which was initiated in 2021 and signed by 160 states including Germany (DEU), New Zealand (NZL) and United States of America (USA) (CCAC, n.d.). The participants commit to emit 30 % less methane than they did in 2020 and 40 % less than in 2010, without any penalties (Appelhans et al., 2025).

However, right now methane is still accumulating in the atmosphere (Saunois et al., 2025; Thoning et al., 2022). Globally the three biggest sources for anthropogenic CH₄ are the agriculture- (approx 40 %), energy- (36 %) and waste-sector (22 %) (Appelhans et al., 2025; Mundra & Lockley, 2024). The emissions derived from the energy and waste sectors can be substantially reduced using existing technical knowledge and available mitigation options (Appelhans et al., 2025; IPCC, 2022a). Emissions from agriculture, on the other hand, are counted as "hard-to-abate", because the world is lacking technologies and strategies that are socially and political accepted (Edelenbosch et al., 2024; IPCC, 2022a; Thoning et al., 2022). This is due to the fact that methane arises naturally in agriculture from biological processes, primarily through enteric fermentation, especially in cattle (77 %), and manure management. Together they make up for 31 % of global methane emission (Appelhans et al., 2025; Smith et al., 2021). The only option for reducing methane emissions for this right now is to reduce the

number of live stock dramatically, which is contrary to the global rise of demand for livestock products and globally rising numbers of livestock (Edelenbosch et al., 2024; FAO, 2009).

This bachelor thesis wants to tackle this lack of options by further investigating a mitigation strategy for the largest single source of methane emissions in the agricultural sector: "immunization of cattle" against its methanogenic microbiome in the rumen. For this purpose, an interdisciplinary approach was chosen with a biochemical and geographical part, which allows uniting two scales of investigation in one work.

The biochemistry part investigates new approaches to reduce methane production on cellular or organism level, while the geographical part places the results in the context of multifactorial climate change and assess the extent to which the biochemical approach can play a role to mitigate.

1.1 Methane production in cattle farming and a possible mitigation

Cattle as a ruminant has a rumen which enables them to break down and digest long fibers of carbohydrates such as plant cells via microbial fermentation (EPA, 2024; Moss et al., 2000). These fermentation processes carried out by a wide variety of microbes produce H_2 and CO_2 , which must be removed as quickly as possible due to the negative feedback effect on fermentation. This is done in the rumen by strictly anaerobic microbes, which utilize the H_2 and CO_2 as substrates and in the process produce a variety of substances (Moss et al., 2000). Two relevant groups are acetogenic bacteria, which produce short-chain hydrocarbons, or methanogenic archaea, which produce CH_4 (Boadi et al., 2004; Garnett & Eckard, 2024). In the rumen, methanogenic archaea are often the dominant group, as they can metabolize H_2 and CO_2 more efficient (Moss et al., 2000). This also means that completely eliminating methanogenic archaea would not be advisable since, according to Morvan et al. (1994), H_2 would accumulate, making digestion in cattle less efficient.

However, the goal of the immunization strategy is to shift the microbiome toward a higher proportion of acetogenic bacteria by training the cattle's immune system to recognize and attack methanogenic archaea (Malyugina et al., 2025). This would result, on the one hand, in less methane production and, on the other hand, in the production of more short-chain hydrocarbons as acetate or butyrate that can still be digested and absorbed by the cattle. In theory, this would also lead to a more efficient digestion (Boadi et al., 2004; Malyugina et al., 2025; Moss et al., 2000). Already, there were some experiments trying to immunize cattle against methanogenic archaea with an inactivated vaccine. But the success was low and inconsistent, mainly due to high surface variability of methanogens (Malyugina et al., 2025;

Wright, 2004). To solve this problem a robust and specific antigen has to be found for the dominant methanogen species. The problem here, however, is that the individual anaerobic, methanogenic archaea in the rumen are poorly understood and characterized, even though some of them have already been isolated and the general composition of these archaea in the rumen is now somewhat better understood. Four main groups of rumen archaea were identified of which the dominant genus is *Methanobrevibacter* accounting for approximately 60 % (Baca-González et al., 2020).

To address this knowledge gap, two strains in particular - *Methanobrevibacter thaueri* CW^T and *Methanobrevibacter boviskoreani* AbM4 - will be further investigated here, which were originally isolated from the rumen of cattle (Lee et al., 2013; Miller & Lin, 2002). For this purpose, the investigation will have two sites. First, investigate growth behavior of the cell under various conditions. Second, trying to implement for the first time a genetic tools for *Methanobrevibacter*. Following the approach of Fink et al. (2021) and Klein et al. (2025) for interdomain conjugation and markerless mutagenesis mutagenesis of *Methanothermobacter*, the transformation of *Methanobrevibacter* was performed via conjugation with *Escherichia coli*. Thereby a suicide plasmid was used, with four main functional sites to allow the selection of transformed *Methanobrevibacter* and also the counterselection to enable genome editing (Figure 2).

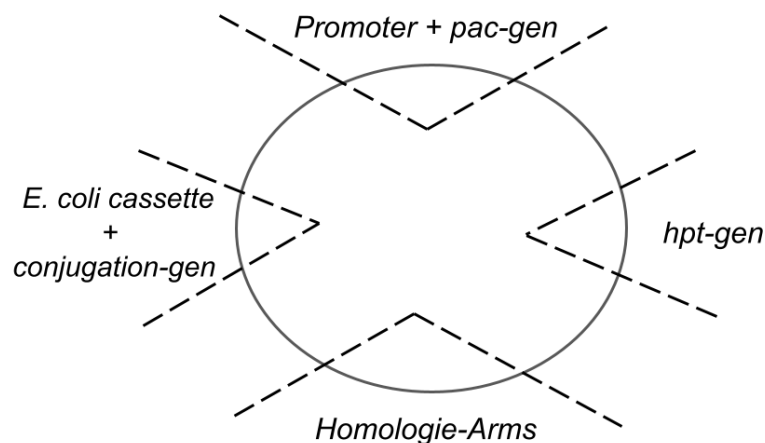


Figure 2: Sketch of the general structure of the suicide plasmids for the transformation of *Methanobrevibacter thaueri* CW^T and *Methanobrevibacter boviskoreani* AbM4 via conjugation using *E. coli*. The plasmid contains the ORI for replication in in gram negative bacteria (*E. coli*) and *catP* as a selection marker against chloramphenicol (*E. coli* cassette), as well as the conjugation gene *traJ*. In addition, all plasmids contain the *hpt* gene to enhance sensitivity against the base-analogue 8-Azahypoxanthin, which is used for counterselection, and the two homology arms that flank the genomic *hpt* gene of the corresponding organism. The final important region is the one containing the *pac* gene and its associated promoter, which serves as a selection marker.

The first functional site is the "*E. coli* cassette + conjugation-gene". This is important for preparing the donor organism (*E. coli*) for the conjugation to transform the *Methanobrevibacter* species. Therefore the plasmid contains for the transformation of *E. coli* an ORI and a selection marker (*catP*) against chloramphenicol. In addition, the *traJ*-gene is needed for the conjugation. The remaining sites on the plasmid are for the *Methanobrevibacter* species. So the *pac*-gene with its promoter is the selection marker for transformed *Methanobrevibacter* against puromycin (Puro). The "Homology arms" site contains the segments of the genomic DNA (gDNA) from the transforming *Methanobrevibacter*. The homology regions thereby flank the gene to be deleted. Through single homologous recombination events, the plasmid is first integrated and then removed again via counterselection, whereby the gene flanked by the homology arms is also excised (Klein et al., 2025). The last of the four sites is hypoxanthine phosphoribosyltransferase (*hpt*) gene, which enhances sensitivity against base-analogue 8-Azahypoxanthin (8-Aza) for the counterselection.

In the following experiments, regions on the homology arms are selected that flank the *hpt* of the respective organism, resulting in a mutant lacking genomic *hpt*. This makes it possible to target any other gene in the genome by selecting the appropriate homology regions and removing them via markerless mutagenesis counterselection (Klein et al., 2025).

1.2 Evaluation based on calculated CH₄ emission

Based on a closer examination of three countries (DEU, NZL, USA), the mitigation strategy "immunization of cattle" will be evaluated. First, it has to be checked if the global methane emission patterns earlier in the introduction fit on the national scale and cattle is a valid target for mitigation. Second, the calculation of methane emission is based on the guideline developed by the intergovernmental panel on climate change (IPCC) in the year 2006 (IPCC, 2006a) and its refinement in 2019 (IPCC, 2019). This guideline standardized the international calculation of national GHG emission for all emission sources.

To calculate methane emissions resulting from enteric fermentation processes in cattle (IPCC, 2006b, 2019), cattle are typically divided into two subcategories: "dairy" and "non-dairy" cattle. Emissions are first calculated separately for these two categories and then summed. There are three different calculation methods (Tier 1 through 3), which differ in their complexity and the amount of data required.

The Tier 1 approach is quite straightforward. The number of animals in the corresponding animal category is multiplied by the appropriate emission factor (EF), which is provided as a default value by the IPCC and specifies how much methane is emitted per animal (IPCC,

2006b, Eq. 10.19). In Tier 2, this EF is now calculated for each animal category and country based on specific local activity data. Here, the gross energy intake (GE) per animal is calculated from energy consumption and the average digestibility of food (DE). The EF is then obtained by multiplying the GE by the methane conversion factor (Ym), which can be taken as the default value from the IPCC or has been scientifically determined for the respective country and animal category (IPCC, 2006b, Eq. 10.21). Tier 3 involves even more complex calculations and uses detailed, mechanistic models or direct measurements at the farm or animal level (e.g., individual feeding data, respiratory gas measurements of individual animals) (IPCC, 2006a).

The mitigation strategy "immunization of cattle" influences clearly impact two values of the Tier 2 approach. First the Ym, whereas a shift in the rumen microbiome towards non-methanogenic microbes leads logically to a lower energy conversion into methane. Second the DE, as described in Section 1.1, the microbiome shift is expected to lead to increased digestive efficiency, since less energy is converted into methane, which cannot be utilized, and more into short-chain hydrocarbons that can be absorbed by the cattle. The performed calculation of methane emission followed the Tier 2 approach (Section 2.2 and Section 3.2.2).

2 DATA AND MATERIALS

2.1 Biochemistry

2.1.1 Organism

Methanobrevibacter thaueri CW^T and *Methanobrevibacter wolinii* SH^T were provided by Dr Graeme Attwood (Bioeconomy Science Institute Ltd (BSI), Palmerston, New Zealand). *Methanobrevibacter boviskoreani* AbM4 and *Methanobrevibacter millerae* SM9 were provided by Lucidome Bio Limited and Dr Graeme Attwood (Bioeconomy Science Institute Ltd (BSI), Palmerston, New Zealand). *Escherichia coli* WM6026 and WM6029 were provided by Prof. William Metcalf (University of Illinois at Urbana-Champaign, IL, USA). *Escherichia coli* DH5 α from Thermo Fisher Scientific Inc. (MA, USA).

2.1.2 Chemicals

All chemicals are from Carl Roth GmbH + Co. KG (Germany) unless explicitly labeled otherwise. Following chemicals were used in the experiments: Brain heart infusion (Merck KGaA, Germany), Proteose peptone, Yeast extract (Oxoid, UK), Na-acetate (Sigma-Aldrich, Merck KGaA, Germany), Na-formate (Sigma-Aldrich, Merck KGaA, Germany), Sodium resazurin, NaHCO₃, KH₂PO₄, MgSO₄ x 7 H₂O, NaCl, NH₄Cl, CaCl₂, HCl, FeCl₂ x 4 H₂O, ZnCl₂, MnCl₂ x 4 H₂O, H₃BO₃, CoCl₂ x 6 H₂O, CuCl₂ x 2 H₂O, NiCl₂ x 6 H₂O, Na₂MoO₄ x 2 H₂O, Vitamin B₁₂, p-Aminobenzoic acid (Aldrich, USA), D-(+)-biotin, Nicotinic acid (Sigma-Aldrich, Merck KGaA, Germany), Calcium pantothenate, Pyridoxine hydrochloride, Thiamine-HCl x 2 H₂O, dithiothreitol, Coenzyme-M (Sigma-Aldrich, Merck KGaA, Germany), FeSO₄ x 7 H₂O, H₂SO₄, Isobutyric acid, DL-2-Methylbutyric acid, Valeric acid, Isovaleric acid, Milli-Q H₂O, Agar-Agar Kobe I, N₂-gas (Air Liquide S.A., France), N₂-CO₂ gas mix (Air Liquide S.A., France), H₂-CO₂ gas mix (Air Liquide S.A., France), LB-Media, glycerol, agarose, TAE-buffer, Gel Loading Dye Purple (6X) (New England Biolabs, USA), GelRed (Biotium, USA), GeneRuler 1 kb DNA Ladder (Thermo Fisher Scientific, USA), ddH₂O, ethanol $\geq$ 99.5 %, puromycin (Fisher BioReagents from Thermo Fisher Scientific, USA), chloramphenicol (CAP) (Sigma-Aldrich, Merck KGaA, Germany), cycloserine, 8-Azahypoxanthin (Thermo Fisher Scientific, USA), dimethylsulfoxid, diaminopimelic acid. Restriction enzymes NotI, AscI, SmaI, and FseI, as well as Q5[®] High-Fidelity DNA Polymerase and OneTaq[®] DNA Polymerase, were obtained from New England Biolabs (USA).

All primers used in this study are listed in Table B7, including their target sequence and corresponding annealing temperature (T_a) for Q5 and OneTaq polymerase. Primers were designed using the NEB® Primer Design Tools (New England Biolabs, USA) and synthesized by IDT Biologika (Germany).

2.1.3 Materials

Anaerobic cultivation was performed in serum bottles sealed with butyl rubber stoppers. Syringes (1 mL and 2 mL, Luer-Lock; Chirana T. Injecta, Slovakia) as well as 5 mL and 10 mL syringes (CODAN, Denmark) were used. Cannulas (Sterican®, B. Braun, Germany; 0.60 x 30 mm) were used for culture handling. Sterile filtration of syringe-applied solutions was performed using Chromafil® CA-20/25 S syringe filters (Macherey-Nagel, Germany). Anaerobic jars (Raff + Grund GmbH, Germany) were used for anaerobic incubation. Plasmid DNA was extracted using the Monarch® Spin Plasmid Miniprep Kit (New England Biolabs, USA). DNA fragments were purified from agarose gels using the Monarch® Spin DNA Gel Extraction Kit (New England Biolabs, USA). Genomic/microbial DNA was extracted using the NucleoSpin® Microbial DNA Kit (Macherey-Nagel, Germany).

2.2 Geography

2.2.1 Study area

The geodata analyses of methane reduction potential in rumineszenz will focus on the impact of the mitigation strategy "immunization of cattle". Two emission sectors of agriculture discussed in the IPCC (2019) could be effected by this. On the one hand methane resulting from enteric fermentation and on the other hand from manure management. However, in this bachelor thesis the manure management sector will not be investigated, primarily due to as-yet-unclear links to a possible microbiome shift, as well as time constraints. All calculations are based on the methodology described in IPCC (2019) for calculating CH₄ emission from enteric fermentation in cattle. Furthermore, is the calculation done for only three countries: DEU, NZL and USA. These three countries were picked, because of three main reasons.

First, the project partners for the biochemical research project to develop an immunization method of cattle against methanogen archaea are located in those very countries. Second, all three countries belong of the ANNEX-I states. This means that, due to their larger economies and higher levels of consumption, they emit a notable amount of GHGs, including CH₄, and therefore bear a greater responsibility with regard to climate change and its mitigation. In

addition, they also have the financial resources to implement and research the mitigation strategy. Third, although all three are categorized as ANNEX-I states, there are some clear difference between those countries. For example, their size in terms of area, population, and economy (the USA and DEU as leading economic nations of the world based on GDP (International Monetary Fund, [n.d.](#))) or in their predominant way to raise animals, particularly cattle (in NZL pasteur grazing is dominant in Germany loose housing (MfE, [2026](#), p. 155; Niebuhr et al., [2026](#), Table 510)).

The time period considered in this bachelor's thesis is determined by the data series published in the common reporting tables (CRTs) - 1990–2024 for DEU and NZL, and 1990–2022 for the USA - as well as by the reference data used to evaluate the "immunization of cattle". For the latter, the decade 2010–2019, as used by Saunois et al. ([2025](#)) for the calculation of an averaged yearly growth of methane, is relevant, along with 2030 as the target year for the emission reduction goals of the GMP.

2.2.2 CRT Data

CRT is the data source for the calculations performed. The CRTs were downloaded from the UNFCCC ([2026](#)). For DEU and NZL, the published data range from 1990 to 2024, and for the USA, from 1990 to 2022. Where is one CRT published per year and the values represent thereby the average of an entire year. For the analysis of methane emissions from enteric fermentation processes involving cattle, the "Table3" and "Table3.A" sheets of the .xlsx files were used. These show only the values for the year it was published for. The "TableS3" sheet serves as the basis for analyzing methane emissions from the various sectors in each country as well as total emissions, so it summarizes the previous detailed tables of the .xlsx file. This sheet lists not only the values for the current year but also all past years, so the CRT for 2024 and 2022 for the U.S. also includes all values for the remaining years.

DEU and NZL have defined two subcategories for cattle according to IPCC ([2019](#)): dairy cattle and non-dairy cattle. The USA, on the other hand, chose to define its own subcategories for cattle, so that dairy cattle and non-dairy cattle were further subdivided (EPA, [2024](#)). Section [3.2.1](#) explains how the data was processed to ensure that all three countries are comparable with one another. The used variables from the CRT table are given in following units for each animal category:

- CH₄ emission in kt CH₄
- Ym in %
- Population size in 1000s
- DE in %
- GE in MJ head⁻¹ year⁻¹

2.2.3 Reference data

In this bachelor thesis, two different scaled data for methane emission are taken into account as a reference to evaluate the possible impact of a mitigation strategy against methane emissions derived from cattle enteric fermentation processes. On the global scale Saunio et al. (2025, fig. 7) calculated the average yearly growth rate of methane 20.9 Tg CH₄ for the decade 2009–2010. Here they also used the tier 1 and tier 2 calculation guidelines of the IPCC (2006a). On the national scale the global methane pledge (GMP) was used. The GMP calls for a methane emission reduction of each country until 2030. The aim is hereby to emitted 30 % less methane over all sectors than in 2020 and 40 % less than in 2010. DEU, USA and NZL signed this pledge in the year 2021 (Appelhans et al., 2025; CCAC, n.d.). Because of that this serves as a reference on a country scale.

In this thesis the focus is on the strategy of immunization of cattle and not possible others, as e.g. food additives. These are used only selectively to compare what makes immunization unique and how it measures up against them.

2.3 Software

All analyses and processing of the data, as well as the visualization, were performed in R (version 4.6.1). The main packages used for processing and analysis were `tidyverse` and `purrr`. Visualization was performed using `ggplot2`. For more information, please refer to the R scripts. One exception is that some tables were written directly in $\text{\LaTeX}$, and the plasmid maps were created in Google Slides. Where other software was used, this is explicitly mentioned in the text.

3 METHODS

3.1 Biochemistry

3.1.1 Liquid media preparation for Methanobrevibacter

For DSM 1523, chemicals were mixed in MilliQ-H₂O as listed in Table 2, and for DSM 119 following Table 3, except NaHCO₃. The needed trace element solution and solution A for DSM 1523 and 119 were premixed as shown in Table B3 and Table B1. FeSO₄ x 7 H₂O solution (0.1 % w/v) and fatty acid mixture were premixed as listed in Table B6 and Table B5.

Table 2: Composition of the starting solution for the preparation of DSM 1523. The quantities listed are for 1 L. Final concentration of defined chemicals are shown as well. The final pH should be between 6.8 and 7.0. Solutions that have already been premixed are shown in bold. Composition of **solution A** is listed in Table B1 and for **Trace element solution SL-10** in Table B3.

ingredient	quantity	concentration
Solution A	10.00 mL	1 %
Trace element solution SL-10	1.00 mL	0.1 %
Brain heart infusion	6.00 g	–
Proteose peptone	6.00 g	–
Yeast extract	2.00 g	–
Na-acetate	1.00 g	12.19 mM
Na-formate	2.00 g	29.41 mM
Sodium resazurin (0.025 % w/v)	4.00 mL	0.4 %
NaHCO ₃	4.00 g	47.62 mM
Milli-Q H ₂ O	985.00 mL	–

Table 3: Composition of the starting solution for the preparation of DSM 119. Final concentrations for defined chemicals were calculated as well. The quantities listed are for 1 L. The final pH should be between 6.8 and 7.0. Solutions that have already been premixed are shown in bold. Composition of **solution A** is shown in Table B1, of **Trace element solution SL-10** in Table B3, of **FeSO₄ x 7 H₂O solution (0.1 % w/v)** in Table B6, of **Fatty acid mixture** in Table B5.

ingredient	quantity	concentration
Solution A	10.00 mL	1 %
Trace element solution SL-10	1.00 mL	0.1 %
Yeast extract	1.00 g	–
Na-acetate	1.20 g	14.63 mM
Na-formate	2.00 g	29.41 mM
Sludge fluid	50.00 mL	5 %
FeSO₄ x 7 H₂O solution (0.1 % w/v)	2.00 mL	0.2 %
Fatty acid mixture	20.00 mL	2 %
Sodium resazurin (0.025 % w/v)	4.00 mL	0.4 %
NaHCO ₃	4.00 g	47.61 mM
Milli-Q H ₂ O	918.00 mL	–

After boiling, the mixture was sparged with N_2 or N_2/CO_2 . Once the solution was cool enough, $NaHCO_3$ was added, and the solution was aliquoted into serum bottles, typically 20 mL for routine cultivation. The bottles were also gassed and then sealed with butyl stoppers and aluminum crimp seals. The gas was exchanged to H_2/CO_2 by evacuating four to six times down to -1 bar and refilling with the gas to 1 bar. Autoclaving (121 °C for 20 min) was then performed, after which the solution should no longer show any red discoloration.

Before inoculating with 2 mL of a pre-grown *Methanobrevibacter* culture per 20 mL of medium, 0.2 mL of a sterile DTT solution (30 g/L; 194.55 mM), sterile solution B (mixed as listed in Table B2), and 0.4 mL of a sterile CoM solution (25 g/L) were added. For solution B, the premixed seven-vitamin solution (Table B4) was sterile-filtered and added after $CaCl_2$ was mixed, aliquoted into a serum bottle, gassed with N_2 or N_2/CO_2 , and autoclaved. In addition, when needed, antibiotics were added. Furthermore, for DSM 119 Wolin's vitamin solution, prepared according to DSMZ specifications, was sterile added instead of the seven-vitamin solution. The cultures were then incubated on a shaker at 37 °C, 120–140 rpm.

3.1.2 Solid media preparation for *Methanobrevibacter*

Spread Plating

The media preparation generally followed the instructions for DSM 1523 (Section 3.1.1). However, here 50 mL were aliquoted into a serum bottle containing 0.75 g agar (1.5 %). Another deviation is that here the headspace was not exchanged to H_2/CO_2 , but instead evacuated and refilled four to six times with N_2 or N_2/CO_2 . Before pouring the plates, they were heated in a water bath and transferred into the glove-box, where 0.5 mL DTT (30 g/L; 194.55 mM) and solution B, as well as 1 mL CoM (25 g/L), were added. 100 µL of a pre-grown culture then were spread plated. The plates were then placed into a jar, which was sealed before being taken out of the glove-box. After changing the atmosphere in the jar to H_2/CO_2 , the plates in the jar were incubated at 37 °C. If the pressure dropped below 0.7 bar, the gas was refilled.

Roll plating

10 mL of DSM 1523 is prepared in a serum vial that is usually used for 20 mL of medium, following the procedure described above. Here, the vial contains 0.15 g agar before the prepared medium is aliquoted. Before autoclaving, the gas is exchanged to H_2/CO_2 by evacuating four to six times to -1 bar. For Plating, the serum bottle is heated in a water bath until the agar is liquid, then, after brief cooling, 0.2 mL DTT and solution B, as well as 0.4 mL CoM, are added. The concentration of these three ingredients is thus double the usual amount. Either

200 µL or 400 µL of a pre-grown culture were added, at which point the medium should still be completely liquid and at a temperature of approximately 50 °C. Once inoculated, the still completely liquid culture is spun in a horizontal position on an ice bath. It is important that the bottle is rotated as fast as possible in the same horizontal position in the ice bath, so that a clean, thin layer forms around the inside of the serum bottle. Otherwise, the agar becomes chunky, which considerably hinders the identification of growing colonies. Afterwards, the cultures are incubated in a vertical position at 37 °C, and the gas has to be replenished regularly (every 2 to 4 days).

Pour Plating

Here, 50 mL DSM 119 was aliquoted into a serum bottle containing 0.5 g agar (1 %). After transferring the heated DSM 119 (1 % agar) in the glove-box 0.1 mL dithiothreitol (DTT), 0.5 mL solution B and 1 mL CoM was added. After 200 µL of a pre-grown culture were put in the empty petri dish, the still liquid DSM 119 were poured onto it. Following that, the petri dishes were moved in an eight shaped form to distribute the organism. Plates were then transferred outside of the glove-box in a gas tight jar. Before incubated by 37 °C, the gas was exchanged to H₂/CO₂. If the pressure dropped below 0.7 bar, the gas was refilled.

3.1.3 Cell counting

Cell counting was performed with a phase-contrast microscope (Carl Zeiss, Germany) at 400 x magnification and a Neubauer improved counting chamber, except the *M. wolinii* sample with an optical density at 600 nm (OD₆₀₀) of 0.387 was counted using a Bürker counting chamber. In the Bürker chamber, the squares were counted, whereas in the Neubauer improved chamber, the group squares were counted. Both have a volume of $4 \cdot 10^{-6}$ mL, as can be seen in LO - Laboroptik GmbH (2026). It is important to note that counting was not restricted to a single focal plane. Instead, all planes were scanned through and the cells counted accordingly. In addition, the top and left edges of the squares were included in the count, but the bottom and right edges were not. Before cell counting, the OD₆₀₀ was measured with a 20 Genesys™ photometer (Spectronic Instruments, USA). Samples were diluted 10- or 20-fold in DSM 1523. Four chambers were filled for counting, and for each chamber, four squares were counted, except for *M. millerae*, where five squares were counted for three chamber fillings, and only two squares for the remaining chamber filling.

3.1.4 Sensitivity experiments

Puromycin was tested at five different concentrations (1, 2, 5, 10, and 20 µg/mL) in triplicates using 20 mL DSM 1523 to assess its effect on *Methanobrevibacter boviskoreani* AbM4 and *M. thaueri* growth. For the three lowest concentrations, a sterile stock solution of 228 µg/mL puromycin in MilliQ-H₂O was prepared; for the two highest concentrations, a stock solution of 2280 µg/mL was used. A control without any additives was included. Cultures were incubated at 37 °C and 120–140 rpm. The OD600 was measured once a day, using DSM 1523 medium as blank, and dilutions, if necessary, were also performed with DSM 1523.

The effect of 8-Aza on *M. boviskoreani*, *M. thaueri*, and *M. millerae* was tested at five different concentrations (100, 250, 500, 750, and 1000 µg/mL, including a positive control without any additives and a solvent control with 1 mL of DMSO. The experiment was performed in triplicates in 20 mL DSM 1523. A 22.8 µg/mL 8-Aza stock solution in DMSO was prepared. Cultures were incubated at 37 °C and 120–140 rpm. The OD600 was measured once a day, using DSM 1523 medium as blank, and dilutions, if necessary, were also performed with DSM 1523. In addition, an uninoculated control was prepared for the 8-Aza sensitivity assay of *M. thaueri* to assess the influence of DMSO on the medium. For this, 20 mL DSM 1523 was supplemented with 1 mL DMSO. This preparation was incubated under the same conditions (37 °C, 120–140 rpm), and the OD600 was measured once a day.

3.1.5 Plasmid construction

In-silico design for suicide plasmids used for *hpt*-gene knockouts in *M. thaueri* and *M. boviskoreani* was performed with SnapGene (Dotmatics, USA) according to the scheme shown in Figure 2. All created plasmids are listed in Table B15. For the plasmids, the original *pac* sequence from *Streptomyces alboniger* was codon-optimized for the respective archaea (*pac_opt*). All primers used in this study are listed in Table B7. All restriction digestions were performed following the Optimizing Restriction Endonuclease Reaction protocol from NEB (USA). PCR (Mastercycler nexus gradient with flexlid, Eppendorf, Germany) of inserts was performed with Q5[®] High-Fidelity DNA Polymerase following the instructions from NEB. Digested vector and PCR DNA fragments were purified on a 1.5 % agarose gel (100 V, 35 min) and extracted using the Monarch Spin DNA Gel Extraction Kit following the manufacturer's instructions, excluding the elution step. Here, a dry centrifugation step (Mikro 22 centrifuge, Andreas Hettich GmbH & Co. KG, Germany) was added before elution was performed with autoclaved ddH₂O. In addition, a Biometra TS1 Thermoshaker (Analytik Jena, Germany) was used for heating. All plasmids were assembled via Gibson assembly following the NEB[®] HiFi

DNA Assembly Reaction protocol. Assembled plasmids were used to transform chemically competent *E. coli* DH5 α cells and were harvested using the Monarch Spin Plasmid Miniprep Kit, following the kit instructions with two modifications. First, for the lysis step, after addition of lysis solution B2, the mixture was inverted 4 to 5 times, and the neutralization solution was then added directly. The mixture was then inverted until the solution turned completely yellow. Second, elution was not performed with the prescribed elution buffer but with autoclaved ddH₂O, and a dry centrifugation step was performed once before elution. All plasmids were verified via Nanopore sequencing (Eurofins Genomics GmbH, Luxembourg).

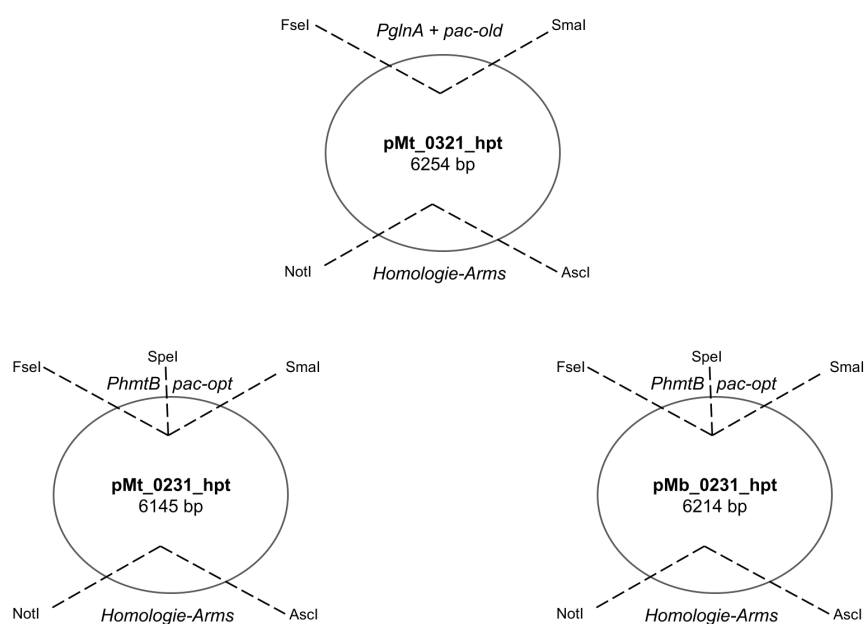


Figure 3: Suicide plasmid maps for transformation by conjugation for *M. thaueri* and *M. boviskoreani*. pMt_0321_hpt (6254 bp) is the parent plasmid used to construct the other two, pMt_0231_hpt (6145 bp) and pMb_0231_hpt (6214 bp). The maps show only the gene regions (between the dashed lines) that have been modified, as well as the restriction site sites (at the end of the dashed lines) used or the newly created restriction site sites. The new produced plasmids were verified via nanopore sequencing.

Two of these plasmids (Table B15) were created: pMt_0231_hpt and pMb_0231_hpt. First, the initial plasmid pMr_0321_hpt (Flaiz, unpublished) was digested with NotI and AscI to remove the homology arm region. Simultaneously, the left and right homology arms for *M. thaueri* were amplified using primer pairs 48/49 and 50/51, respectively, with genomic DNA (gDNA) from *M. thaueri* as template. The gDNA was extracted from a grown culture using the NucleoSpin Microbial DNA Kit, following the manufacturer's instructions except that autoclaved ddH₂O was used for elution instead of the kit elution buffer, with an additional dry centrifugation step before elution. After purifying the DNA fragments and measuring their concentration with the EzDrop 1000 (Blue-Ray Biotech, Taiwan), the pMr_0321_hpt backbone and the amplified homology arms were assembled into pMt_0321_hpt.

pMt_0321_hpt served as the precursor for the target plasmids (Figure 3) and was digested with FseI and SmaI to remove the *pac* gene together with its promoter. For the assembly, an insert containing the codon-optimized *pac* gene for the corresponding *Methanobrevibacter* and the *PhmtB* promoter, with an additional SpeI restriction site between promoter and gene, was designed in-silico with SnapGene and synthesized by IDT. This insert was amplified with primer set 17/63 for *M. thaueri* and 17/71 for *M. boviskoreani*. Gibson assembly yielded two plasmids: the correct pMt_0231_hpt and pMt_0231_hpt containing the codon-optimized *pac* gene for *M. boviskoreani*. The latter was subsequently digested with NotI and AscI, and the *M. boviskoreani* homology arms were amplified with primer sets 52/53 (left arm) and 54/55 (right arm), using extracted *M. boviskoreani* gDNA as template. After purification, the backbone and homology arm regions were assembled to pMb_0231_hpt and cloned into *E. coli* DH5 α .

3.1.6 Transformation of Methanobrevibacter

All 28 preparation lines are listed in Table B16. Figure 4 visualizes the general approach taken in this bachelor thesis to transform three *Methanobrevibacter* species: *M. thaueri*, *M. boviskoreani*, and *M. millerae*. Thereby, DAP-auxotrophic *E. coli* WM6026 and *E. coli* WM6029 were transformed with plasmids (pMt_0231_hpt, pMm_0231_hpt, pMb_0231_hpt) on selective agar plates containing 0.1 mM DAP and 30 μ g/mL CAP.

The day before the mating (Day 0, Figure 4), 5 mL LB medium containing 0.1 mM DAP and 30 μ g/mL CAP was inoculated with each possible combination of *E. coli* WM6026 and *E. coli* WM6029 and the named plasmids, and incubated overnight at 37 °C and 120–140 rpm.

On the day of the mating (Day 1), 20 mL LB medium with 0.1 mM DAP and 30 μ g/mL CAP was inoculated with 1 mL of the overnight *E. coli* culture and cultivated (37 °C, 120–140 rpm) for 4.5 h, except for the preparation with *M. boviskoreani** (Table B16), where 1.5 mL was used for inoculation and the *E. coli* cultures were cultivated for 3.75 h. Final OD600 was measured before the 20 mL *E. coli* cultures were divided into 10 mL aliquots and centrifuged twice (Centrifuge 5910R, Eppendorf, Germany); the supernatant was decanted, and the pellets were washed once with 1 mL of DSM 1523. Centrifugation was performed for 5 min each time, at 6000 rcf the first time and 9000 rcf the second time. In addition, 50 mL DSM 1523 (1.5 % agar) was prepared with 10 mM MgSO₄, 5 mM CaCl₂, and 0.1 mM DAP. *E. coli* pellet, the prepared liquid agar medium, as well as a full-grown *Methanobrevibacter* culture were transferred inside the glove-box. Colonies that had grown on that same day were marked with an asterisk.

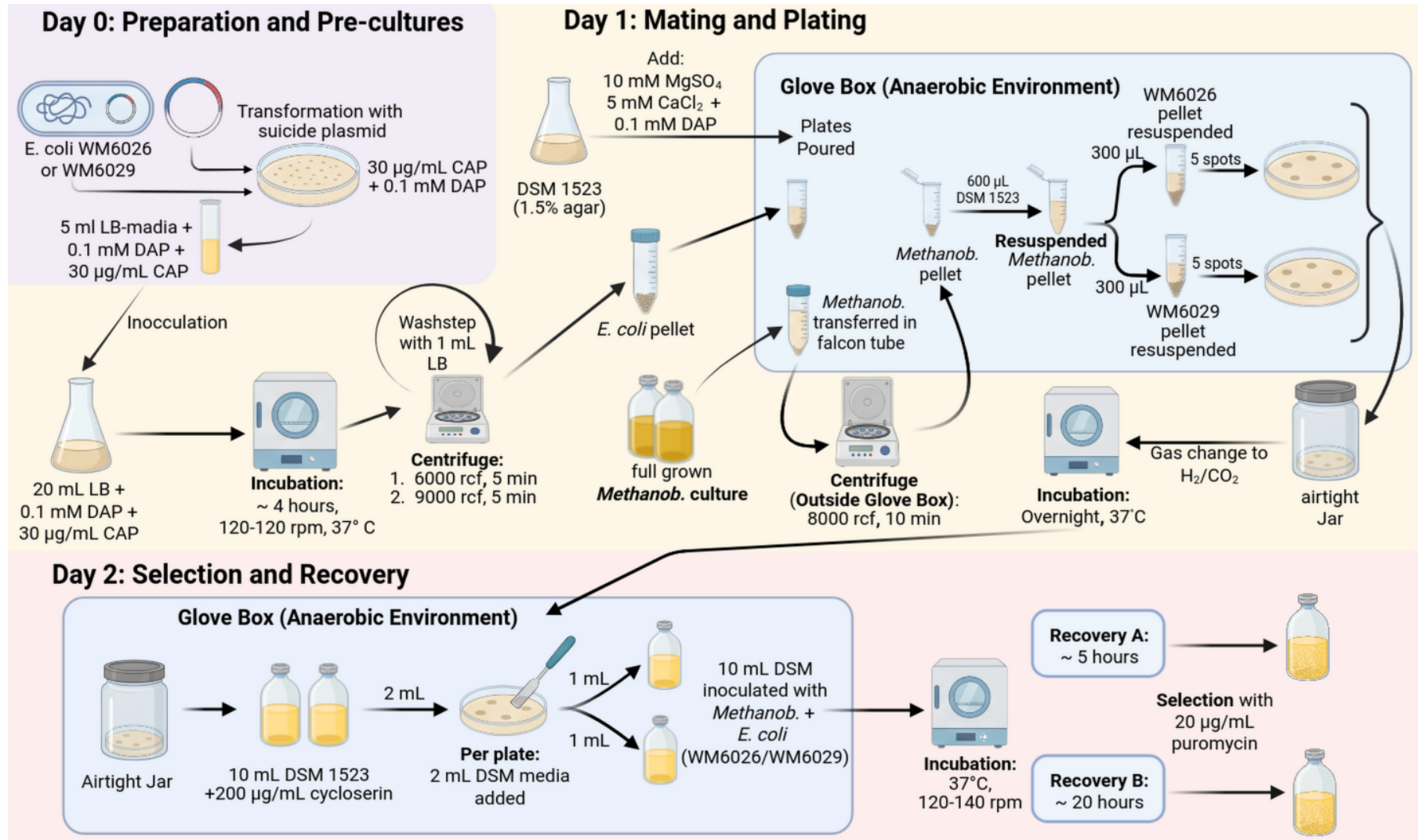


Figure 4: Methodological approach to transform a *Methanobrevibacter* (*Methanob.*) species via interdomain conjugation using two diaminopimelic acid (DAP) auxotrophic *E. coli* strains, WM6026 and WM6029. Suicide plasmids corresponding to the targeted *Methanob.* species were used, containing chloramphenicol (CAP) as a selection marker for *E. coli* and puromycin as a selection marker for the transformed *Methanob.* Two recovery periods were tested: Recovery A (approx. 5 h) and Recovery B (approx. 20 h).

Inside the glove box, *Methanobrevibacter* culture was transferred into 50 μ L falcon tubes, centrifuged (10 min, 8000 rcf) outside the glove box, and transferred back into the glove box. The *Methanobrevibacter* pellet was resuspended in 600 μ L DSM 1523. Aliquots of 300 μ L of this resuspended culture were used to separately resuspend the *E. coli* WM6026 and WM6029 pellets carrying the corresponding plasmid. These 300 μ L mixtures of *Methanobrevibacter* and *E. coli* were spot-plated (4 to 5 spots per plate) on the previously described DSM 1523 (1.5 % agar). The plates were transferred in an airtight anaerobic jar out of the glove box, the gas was changed to H₂/CO₂, and the jar was incubated overnight at 37 °C.

On the following day (Day 2), the incubated plates and two bottles per plate containing 10 mL DSM 1523 medium supplemented with 200 μ g/mL cycloserin were transferred into the glove-box. The spots were gently scraped off the plate with a spatula together with 2 mL of medium, and 1 mL of this suspension was then transferred back into each of the 10 mL bottles. The bottles were transferred back outside and incubated at 37 °C and 120–140 rpm for different recovery time periods (t_p) before adding 20 μ g/mL Puro. The 10 mL bottle per plate with the shorter recovery time was marked A (approximately 5 h), and the other B (approximately 20 h). Figure 4 visualizes the approach up to this point.

All cultures were subsequently subcultured two more times into fresh selective DSM 1523 medium containing 200 μ g/mL cycloserin and 20 μ g/mL Puro. The time between these two transfers (t_{2_1}) as well as the time from the end of recovery to the first transfer (t_{1_t}) were recorded. The gDNA of growing cultures was extracted using the NucleoSpin® Microbial DNA Kit following the manufacturer's instructions, except for the elution step, where autoclaved ddH₂O was used instead of the kit's elution buffer, and a dry centrifugation step was added before elution. For cell lysis, a bead beater (MP Biomedicals) was used. Using the gDNA as template, the 16S rDNA of archaea was amplified with primer set 22/23 (Table B7), and the bacterial 16S rDNA with primer set 1/2; a PCR with OneTaq® DNA Polymerase was performed following the instructions from NEB (USA). 16S-Sequencing was performed via Sanger sequencing (Eurofins Genomics GmbH, Luxembourg). In addition, a Q5 polymerase PCR was performed with primer set 38/39 to amplify an 838 bp fragment of the suicide plasmids used. Primer set 38/39 was checked for off-target products smaller than 4000 bp in *M. boviskoreani*, *M. millerae*, *E. coli*, and *Staphylococcus epidermidis* via NCBI Primer-BLAST (USA), using default settings for primer specificity stringency and the RefSeq reference genomes as the database.

A gel (1.5 % agarose) image was taken of this 838 bp PCR product and of two independently performed 16S rDNA amplifications using different, older primer preparations, which had previously been verified using a negative control. GelRed, Gel Loading Dye Purple (6X), and 1 kb GeneRuler were used as the ladder for the 1.5 % agarose gel.

3.1.7 Ethanol as alternative substrate

5 concentration of EtOH (10, 20, 50, 100 and 200 mM) were tested in media, where the headspace was sterile changed to N₂/CO₂ before inoculation with *M. boviskoreani*, *M. thaueri* or *M. wolinii*. For *M. boviskoreani* and *M. wolinii* DSM 1523 was used, for *M. thaueri* DSM 119. Incubation was done at 37 °C and 120–140 rpm.

3.2 Geography

The general procedure for the processing, analyses and visualization of the CRT data in R is described in the part below. For a more detailed overview, please check the R scripts.

3.2.1 Processing CRT-Data

Cattle enteric fermentation

For DEU and NZL all given data for cattle and its subcategories dairy cattle and non-dairy cattle were imported from the "Table3.A" sheet of the CRTs for each year. Then they were processed, that for each year of each country a table was created with one column with the year, one column with the cattle type and the following columns are for the measured or calculated values. Afterwards, all one year tables of one country were merged, so that one country has one table, where the values for the cattle categories from 1990 to the last published year are present.

For the USA there are no direct dairy cattle and non-dairy cattle categories, but more detailed one. So after the import of the same value columns for each cattle category as described before, these more detailed subcategories have to be merged in the same 2 as for DEU and NZL. For this purpose, the official definition and classifications from the USA (EPA, 2024, Chapter 5, p. 6) were used. DE, GE and Ym were merged as a weighted mean with the populations as weight. The methane emission were summed.

Methane emission for each sector

For sector-specific emissions, only the “TableS3” sheet was used for the most recent CRT (for the USA, the year 2022; for DEU and NZL, the year 2024). Here, all sectors and their subcategories were imported for all years, as were the total emissions. After processing the data, only the methane emission values for the sectors in general were retained, as well as the total methane emissions for each country.

3.2.2 Calculation of methane emissions from enteric fermentation processes in cattle

The total methane emission for cattle and its subcategories dairy and non-dairy was calculated following the Tier 2 approach from IPCC (2006a) via the indicators Y_m , GE and the population number of the animal categories. Therefore, the CH_4 emissions were calculated for dairy and non-dairy cattle following the eq. (1). The methane emission for cattle than is the sum of dairy and non-dairy.

$$CH_{4,T} = \frac{GE \cdot \left(\frac{Y_m}{100}\right) \cdot 365}{55.65} \cdot \frac{N_T}{10^6} \quad (1)$$

where:

- $CH_{4,T}$ = yearly methane emission of an animal category [kt CH_4 year⁻¹]
- GE = gross energy intake [MJ head⁻¹ day⁻¹]
- Y_m = methane conversion factor [%]
- 55.65 = energy content of CH_4 [MJ kg⁻¹]
- N_T = population-size of animal category
- 10^6 = conversion-factor kg to kt

3.2.3 Share of cattle methane emissions from enteric fermentation

The share of enteric fermentation was calculated for its fraction on the total amount of anthropogenically produced CH_4 in a country and on the methane emissions resulting from the agriculture sector. For this purpose, the published data in the CRT was used for each country.

3.2.4 Calculation of saved methane

To calculate the potential methane savings resulting from the mitigation strategy under investigation (immunization of cattle), various success scenarios are defined: 10, 20, 30, 40, and 50 % shifts in the rumen microbiome of cattle from methanogens toward non-methane-producing microbes. These will have an impact on two indicators used in IPCC (2019). Y_m , which represents the proportion of energy or feed consumed by cattle that is converted into methane by that microbiome, and DE , assuming that the reduced amount of feed energy converted into methane now benefits the animal during digestion, meaning that the percentage of feed digestibility for the animal increases. The values of the resulting Y_m ($Y_{m_{Red}}$) and DE (DE_{Red}) are calculated as in eq. (2) and eq. (3).

$$Y_{m_{Red}} = Y_m \cdot \left(1 - \frac{Red}{100}\right) \quad (2)$$

where:

- $Y_{m_{Red}}$ = adjusted Y_m for the investigated reduction scenario [%]
- Y_m = original methane conversion factor from CRT [%]
- Red = investigated reduction scenario [%]

$$DE_{Red} = DE + \left(Y_m \cdot \frac{Red}{100}\right) \quad (3)$$

where:

- DE_{Red} = adjusted DE for the investigated reduction scenario [%]
- DE = Digestibility of feed as a fraction of gross energy intake [%]
- Red = investigated reduction scenario [%]

The eq. (1) presents the linear relation of Y_m and methane emission of cattle already. DE , on the other hand, is named in three different equations in the calculation of IPCC (2019), shown here with little adjustments in eq. (4) through (6). It has his impact on produced methane through his role as value for the calculation of GE , which has a linear relation towards methane emission (eq. (1)).

$$GE = \frac{\left(\frac{NE_x}{REM}\right) + \left(\frac{NE_y}{REG}\right)}{\left(\frac{DE}{100}\right)} \quad (4)$$

where:

- GE = gross energy intake [$\text{MJ head}^{-1} \text{ day}^{-1}$]
- NE_x = sum of the net energy required for maintenance, activity, lactation, work and pregnancy of the animal [$\text{MJ head}^{-1} \text{ day}^{-1}$]
- REM = ratio of net energy available in a diet for maintenance (Equation (5))
- NE_y = sum of the net energy required for growth and for wool production [$\text{MJ head}^{-1} \text{ day}^{-1}$]
- REG = ratio of net energy available for growth in a diet (Equation (6))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REM = \left[1.123 - (4.092 \cdot 10^{-3} \cdot DE) + (1.126 \cdot 10^{-5} \cdot DE^2) - \left(\frac{25.4}{DE}\right) \right] \quad (5)$$

where:

- REM = ratio of net energy available in a diet for maintenance
- DE = Digestibility of feed as a fraction of gross energy intake [%]

$$REG = \left[1.164 - (5.16 \cdot 10^{-3} \cdot DE) + (1.308 \cdot 10^{-5} \cdot DE^2) - \left(\frac{37.4}{DE}\right) \right] \quad (6)$$

where:

- REG = ratio of net energy available for growth in a diet
- DE = Digestibility of feed as a fraction of gross energy intake [%]

To calculate the new methane emission for cattle corresponding to the reduction scenario, GE has to be recalculated with the adjusted DE . Afterwards eq. (1) can be used again to calculate the adjusted CH_4 emission with the adjusted GE (GE_{Red}) and Y_{mRed} . For calculation the GE_{Red} , DE_{Red} can not just simply be put in the eq. (4), because of the missing values for NE_x and NE_y . Although the values for the individual net energy intake categories (eq. (4)) should not change, these are not given in the published CRTs of the countries. To solve this problem, two things are assumed:

1. The net energy required for wool production is an irrelevant category for the investigation of cattle.
2. Most of the animal in the population of dairy and non-dairy cattle are fully grown or close to it. Because of that the net energy needed for growth will be overlooked and set to 0.

Then eq. (4) can be slimmed down and rearranged to calculate a fix value for NE_x for the corresponding year, because the part with NE_y turns zero (eq. (7)). For this purpose, GE_{Red} is calculated for a given year using DE_{Red} and the fixed values for NE_x , all due to the assumption that NE_y is 0 (eq. (8)). The calculated gross energy intake is then substituted into eq. (1) along with the corresponding Ym_{Red} value. This yields the methane emissions ($CH_{4,R}$) for a cattle category under a reduction scenario for a specific year.

$$NE_x = GE \cdot REM \cdot \left(\frac{DE}{100} \right) \quad (7)$$

$$GE = \frac{\left(\frac{NE_x}{REM} \right)}{\left(\frac{DE}{100} \right)} \quad (8)$$

where:

- NE_x = sum of the net energy required for maintenance, activity, lactation, work and pregnancy of the animal [$MJ \text{ head}^{-1} \text{ day}^{-1}$]
- GE = gross energy intake [$MJ \text{ head}^{-1} \text{ day}^{-1}$] (eq. (4))
- REM = ratio of net energy available in a diet for maintenance (Equation (5))
- DE = Digestibility of feed as a fraction of gross energy intake [%]

The saved methane emissions ($CH_{4,S}$) for a specific reduction scenario are calculated by subtracting the original value for methane emissions from the value adjusted for the scenario (eq. (9))

$$CH_{4,S} = CH_{4,T} - CH_{4,R} \quad (9)$$

where:

- $CH_{4,S}$ = saved CH_4 through reduction scenario [$kt \text{ CH}_4 \text{ year}^{-1}$]
- $CH_{4,T}$ = yearly methane emission of a animal category [$kt \text{ CH}_4 \text{ year}^{-1}$]
- $CH_{4,R}$ = yearly methane emission of a animal category for a corresponding reduction scenario [$kt \text{ CH}_4 \text{ year}^{-1}$]

3.2.5 Prediction of CH₄ emission until 2030

Three steps are required to calculate total methane emissions, as well as methane emissions for three reduction scenarios (10, 30, and 50) through the year 2030. First, total methane emissions without emissions from enteric fermentation (EnterFe) processes in cattle are calculated and then extrapolated. Second, to calculate the methane emission from EnterFe in cattle, the values for DE, Y_m, Population size and NE_x were predicted for each subcategory (dairy and non-dairy cattle) until 2030. For that, NE_x had to be first calculated (eq. (7)). From the predicted values for these indicators the cattle EnterFe CH₄ emission could be calculated (via eq. (8) and (1)). Then the methane emission values for three different reduction scenarios (10, 30 and 50) were calculated with these values using the eq. (2), (3), (8) and (1). Afterwards, the emissions of the subcategories were summed to the CH₄ emission for cattle resulting from enteric fermentation processes. Third, the predicted emission values for total without cattle enteric fermentation were added with the calculated emission values for cattle enteric fermentation to achieve the total methane emission of a country.

For this purpose, the forecast was performed by using a linear regression for each relevant value until 2030. However, the regression was not based on the entire time period, but rather on three different time intervals (the last 5, 10, and 15 years). For each linear model, both the confidence interval (CI) with a confidence level of 95 % and the total emissions were calculated. In addition, the R² value was determined for each model. The extrapolation extends from 2025 to 2030 for NZL and DEU, and from 2023 to 2030 for USA. All models with their corresponding equation and R² for each country are listed in Table B18 through Table B20.

4 RESULTS

4.1 Biochemistry

All following experiments were done with *Methanobrevibacter boviskoreani* AbM4 and *Methanobrevibacter thaueri* CW^T, in some *Methanobrevibacter wolinii* SH^T and *Methanobrevibacter millerae* SM9 were investigated as well.

4.1.1 Cultivation

M. thaueri, *M. boviskoreani* and *M. millerae* were cultivated in DSM 119 and DSM 1523. For DSM 119 no growth curve was measured, but no visual difference was detected compared to cultivation in DSM 1523. For DSM 1523 growth curves were recorded for *M. thaueri* (Figure 8 and 10), for *M. boviskoreani* (Figure 7 and 9) and for *M. millerae* (Figure 11) as a positive control of the sensitivity experiments. In general, it can be said, based on these growth-curves, that the used *Methanobrevibacter* strains reached their growth maximum within 2 to 4 days. This maximum was usually around a optical density at 600 nm (OD600) of 0.3 and does not exceed 0.4. Afterwards continuous, usually slightly decline began, so after 6 to 10 days the cultures had an OD600 below 0.1 (Figure 7, 9 and Figure 11). Further growth parameter such as generation time or growth rate were not calculated due to low time resolution (at most one measurement per day).

Nevertheless, some specific growth behavior for *Methanobrevibacter* species could be detected. So *M. millerae* and *M. boviskoreani* grew reliably in the described manner, but latter formed precipitates, which were not completely dissolvable through shaking (Figure A1). This was seen in DSM 1523 and DSM 119. *M. thaueri* on the other hand differed a bit in growth from the described manner. First, it did not grow as densely, with a maximal measured OD600 around 0.2 and clearly below 0.3 (Figure 8 and 10). Second, *M. thaueri* lysed notably faster than the other species, within 3 days the OD600 was for the positive control below 0.1 and within 5 to 6 days after maximal growth below 0.3 (Figure 8). The growth curve of positive control for the other species (Figure 7, 9 and 11) did not drop below 0.3 for the investigated time period of up to 14 days. This lysing behavior of *M. thaueri* was also visible in DSM 119 but less pronounced. In addition, *M. thaueri* grew much more unreliably. For example, in Figure 8, the maximal OD600 was reached within 3 days, whereas for the positive control in Figure 10, this was the case only after 9–15 days.

M. boviskoreani and *M. thaueri* showed no visual growth with ethanol (EtOH) as alternative substrate to H₂. EtOH concentration ranging from 10 mM to 200 mM were tested.

4.1.2 Cell count

Cells were counted for *M. boviskoreani*, *M. millerae*, *M. wolinii*, and *M. thaueri* with a phase-contrast microscope at 400 times magnification (Table B8 to Table B13). Thereby, the archaea appeared as little black dots, who tended to link together in a row or cluster (Figure A2). The cellnumber per mL for the cultures then were determined using the following calculation:

$$\text{cellnumber} \cdot \text{mL}^{-1} = \frac{\varnothing \text{ cellnumber} \cdot \text{dilution factor}}{\text{Volume of the counted Square}} \quad (10)$$

To calculate the cellnumber per mL, the arithmetic mean was determined for each culture (tab. B14) and substituted into the equation 10 along with the corresponding dilution factor (Table B14). The results of the calculation were recorded (Table 4). The calculation was performed as an example using *M. wolinii* (OD600 = 0.387):

$$\text{cellnumber} \cdot \text{mL}^{-1} = \frac{216.375 \text{ cells} \cdot 20}{4 \cdot 10^{-6} \text{ mL}} = 1,081,875,000 = 10.819 \cdot 10^8$$

Table 4: Cellnumber per mL of different *Methanobrevibacter* (*Methanob.*) culture for corresponding optical density at 600 nm (OD600). For counting the cells a phase-contrast microscope was used and a Bürker or Neubauer improved counting chamber at a 400 times magnification. The cellnumber per mL was calculated using the eq. (10), and the values were taken from the table B14.

culture	OD600	cellnumber · mL ⁻¹
<i>M. wolinii</i>	0.387	10.819 · 10 ⁸
<i>M. wolinii</i>	0.396	10.672 · 10 ⁸
<i>M. thaueri</i>	0.235	5.603 · 10 ⁸
<i>M. thaueri</i>	0.116	3.644 · 10 ⁸
<i>M. millerae</i>	0.378	8.880 · 10 ⁸
<i>M. boviskoreani</i>	0.287	9.272 · 10 ⁸

In general, the calculated cellnumber per mL for the *Methanobrevibacter* species studied ranged from 3.644 · 10⁸ cells/mL for *M. thaueri* (OD600 of 0.116) to 10.819 · 10⁸ cells/mL for *M. wolinii* (OD600 of 0.387) (Table 4). A clear trend showing, that a higher OD600 always correlates with a higher cellnumber/mL, could not be established. For example, *M. boviskoreani* (OD600 = 0.287) had a higher cellnumber per mL, at 9.272 · 10⁸, than *M. millerae* (OD600 = 0.378, cellnumber per mL = 8.880 · 10⁸), despite having a lower OD600 by nearly 0.1. Furthermore, both *M. wolinii* cultures and *M. millerae* had nearly similar OD600 values, ranging between 0.378 and 0.396 or 0.387, but the calculated cellnumber per mL of the *M. wolinii* cultures was around 2 · 10⁸ cells per mL higher (10.819 or 10.672 · 10⁸ versus 8.880 · 10⁸ cells per mL). The two *M. wolinii* cultures were quite similar to each other in terms of cellnumber

per mL and OD600, with the higher cellnumber/mL value being calculated for the culture with the lower OD600.

Besides *M. wolinii*, two cultures of *M. thaueri* were also examined to determine cell count. Here, one culture had an OD600 slightly more than twice as high as the other (0.235 versus 0.116), but a cell count per mL only slightly less than twice as high ($5.603 \cdot 10^8$ to $3.644 \cdot 10^8$ cells per mL).

4.1.3 Plating

Spread plating

M. boviskoreani and *M. thaueri* were cultivated on solid medium (DSM 1523 with 1.5 % agar) as duplicates by spread plating 100 μ L of a full-grown pre-culture. Results of the duplicates are shown in Figure 5.

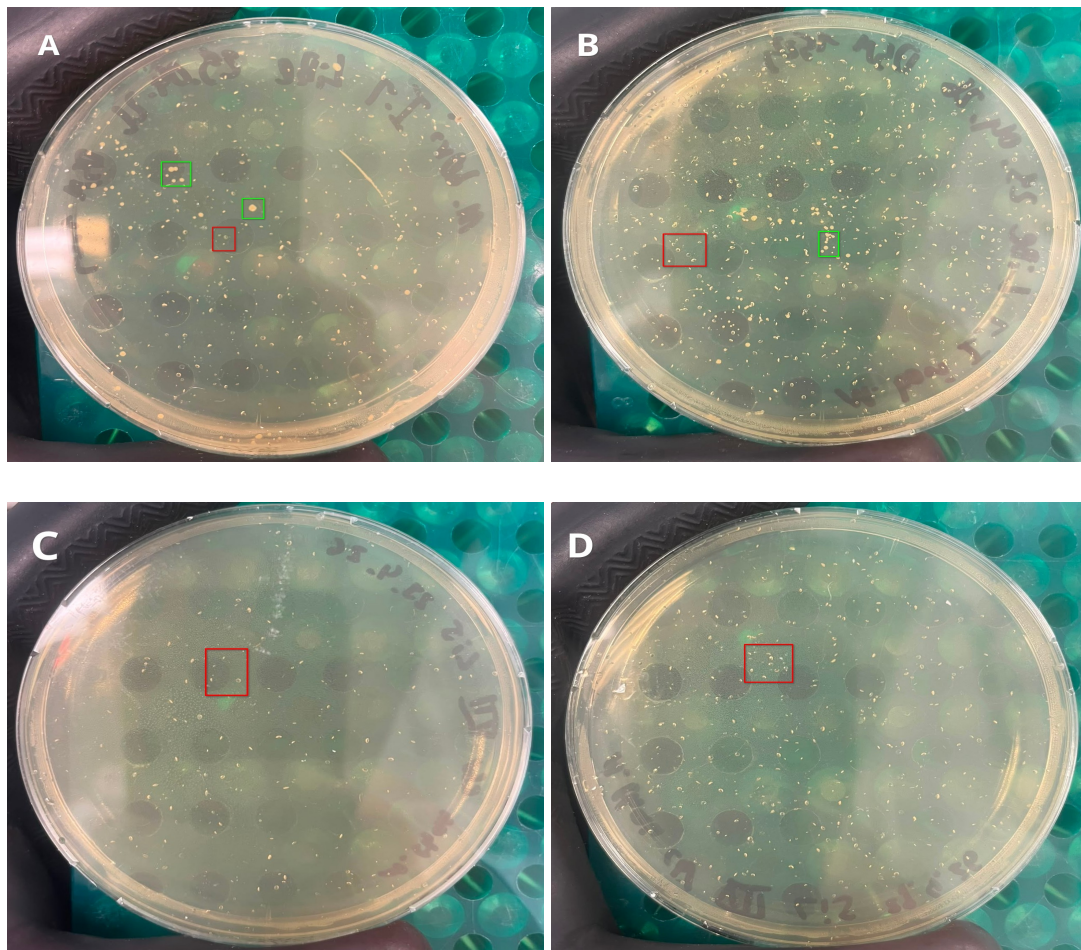


Figure 5: Spread plating results of *M. thaueri* and *M. boviskoreani*. 100 μ L of a full-grown culture was plated on DSM 1523 (1.5 % agar) in duplicates. Red rectangles highlight gas bubble spots, green rectangles grown colonies. In image A and B *M. boviskoreani* was plated and in image C and D *M. thaueri*. For plate A 47 colonies were counted and for plate B 56, on plate C and D 0 colonies were counted.

Various spots were identified on all 4 plates. Most of these were gas bubbles, as indicated by the smaller size and transparency of the spots (red rectangles in Figure 5). In contrast, a few spots were opaque, larger, round, slightly milky, and yellow (green squares Figure 5). These were grown colonies. Cultivation of *M. thaueri* with spread plating was not successful (image C and D in Figure 5). *M. boviskoreani*, on the other hand, grew on both plates with 47 counted colonies for one duplicate and 56 for the other. Single colonies were picked from the plates, propagated in liquid DSM 1523, and verified as *M. boviskoreani* via 16S-Sequencing.

Pour plating

The pour plating was not successful for 200 μ L of a full-grown *M. thaueri* culture with DSM 119 (1 % agar) as you can see in fig. A3. This experiment was also performed as duplicate.

Roll Plating

M. thaueri, *M. wolinii* and *M. boviskoreani* were roll plated in DSM 1523 (1.5 % agar) with once 200 μ L and once 400 μ L inoculation of a full-grown pre-culture. The headspace was regularly refilled for the 200 μ L preparations over a month, but not for the 400 μ L preparation. For the 400 μ L there were no visible colonies for *M. thaueri*, but tiny potential colonies for *M. boviskoreani* and *M. wolinii*. The 200 μ L, on the other hand, resulted in colonies for all three microbes used (Figure 6).

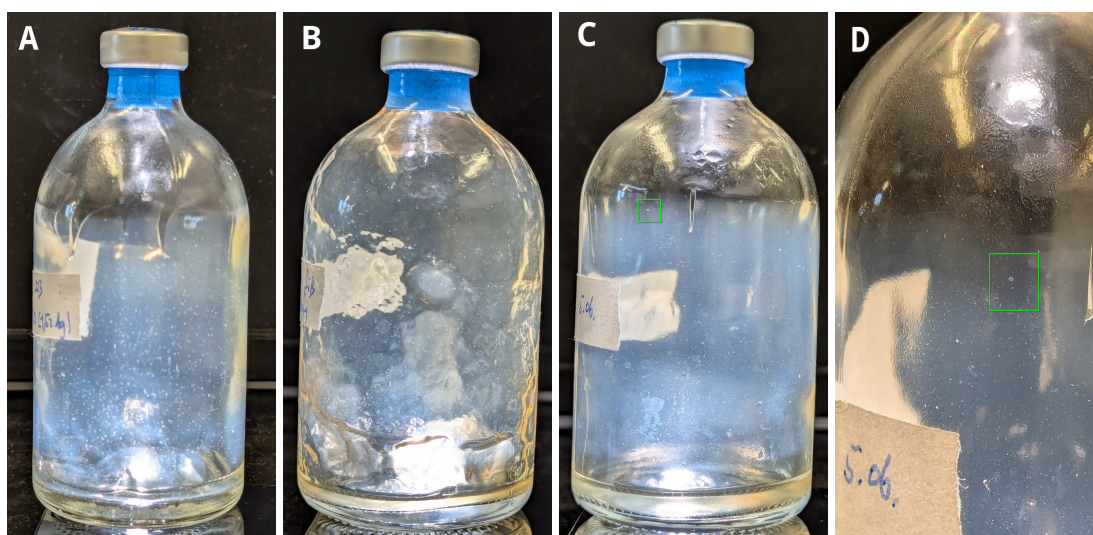


Figure 6: Roll plating of *M. thaueri*, *M. wolinii* and *M. boviskoreani* in 10 mL DSM 1523 (1.5 % agar). 200 μ L of a full-grown culture were used. Differences in the agar structure are due to methodological variations. The green rectangle marks the same colony in C and D. A: *M. boviskoreani*, B: *M. wolinii*, C: *M. thaueri*, D: *M. thaueri* zoomed in on colony.

Although in all three bottles colony forming unit (CFU) were formed (Figure 6) for *M. thaueri* were notable less colonies counted as for the other species. So for *M. thaueri* 15 colonies were counted while for *M. wolinii* 2432 and for *M. millerae* 3536 were detected, but the colonies of *M. thaueri* were larger (Figure 6). During cultivation the partial pressure of the headspace in the bottles dropped. So approximately every two to four days the gas pressure in the bottle dropped from about 1.1 bar to below 0.8 bar and had to be refilled. Verification via 16S-Sequencing, as was done with spread plating, is still pending.

Even though the OD600 of the cultures used was not measured prior to plating, the plating efficiency was estimated. This was done on the assumption that the cultures, which appeared visually full-grown, had an OD600 approximately equal to those listed in Table 4, with the higher value for cellnumber per mL always being used.

$$\text{Plating efficiency [ppm]} = \frac{\text{counted CFU}}{\text{number of used cells}} \cdot 10^6 \quad (11)$$

The calculation was based on eq. (11) and was performed as follows, for example with *M. wolinii*:

$$\text{Plating efficiency} = \frac{2432}{0.2 \text{ mL} \cdot 10.819 \cdot 10^8 \text{ cells} \cdot \text{mL}^{-1}} \cdot 10^6 = 11.24 \text{ ppm}$$

Table 5: Counted colony forming unit (CFU) and estimated plating efficiency of different *Methanobrevibacter* (*Methanob.*) cultures for the roll plating technique. 200 µL of a full-grown culture was used for plating. For *M. wolinii* and *M. boviskoreani*, only a sixteenth of the bottle was counted. The higher of the two cell number values per mL from table 4 was always used as the reference for the cell count in the initial solution. Plating efficiency was then calculated following eq. (11).

culture	counted CFU	plating efficiency [ppm]
<i>M. wolinii</i>	2432	11.24
<i>M. thaueri</i>	15	0.13
<i>M. boviskoreani</i>	3536	19.07

Table 5 lists the CFU counts per bottle as well as the corresponding plating efficiency. For *M. wolinii* and *M. boviskoreani*, the efficiency was in the lower two-digit ppm range with 11.24 ppm and 19.07 ppm. The efficiency for *M. thaueri* was notable less, at 0.13 ppm.

4.1.4 Antibiotic sensitivity

Puromycin

Growth experiment for *M. thaueri* and *M. boviskoreani* was performed in DSM 1523 media with various concentration of Puro: 0 µg/mL, 1 µg/mL, 2 µg/mL, 5 µg/mL, 10 µg/mL and 20 µg/mL. The experiment was performed in triplicates, with each antibiotic (AB) concentration

series inoculated with one full-grown pre-culture. For plotting negative measured OD600 values were set to 0.001. Figure 7 shows the results for *M. boviskoreani* and Figure 8 for *M. thaueri*.

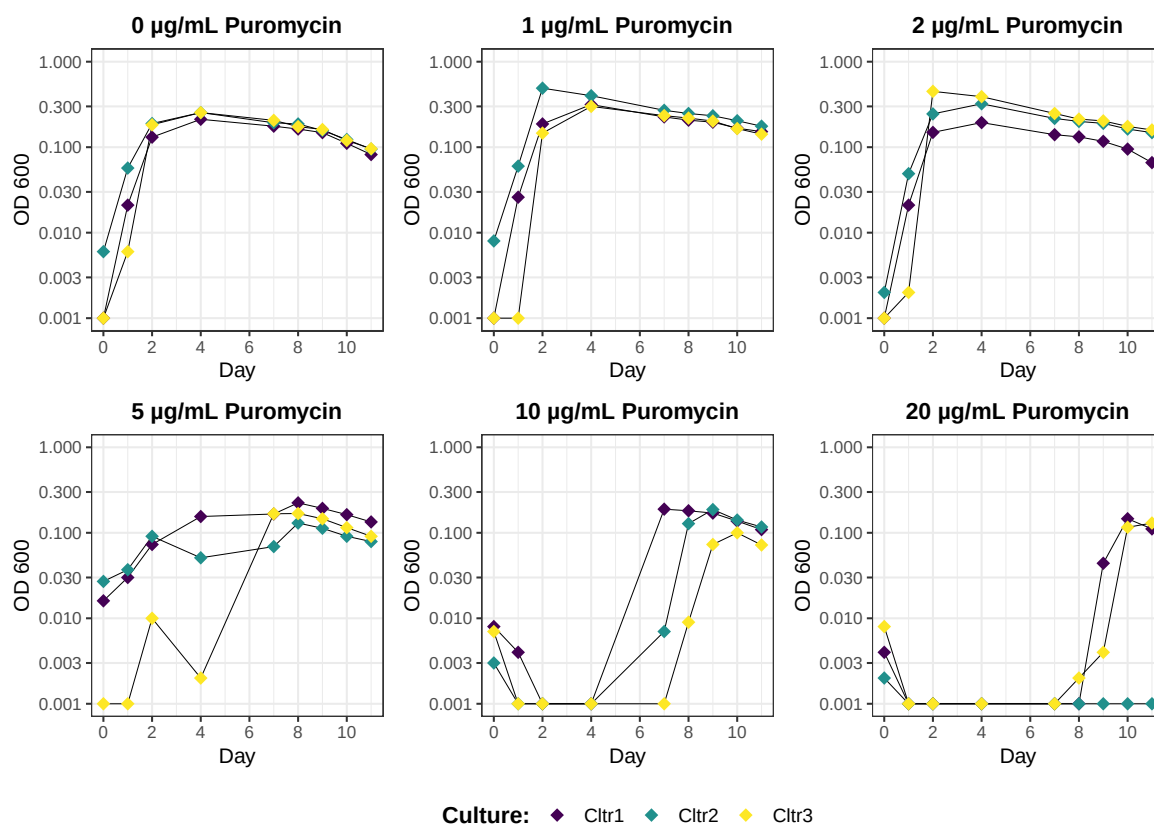


Figure 7: Growth curve in DSM 1523 for *M. boviskoreani* for various concentration of puromycin (Puro): 0 µg/mL, 1 µg/mL, 2 µg/mL, 5 µg/mL, 10 µg/mL and 20 µg/mL. Experiment was performed in triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The optical density at 600 nm (OD600) of the pre-culture were 0.110 for Cltr 1, 0.091 for Cltr 2 and 0.034 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

Fig. 7 shows that *M. boviskoreani* exhibited a very similar growth pattern for Puro concentrations of 0, 1, and 2 µg/mL. Exponential growth occurred within the first two days. This was followed by a delayed growth phase, such that the growth maximum was reached between days two and four at an OD600 of around 0.3. Until the final measurement day (day 10), the OD decreased continuously to an OD600 value of around 0.1. One difference that can still be noted here, is that the variation between the triplicates at 0 µg/mL was smaller than observed in the preparation containing Puro. In the latter case, Cltr 3 did not begin exponential growth until one day later as the other triplicates.

When *M. boviskoreani* was cultivated in the presence of 5 µg/mL Puro, growth was inhibited as the maximal OD600 was reached only after 8 days. Nevertheless, the maximum OD600 values differ between the triplicates with 0.224 for Cltr 1, 0.130 for Cltr 2 and 0.168

for Cltr 3. They are also slightly lower than in the previous growth curves. In addition, the triplicates did not show comparable growth curves: Cltr 3 only started growing after 4 days, whereas the other two started growing immediately but rather slowly, and Cltr 2 dipped in between. After the maximal OD600 was reached the values decreased continuously.

At 10 and 20 $\mu\text{g/mL}$ Puro concentrations, exponential growth began with a notable delay of 4 to 7 days (10 $\mu\text{g/mL}$) and 7 to 8 days (20 $\mu\text{g/mL}$). In addition, at 20 $\mu\text{g/mL}$ one replicate (Cltr 2) showed no growth at all after 11 days. For 10 $\mu\text{g/mL}$ the max. OD600 was reached between 7 and 10 days, with maximum OD600 values ranging from 0.100 (Cltr 3) to 0.189 (Cltr 1). Afterwards the values declined continuously. Triplicates Cltr 3 and Cltr 1 of the 20 $\mu\text{g/mL}$ Puro preparation reached their maximal OD600 of around 0.14 on day 10. By day 11, a slight decrease in the OD600 values could already be observed.

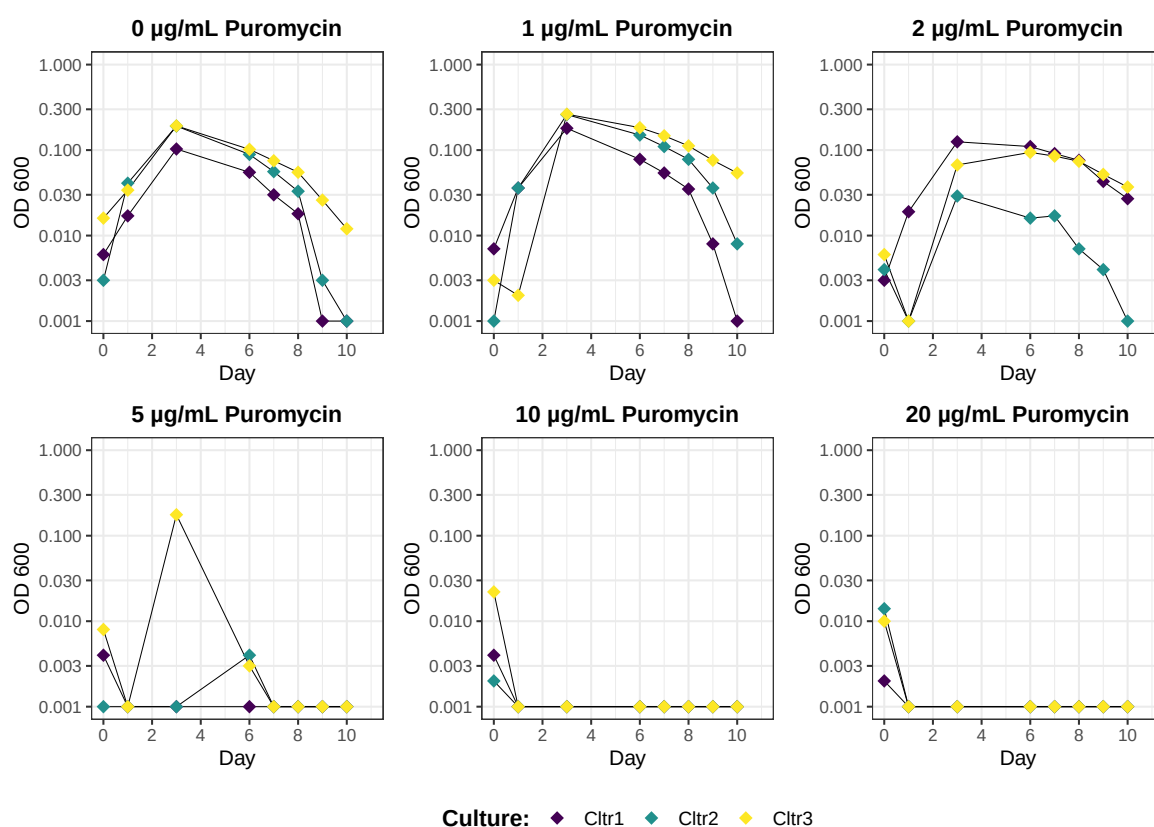


Figure 8: Growth curve in DSM 1523 for *M. thaueri* for various concentration of puromycin (Puro): 0 $\mu\text{g/mL}$, 1 $\mu\text{g/mL}$, 2 $\mu\text{g/mL}$, 5 $\mu\text{g/mL}$, 10 $\mu\text{g/mL}$ and 20 $\mu\text{g/mL}$. Experiment was performed in triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The OD600 of the pre-culture were 0.154 for Cltr 1, 0.173 for Cltr 2 and 0.120 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

The growth curves of *M. thaueri* for Puro concentrations of 0 and 1 $\mu\text{g/mL}$ reached their OD600 maximum after 3 days, similar to *M. boviskoreani* (Figure 8). However, the maximum did not exceed 0.3 and ranged from 0.103 (Cltr 1, 0 $\mu\text{g/mL}$) to 0.263 (Cltr 3, 1 $\mu\text{g/mL}$). After

that, the OD600 decreased rapidly again. For example, at 0 µg/mL Ctr 1 and 2 already had an OD below 0.003 after 9 days, and Ctr 3 was also just barely above 0.01 after 10 days. For 1 µg/mL, the final measured OD values ranged between 0.001 and 0.05.

For 2 µg/mL Puro, Figure 8 showed growth for all three triplicates, reaching its maximum on day 3. However, the increase in OD600 values is notably lower. Thus, the maximum for Ctr 2 was below 0.03, for Ctr 3 below 0.1 and for Ctr 1 bit more than 0.1. This was followed by another decrease in the values, so that the final OD600 ranged between 0.001 and 0.03. For the other measured Puro concentrations, in contrast to *M. boviskoreani*, no growth above an OD600 of 0.003 was observed with one exception (Ctr 3 at 5 µg/mL) (Figure 7 and Figure 8). For Ctr. 3 at a Puro concentration of 5 µg/mL, an OD600 of more than 0.1 was measured on day 3, but this value dropped back to 0.003 by day 6.

Cycloserin

M. thaueri was successfully cultivated in DSM 1523 containing 200 µg/mL cycloserin (Cyc-Se.) as duplicates. Visual growth could be detected after 1 day.

4.1.5 Nucleotide-base analogs sensitivity test

Similar to the AB sensitivity test, a growth experiment was conducted for the base analog 8-Azahypoxanthin (8-Aza) in *M. thaueri*, *M. boviskoreani*, and *M. millerae*. One difference is that the base analog is not dissolvable in H₂O, so it was dissolved in dimethylsulfoxid (DMSO). Furthermore, different concentrations were selected: 0 µg/mL, 100 µg/mL, 250 µg/mL, 500 µg/mL, 750 µg/mL, 1000 µg/mL, and one with 1 mL of DMSO. The OD600 was measured and plotted on a logarithmic scale against time (Figures 9 through 11).

8-Azahypoxanthin

In the absence of 8-Aza, the maximal OD600 of *M. boviskoreani* was reached after 3 to 4 days with exponential growth observed within the first two days (Figure 9), followed by a continuous decline to a final OD600 of just under 0.1 on day 13. The solvent control (DMSO), in which no base analogs were added but only 1 mL of the solvent DMSO, showed a different growth curve. Initially, none of the triplicates began to grow. Instead, the OD600 values first decreased. On day 3, slight growth started, reaching a maximum of 0.1 on days 4 to 6. Afterward, the values declined to approximately 0.03 on day 13.

At the lower concentrations of 8-Aza (100 and 250 µg/mL), *M. boviskoreani* showed a growth curve shape more or less similar to that of the corresponding positive control (Figure 9).

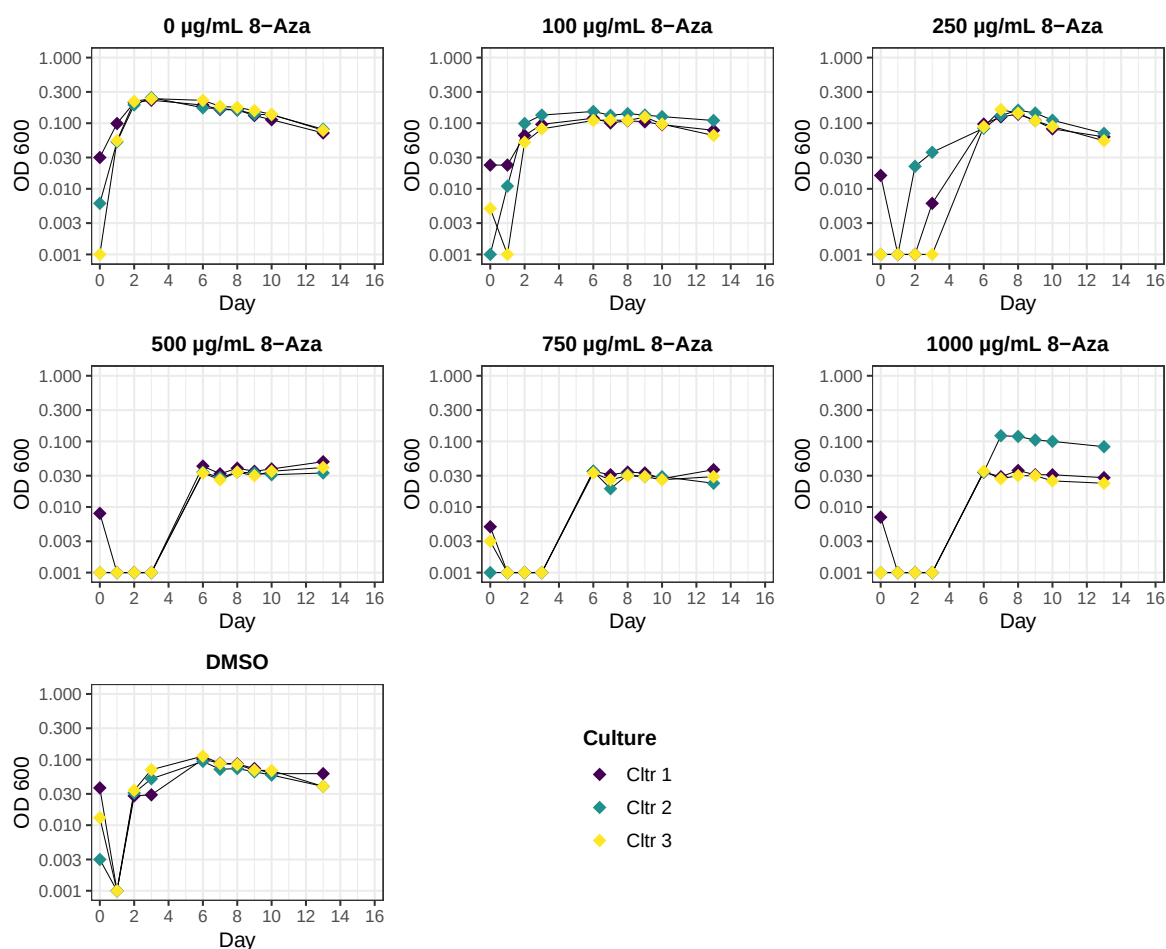


Figure 9: Growth curve in DSM 1523 for *M. boviskoreani* for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/mL, 100 µg/mL, 250 µg/mL, 500 µg/mL, 750 µg/mL, 1000 µg/mL) and one bottle with 1 mL DMSO as solvent of 8-Aza. Experiment was performed in triplicates (Ctr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The OD600 of the pre-culture were 0.096 for Ctr 1, 0.105 for Ctr 2 and 0.152 for Ctr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

However, at 100 µg/mL 8-Aza, the maximum OD600 value was reached within 1 to 3 days, but it was approximately 0.1 instead of just under 0.3. For the 250 µg/mL preparation 0.1 was the maximum OD600 value as well. Here, growth did not begin until 2 to 4 days later, therefore the maximum was not reached until day 7.

For the three remaining 8-Aza preparation lines (500, 750 and 1000 µg/mL) for *M. boviskoreani* an OD600 below 0.01 was measured during the first 4 to 6 days (Figure 9). Then there was an increase to an OD600 of 0.03, which remained constant until the final measurement day (day 13). An exception was Ctr 2 in the 1000 µg/mL 8-Aza preparation. This triplicate grew from day 3 to day 5 to an OD600 of 0.1, which then decreased slightly until the end of the measurement series.

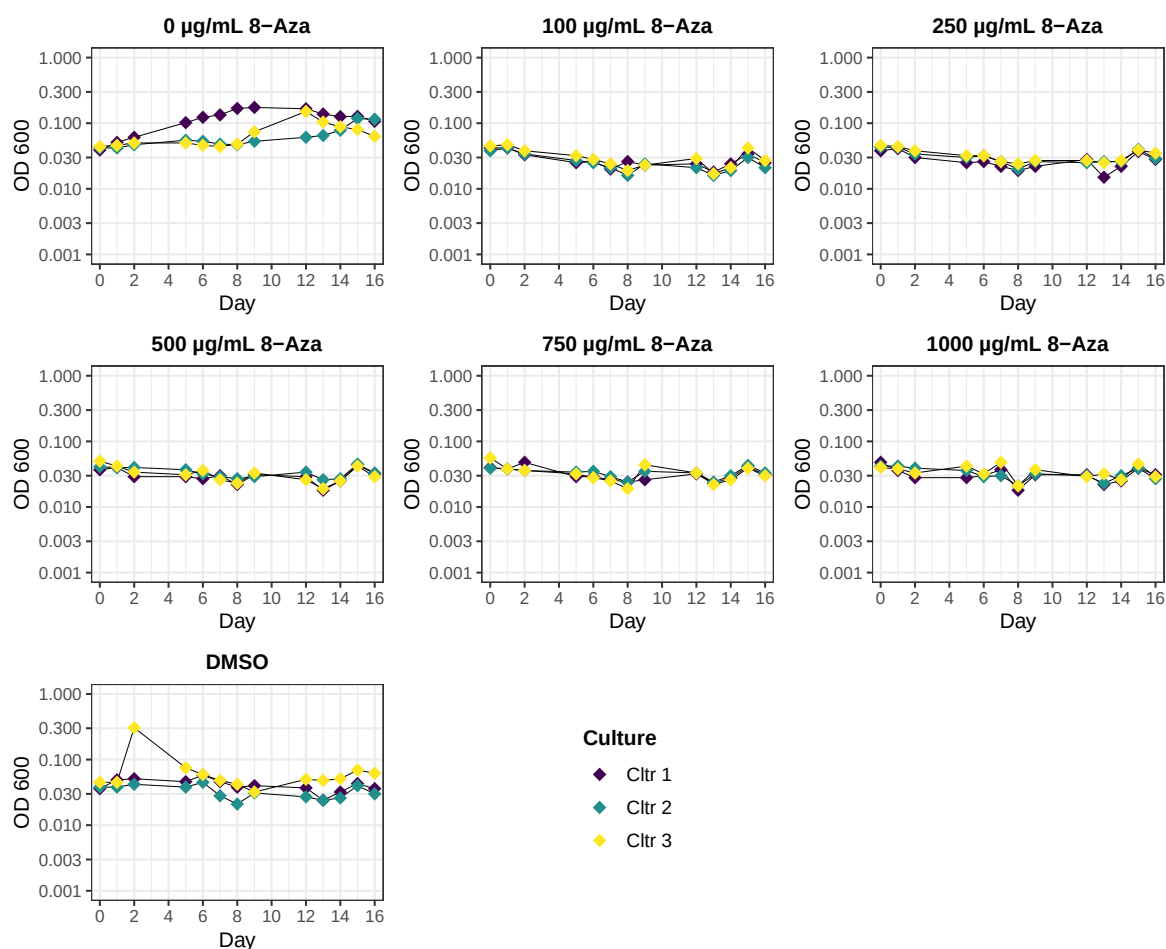


Figure 10: Growth curve of *M. thaueri* for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/mL, 100 µg/mL, 250 µg/mL, 500 µg/mL, 750 µg/mL, 1000 µg/mL) in DSM 1523 and one bottle with 1 mL DMSO as solvent of 8-Aza. Experiment was performed in triplicates (Cltr. 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The OD600 of the pre-culture were 0.125 for Cltr 1, 0.148 for Cltr 2 and 0.210 for Cltr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

M. thaueri growth curves are visualized in Figure 10. Except for the positive control (0 µg/mL 8-Aza), no growth could be detected. An exception here was the triplicate Cltr 3 in the DMSO sample. Here, an OD600 of 0.3 was measured on Day 2, but it dropped to 0.03 by the next measurement day (day 4). In the positive control, slight growth of *M. thaueri* was detected. However, the growth curve differed noticeably from that of the positive control for *M. thaueri* in the Puro sensitivity experiment (Figure 8). Furthermore, there was considerable heterogeneity among the individual triplicates (Figure 10). Cltr 1 began to grow from the first day and reached its maximum OD600 0.174 on day 9 (see Figure 10). By the end of the measurement, the OD600 decreased continuously to 0.1. Cltr 3 began to grow on Day 8, reaching a maximum OD600 of approximately 0.151 on Day 12. Between Days 13 and 14, the last triplicate then started to show slight growth, reaching an OD600 of 0.118 on Day 16.

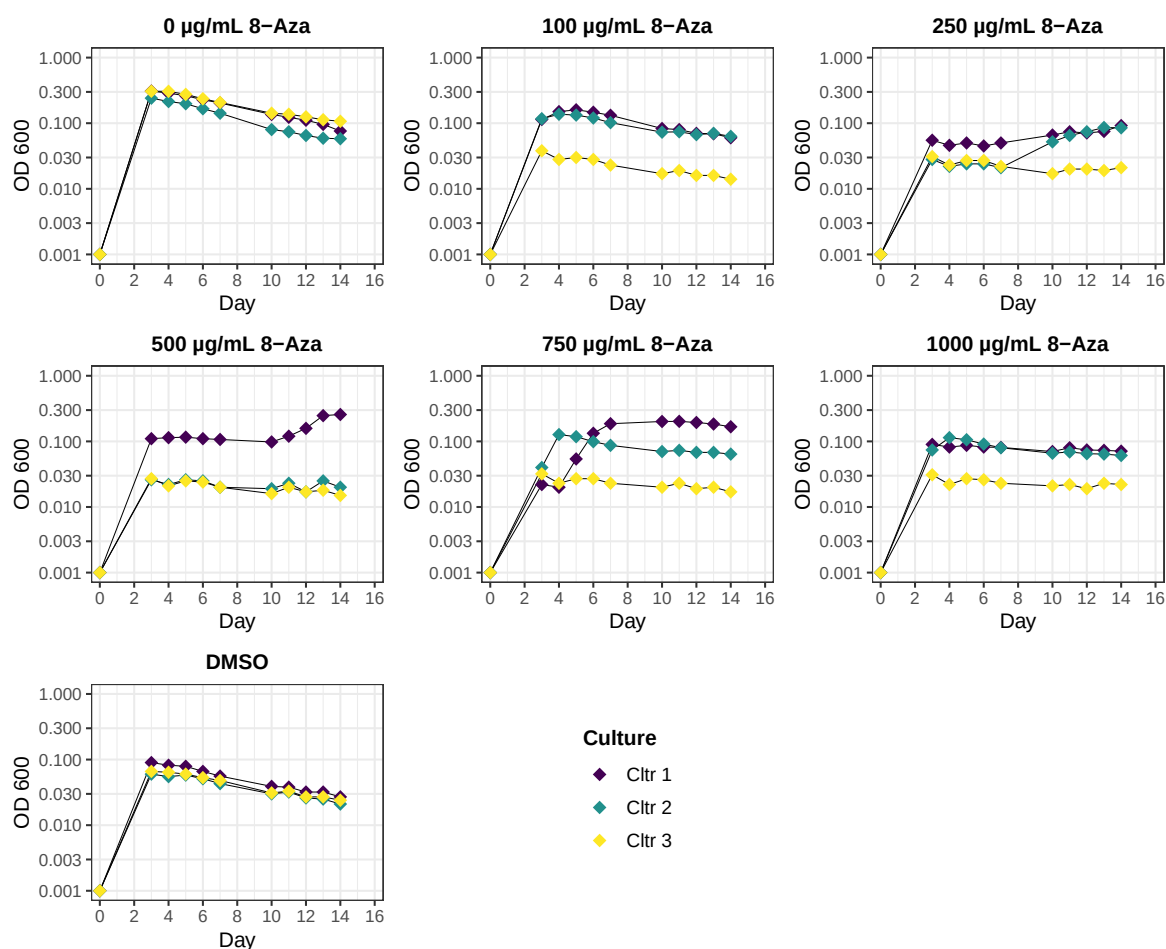


Figure 11: Growth curve of *M. millerae* for various concentration of 8-Azahypoxanthin (8-Aza) (0 µg/mL, 100 µg/mL, 250 µg/mL, 500 µg/mL, 750 µg/mL, 1000 µg/mL) in DSM 1523 and one bottle with 1 mL DMSO as solvent of 8-Aza. Experiment was performed in triplicates (Ctr 1, 2 and 3), with each triplicate concentration series is inoculated with its own full-grown pre-culture. The OD600 of the pre-culture were 0.366 for Ctr 1, 0.339 for Ctr 2 and 0.281 for Ctr 3. The OD600 was plotted on a logarithmic scale against time. OD600 values less than 0.000 were set to 0.001. A non-inoculated DSM 1523 medium solution was used as a blank.

M. millerae growth results are visualized in Figure 11. The positive control (0 µg/mL 8-Aza) and the solvent control (DMSO) had a similar curve shape for all triplicates, with the maximum OD600 being reached on day three, followed by a continuous decline. However, the curves differed in their OD600 values. The positive control reached a maximum value of 0.3, while the solvent control did not exceed 0.1. This resulted in a lower final OD600 measurement for DMSO of below 0.03, and slightly below 0.1 for the positive control. Clear trends for the other samples were generally difficult to discern at first glance. However, most triplicates did not reach an OD600 higher than 0.1.

The 100 µg/mL 8-Aza generally followed the curve shape of the positive and solvent control, although triplicates Ctr 1 and 2 reached their maximum OD600 value on day 4 with a value around 0.15, while Ctr 3 did not exceed 0.05. For the 250 µg/mL concentration, the triplicates had a constant OD600 of approximately 0.03 since day 3, although Ctr 3 showed

a highwe OD600 of aprox. 0.05. While the value remained constant for Cltr 3, Cltr 1 and 2 showed slight growth starting on day 7, reaching approximately 0.1 by the end of the measurement. The triplicates for Cltr 2 and 3 at the 500 µg/mL concentration showed a constant OD600 value of 0.03 since day 3. For Ctr. 1, an OD600 of 0.1 was measured between Days 2 and 10, which increased to just under 0.3 by Day 13. The higher concentrated preparations (750 µg/mL and 1000 µg/mL 8-Aza) showed for triplicate Ctrl 3 a constant value of 0.03 starting on day 2. At 1000 µg/mL, the other two triplicates showed a constant OD600 of around 0.1 starting on day 3, while at 750 µg/mL, only Cltr 2 showed this value since the third day. The Cltr 1 triplicate in the 750 µg/mL solution showed a value of 0.03 on days 3 and 4, then increased to an OD600 between 0.2 and 0.3 by day 7. This value decreased slightly to the end of the measurement series.

DMSO

In general, all preparation with DMSO showed the tendency to be more vulnerable to turn red, when taking a sample for photometric measurement. In addition, the decolorization also took longer than in the positive control or other experimental conditions without DMSO, for example, sometimes over an hour. This interaction of DMSO with the DSM 1523 media was investigated more closely in the *M. thaueri* 8-Aza sensitivity experiment (Figure 10). Here, an extra preparation was prepared with just 1 mL DMSO added in the DSM 1523, but without inoculation (DMSO –, Figure 12).

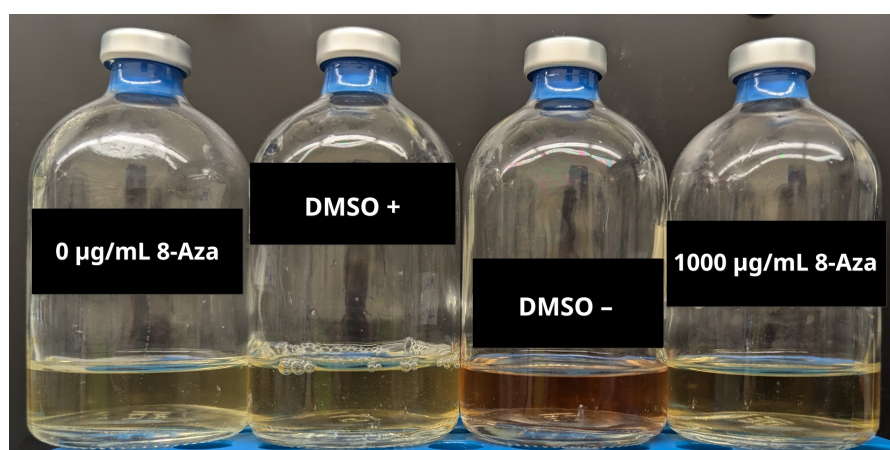


Figure 12: Color differences with and without DMSO after 6 days of incubation. Image of various preparations for *M. thaueri* from the same triplicate (Cltr. 3). The positive control (0 µg/mL 8-Azahypoxanthin (8-Aza)) contains only the microbe (*M. thaueri*) in DSM 1523; 1000 µg/mL preparation 8-Aza *M. thaueri* and the named base analog; DMSO + *M. thaueri* and 1 mL DMSO; DMSO – also contains 1 mL DMSO but was not inoculated.

After the first few days of incubation, a difference in color was observable among the cultures. Specifically, the DMSO – bottle exhibited a red color, in contrast to the yellow color of the other samples in the 8-Azahypoxanthin (8-Aza) measurement series for *M. thaueri*.

This is illustrated in Figure 12 using selected samples from the triplicate Cltr. 3 after 6 days of incubation. The OD600 was also measured for DMSO, but showed no difference from the baseline OD600 values of the other samples ranging from 0.03 to 0.04.

4.1.6 *pac*-gene

The *pac*-gene in the initial plasmid (*pac_old*), that should have been taken for the transformation via conjugation of the archaea, showed a different amino acids (AA) sequence to the original sequence (*pac_org*) from *Streptomyces alboniger* ((a) in Figure 13). They differ in five AA on the C-terminal end ((a) in Figure 13).

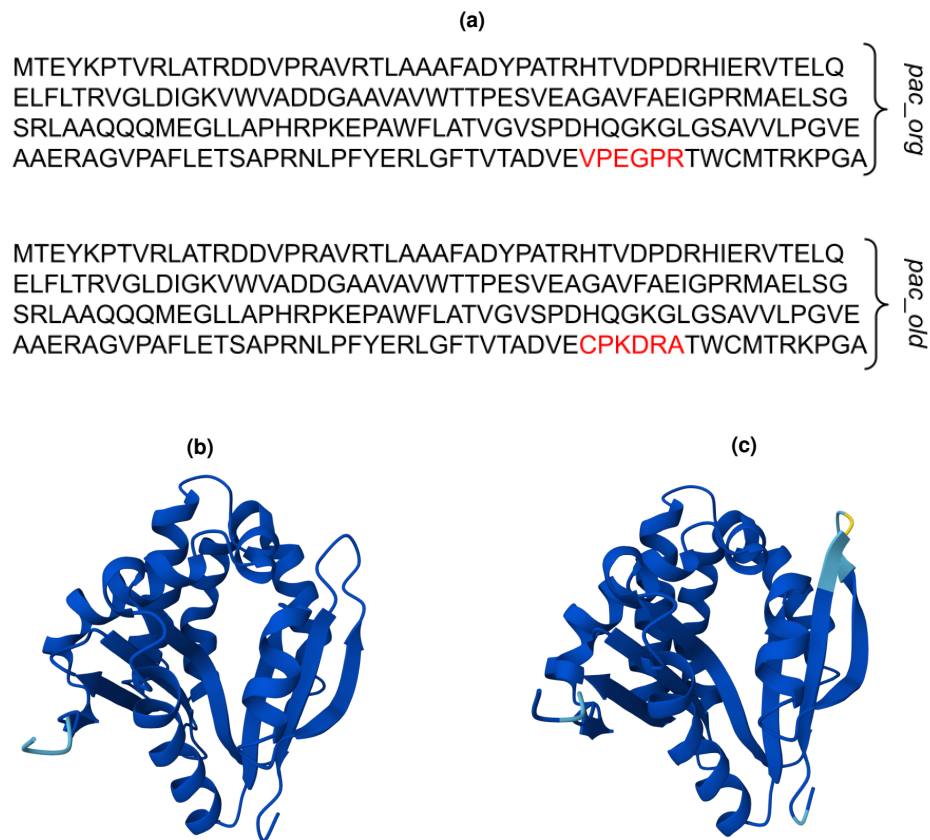


Figure 13: Amino acids (AA) sequence of two *pac* genes and their 3-D structures. (a) shows the AA sequence of the original sequence from *Streptomyces alboniger* (*pac_org*) as well as from the *pac* gene that was located in the plasmid originally intended for transformation (*pac_old*). The differing section is highlighted in red. (b) The projected 3D structure for *pac_org* calculated by AlphaFold. (c) The projected 3D structure for *pac_old* calculated by AlphaFold. The color of the structure in AlphaFold shows the confidence of the predicted structure: blue, very confident, light blue less and yellow not very confident.

These mutation led in AlphaFold to different 3-D structures. The original *pac* had an antiparallel β -sheet with a big loop on the right side of its structure ((b) of Figure 13). The *pac_old* ((c) in Figure 13) showed also an antiparallel β -sheet at this position, but it was longer and had a shorter loop. Interestingly is, was not so confident in its prediction for this region as before (color change between (b) and (c) in Figure 13).

4.1.7 Transformation via conjugation

M. thaueri, *M. boviskoreani*, and *M. millerae* were used for the transformation via interdomain conjugation with two auxotrophic *E. coli* strains, WM6026 and WM6029 (Section 3.1.6). Of the 28 preparations, 3 preparation lines showed visible, sustained growth after being transferred into new selective media containing 20 µg/mL Puro and 200 µg/mL Cyc-Se. (highlighted red in Table B16). It should be noted here that most of the preparations showed growth in the recovery media state, but grew atypically dense for a pure *Methanobrevibacter* culture, and only in the three stated preparation lines could growth still be detected later on. The 6.6A preparation, which was inoculated with *M. boviskoreani** (Table B16), on the other hand, showed the typical precipitates for *M. boviskoreani*. The other two successful preparation lines, 5.6A and 5.6B, contained *M. millerae*.

All three successful preparation lines were inoculated with fresh grown *Methanobrevibacter* culture (*M. boviskoreani** and *M. millerae**) and all had as conjugation strain *E. coli* WM6026. On the other hand, they had different recovery times: two belong to the preparation that had a short recovery time of 4.5 to 5 h (5.6A and 6.6A), and one with a long recovery time 19.5 h (5.6B). Furthermore the OD600 of *E. coli* WM6026 for 5.6A and 5.6B was 1.24 and for 6.6A 1.79 (Table B16).

Specifically, within these three preparation lines, six cultures at different stages of the transfer process showed growth. 2 for 5.6B (1st and 2nd transfer), one for 5.6 A (1st transfer), and three cultures for 6.6A (1st and 2nd transfer plus the recovery culture). All six were transferred two more times, and the 1st and 2nd transfer of 6.6A were transferred a third time after 6 days. For all a similar behavior could be detected. The 2nd and 1st transfer of 6.6A grew very well while the others grew but not as dense. Furthermore, were five of these six cultures within a preparation line with a shorter recovery time (Table B16). Sequencing of the 16S-DNA of archaea revealed for these six cultures, that the cultures contain the expected *Methanobrevibacter* species. The negative control of the used primers did not show any bands. The sequencing of bacterial 16S-DNA showed no sign of *E. coli* WM6026, but a contamination with *Staphylococcus epidermidis* (*S. epidermidis*). A negative check of the original used primer revealed also a contamination of them with *S. epidermidis*. Two new primer sets were used, that did not show a contamination in the negative control, the results are shown in Figure 14 ("bac. 16S fresh" and "bac. 16S extra" lanes). Again a clear band appears at approx. 1000 bp, which was verified as *S. epidermidis* via 16S-Sequencing. The band thereby was pretty thick and tended to smear.

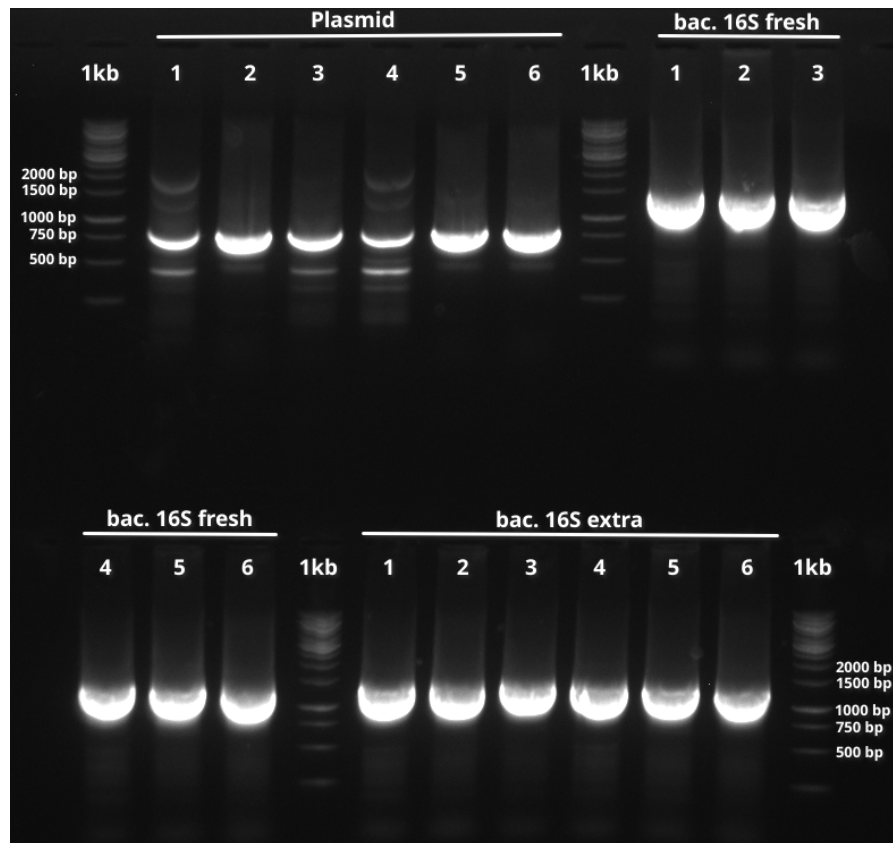


Figure 14: Gel image for different PCR products of growing cultures of *M. boviskoreani* and *M. millerae* after conjugation. For the lanes labeled 1, gDNA from culture 5.6B (1st transfer) was used; for 2, from 6.6A (1st transfer); for 3, from 5.6A (2nd transfer); for 4, from 5.6B (2nd transfer); for 5, 6.6A (2nd transfer); for 6, 6.6A (recovery). For a closer look at the cultures see Table B16. "Plasmid" displays the results of the PCR for a portion of the plasmid used in the conjugation. The expected size would be 838 bp. "bac. 16S fresh" and "bac. 16S extra" show both the PCR product for the bacterial 16S-rRNA. Thereby were different primers used, but both had a clear negativ control before. For "bac. 16S fresh" the primers were freshly aliquoted. For "bac. 16S extra" primers different from those used in the first 16S sequencing were used. As ladder the 1 kb ladder from ThermoFischer was applied ("1kb" lanes).

Figure 14 presents also the result for the PCR with a primer set (38 and 39, Table B7) for a 838 bp part of the suicide plasmid. All six cultures (lane 1 to 6 of the "Plasmid" lanes) showed a clear band at 750 bp, which was a bit less thick as the 1000 bp band of the bacterial 16S-rRNA. For the lane 1,3 and 4, which belonged to the three *M. millerae** (1st (lane 4) and 2nd transfer (lane 1) of 5.6B plus 1st transfer (lane 3) of 5.6A), there was a second smaller band slightly below 500 bp. Furthermore, was the band at 750 bp a bit smaller than the band of *M. boviskoreani** cultures. "Primer Blast" on NCBI did not reveal any off-targets with PCR products smaller than 4000 bp in the gDNA of *M. millerae*, *M. boviskoreani*, *E. coli* and *S. epidermidis*.

A full genome sequencing of the culture is still pending due to the contamination problems.

***E. coli* cultivation test in DSM 1523**

Cultivation of *E. coli* DH5 α was tested under similar settings in DSM 1523 as for *Methanobrevibacter*. Three drops for inoculation was enough to get a dense culture the next day.

4.2 Geography

4.2.1 National methane emission since 1990

Figure 15 shows the development of the total methane emission of a year as well as of the specific sectors from 1990 to 2020s for Germany (DEU), New Zealand (NZL) and United States of America (USA). The first thing that stands out is, that the USA generally produced notably more CH₄, what cumulates in a different scale than for the other two countries. The maximum total emission was 35 Tg CH₄, while for DEU the maximum is 5 Tg and for NZL it was 1.5 Tg.

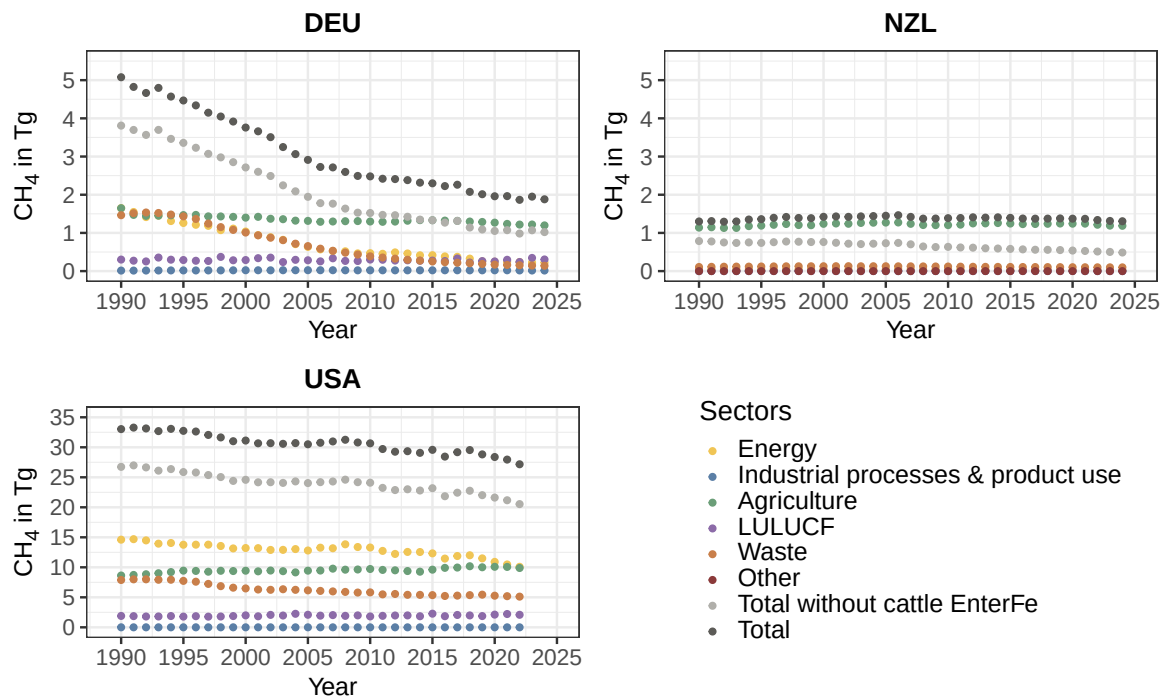


Figure 15: Time series of the anthropogenic methane emission of Germany (DEU), New Zealand (NZL) and United States of America (USA). The different relevant emission sectors are shown with their total sum (Total). Furthermore, the total emission minus the CH₄ produced in enteric fermentation (EnterFe) processes of cattle is plotted as well. Methane is plotted as Tg CH₄ of a year against the time.

For DEU, a clear downward trend was evident in total CH₄ emissions (Figure 15). Emissions fell around 3 Tg CH₄ in the 34-year-long time series from 5 Tg in 1990 to 2 Tg in 2024, representing a decrease of approximately 60 %. Two distinct phases of this trend could be

identified: first, from 1990 to 2006, with a sharp decline of about 2.4 Tg CH₄ (approximately 2.6 Tg CH₄ in 2006), followed by a second phase with a more moderate decline of around 0.6 Tg to 2 Tg of anthropogenically produced methane in 2024. Total emissions excluding those from enteric fermentation (EnterFe) in cattle followed a similar trend, with the curve's value being around 1 Tg CH₄ lower (Figure 15 DEU).

Looking at the sectoral emissions of DEU, it became apparent that in 1990 there were three equally large CH₄-producing sectors: energy, waste, and agriculture each accounting for approximately 1.5 Tg CH₄ (Figure 15). However, these sectors developed differently. The energy and the waste sector followed a similar trend. Following a sharp decline between 1990 and 2006 - with the waste sector not beginning to change until 1994 - from 1.5 to 0.5 Tg, a moderate decline then set in, until approximately 0.025 Tg of CH₄ was emitted in 2024. Agriculture was also changing, though only very slightly compared to the other two sectors: from 1.6 Tg of methane in 1990 to approximately 1.25 Tg CH₄ in 2024, with the largest single change occurring between 1990 and 1991, amounting to a difference of about 0.1 Tg. So in 2024 the anthropogenic methane emission of DEU was dominated by the agriculture sector, which accounts for 60 % of the total emission. In addition to these three sectors, two others are shown in Figure 15 (industrial processes & product use and LULUCF (land use and land use change and forestry)). The industrial processes & product use sector was a negligibly small source of methane emissions. Its CH₄ emissions had remained consistently well below 0.1 Tg over the 35-year period. With CH₄ emissions consistently ranging between 0.3 and 0.4 Tg, the LULUCF sector was the fourth-largest single methane emission sector in DEU during the 1990s, but it surpassed the waste sector in 2016 and has been the second-largest CH₄-emitting sector in DEU after agriculture since 2019.

For the USA the distance between total methane emissions and total emissions excluding cattle EnterFe remained fairly constant at around 10 to 15 Tg CH₄ (Figure 15). In addition, a clear, steady decline could be observed for the total methane emission from 1990 (approx. 33 Tg CH₄) to 2022 (approx. 27 Tg CH₄), although the curve did not follow the trend linearly but rather fluctuated slightly. Over this 32-year period, the U.S. had reduced its methane emissions across all sectors by about 6 Tg CH₄, which corresponded to a reduction of 18 %. As in DEU, the three most relevant methane-producing sectors in the U.S. were energy, agriculture, and waste in 1990. However, here the energy sector was the largest contributor, accounting for nearly half of the emissions in 1990 with 15 Tg CH₄, followed by the waste and agriculture sectors with approximately 8 Tg CH₄ each. The energy and the waste sector showed a downward trend over the time series (energy: from 15 to 10 Tg CH₄ and waste: from 8 to

5 Tg CH₄), while emissions from agriculture rose slightly over time, from 8.5 Tg to 10 Tg CH₄. Thus, in 2022, agriculture and energy - each at 10 Tg CH₄ - were the largest methane emission sectors. Together they accounted for more than two-thirds of total methane emissions in the U.S. in 2022. Two other sectors contributed to U.S. methane emissions: the industrial processes & product use sector, which was relatively small and therefore negligible, and the LULUCF sector, with a constant range of 0.2 to 0.25 Tg CH₄.

Compared to the other two countries, NZL presented a different picture (Figure 15). First, total CH₄ emissions in 2024 were nearly the same as those in 1990, namely about 1.4 Tg CH₄. Furthermore, a slight increase in emissions to 1.5 Tg CH₄ could be observed through 2006, followed by a slight decline. Second, agriculture was the absolute dominant sector in terms of methane emissions and accounted for nearly 90 % of CH₄ emissions from NZL (Figure 15). All other sectors accounted for notably less than 0.1 Tg CH₄ per year, with one exception for the waste sector. The waste sectors methane emissions were slightly above 0.1 Tg CH₄. Third, the difference between total methane emissions and total emissions excluding cattle EnterFe increased by about 60 % (0.5 Tg CH₄ in 1990 and 0.8 Tg in 2024).

4.2.2 Enteric fermentation processes of cattle

Methane emissions

Figure 16 shows the trend in methane emissions from enteric fermentation in cattle, as well as their subcategories dairy cattle and non-dairy cattle for each country.

Cattle emissions for DEU declined from 1990 to 2024 (from 1.25 Tg CH₄ to approximately 0.8 Tg), with a steeper decline in the first year of the time series, followed by a more gradual decline (Figure 16). For dairy and non-dairy cattle, this initial decline was evident as well (Figure 16). Then a slight decline could be observed (dairy cattle: 0.75 Tg CH₄ in 1990 to 0.5 Tg in 2024; non-dairy cattle: 0.55 in 1990 to 0.35 in 2024). Dairy cattle were an exception: their emissions dropped to 0.5 Tg CH₄ by 2006, then rose slightly again through 2015, and subsequently declined once more, returning to 0.5 Tg by 2024. Dairy cattle consistently produced slightly more CH₄ per year than non-dairy cattle (0.1–0.2 Tg CH₄, Figure 16). This contrasts with the population ratio shown in Figure A4, where the dairy cattle population size generally accounted for about half of the non-dairy cattle over the years. Further differences in the indicators were evident for Ym, where dairy cattle had a significantly higher value of 7 % in 1990 compared to 6.25 % in non-dairy cattle. This value, however, declined sharply to 6.1 % by 2024, while the value for non-dairy cattle rose to 6.5 %. GE changed similarly, the value for the non-dairy cattle remained constant at 100 MJ head⁻¹ day⁻¹, while the value for

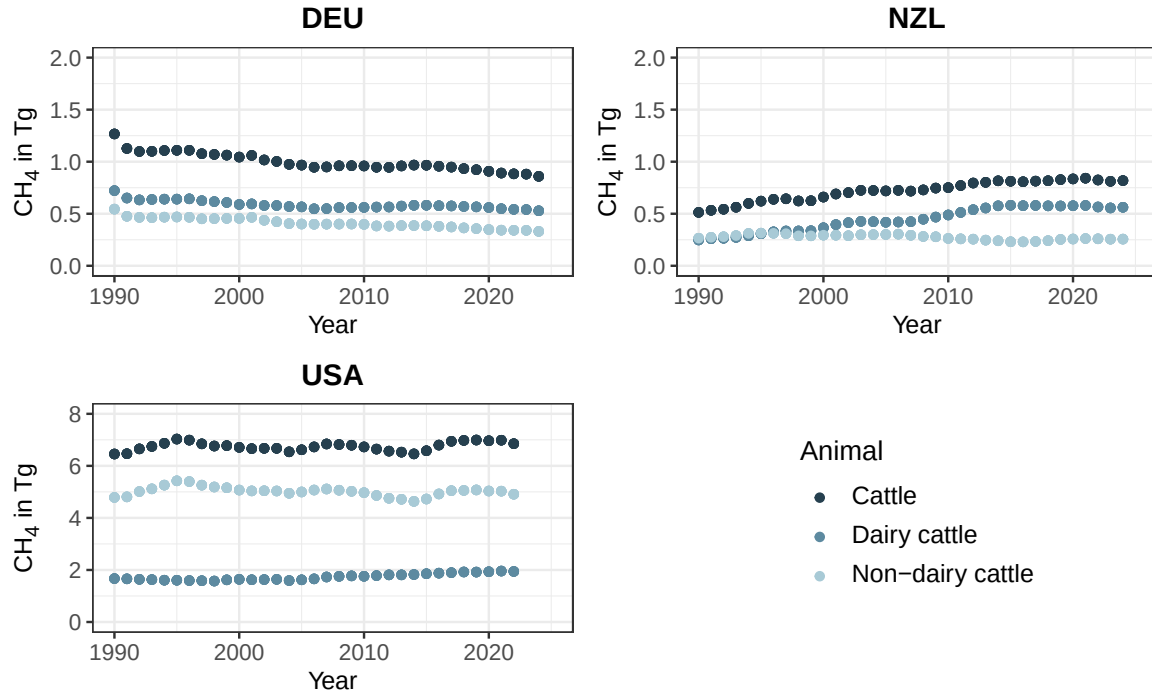


Figure 16: Time series of methane emission derived from enteric fermentation processes in cattle and its subcategories dairy and non-dairy for the countries: Germany (DEU), New Zealand (NZL) and United States of America (USA). Methane emission are measured in emitted Tg CH₄. Emissions are calculated for dairy and non-dairy cattle with the indicators in Figure A4:A6 and the eq. (1). The cattle emission is than the sum of its subcategories.

the dairy cattle population rose from 250 to 380 MJ head⁻¹ day⁻¹. Furthermore, the trend in emissions for cattle and its subcategories aligned very well with the trend in the corresponding population size (Figure A4).

Methane emissions from cattle in EnterFe for NZL increased from 0.5 Tg CH₄ in 1990 to 0.75 Tg in 2024, while the values have remained relatively constant since 2014. An increase was also evident for dairy cattle, which had plateaued since 2014 (0.25 Tg CH₄ in 1990 to 0.55 Tg in 2024). Non-dairy cattle, on the other hand, showed similar values of around 0.5 Tg CH₄ in both 1990 and 2024. Thus, in 1990, dairy and non-dairy cattle contributed equally to methane emissions from the cattle population. However, this share had changed over the years for dairy cattle. In 2024 dairy cattle accounted for about two-thirds of the emissions. This shift in the ratio was similar observable across the subcategory populations size (Figure A5). Furthermore, the GE values for non-dairy cattle remained constant at 150 MJ head⁻¹ day⁻¹, while the value for dairy cattle rose from 160 to 225 MJ head⁻¹ day⁻¹.

In the USA, methane emissions from dairy cattle rose from just under 2 Tg CH₄ to 2 Tg, with the slight increase not beginning until around 2005. Non-dairy cattle methane emission, on the other hand, fluctuated over the time period from 1990 to 2022 ranging from 4.9 to 5.5 Tg. This fluctuation was visible for cattle as well, where the values ranged from 6.5 Tg CH₄

to 7.1 Tg (Figure 16). Contrary to the other two countries, non-dairy cattle here emitted notably more than dairy cattle (2022: 5 Tg CH₄ versus 2 Tg). As with the other countries, the emission curves corresponded fairly well with the change in population size from Figure A6, where non-dairy cattle (80 million head) were kept in clearly greater numbers than dairy cattle (20 million head). The curve for dairy cattle did not match the livestock numbers exactly, as these remain constant. However, the GE for dairy cattle increased steadily from 200 to 260 MJ head⁻¹ day⁻¹ between 1990 and 2022, while the Ym value decreased from 6.5 % to 6.1 % over the same period.

When comparing cattle emissions across countries in Figure 16, it was striking that the USA produced more than three times as much in 2022 as the other two countries combined (6.8 Tg CH₄ compared to less than 2 Tg combined). This was also reflected by the number of animals: in 2022, the US had approximately 100 million cattle, while DEU and NZL each had around 10 million (Figure A4 through A6). DEU and NZL produced similar amounts of methane from cattle farming in 2024 (approx. 0.8 Tg CH₄). In 1990, however, DEU had produced twice as much, but also kept twice as many cattle (approx. 20 million animals in 1990 compared to 10 million in 2024).

Share of CH₄ emissions from EnterFe of cattle

Figure 17 shows the share of methane emissions derived from enteric fermentation processes in cattle relative to both the total amount of anthropogenically produced methane in a country and the agricultural methane emissions.

The share of CH₄ produced through EnterFe processes in cattle relative to methane emissions from the entire agriculture sector declined slightly for DEU (approx. from 78 % in 1990 to 76 % in 2024, Figure 17) and for the USA (from 76 % in 1990 to 74 % in 2022, Figure 17). Contrary to this development, the share of total methane emissions increased nearly linearly for both countries over the investigated time period. However, the increase for the USA was less pronounced, from 20 % (1990) to 25 % (2024), while DEU showed a strong increase, from a share of 20–25 % in 1990 to 45 % in 2024. The NZL time series for the development of the share of enteric fermentation cattle CH₄ emissions looked somewhat different from those described above (Figure 17). Both curves - for the share of total and agricultural methane emissions - showed a similar shape, differing only slightly in their values. The share of agricultural emissions grew from 45 % to 70 % between 1990 and 2024, while the share of total emissions grew from 40 % to just under 65 %.

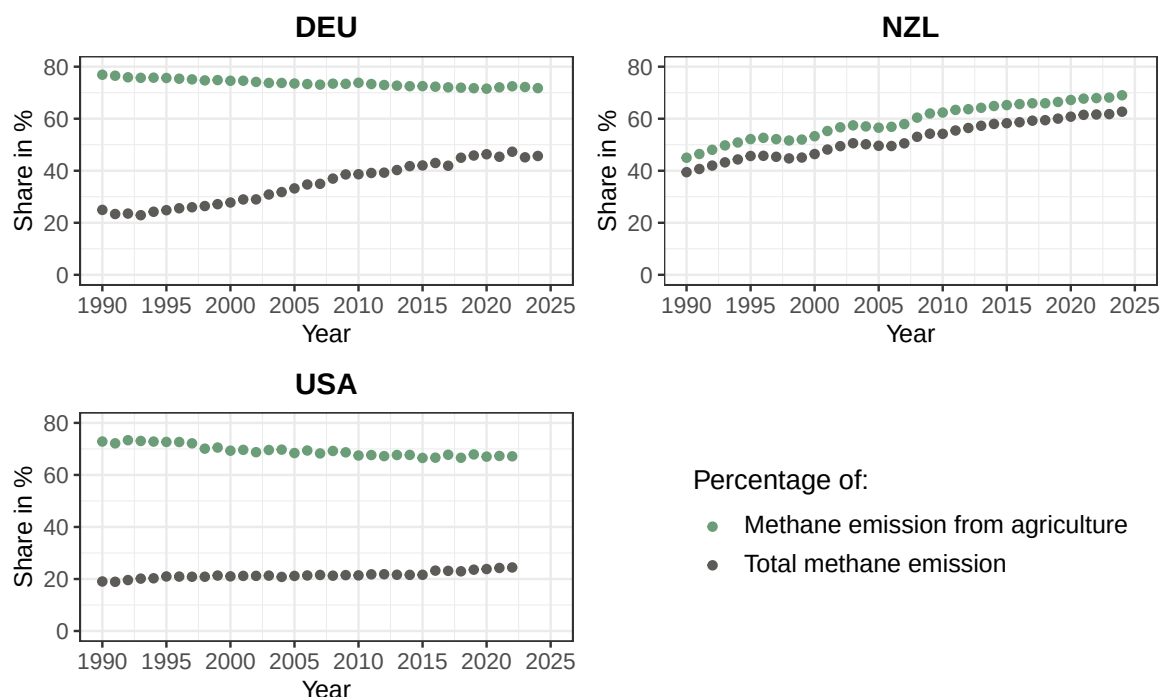


Figure 17: Time series of the share of methane emissions resulting from enteric fermentation of cattle on methane produced in the entire agriculture sector and the total amount of CH₄ per country and year. The calculation of the share are based on the information of the reported CRT.

4.2.3 Reduction potential in enteric fermentation processes of cattle

Figure 18 shows the possible yearly average of saved CH₄ for the decade from 2010 to 2019 in the countries NZL, DEU, and USA, assuming that the mitigation strategy "immunization of cattle" had been implemented. For this purpose, five different success scenarios were assumed and tested, ranging from a 10 % to a 50 % microbiome shift towards non-methane-producing microbes. Table B17 in the appendix lists the precisely calculated amounts of saved CH₄ and CO₂ equivalent (CO₂-eq.).

In the 10 % microbiome shift scenario, the three countries would have saved together approximately 0.9 to 1 Tg CH₄ per year, with about 0.75 Tg attributable to the United States alone. At 30 %, the total reduction in methane emissions would have been approximately 2.8 Tg CH₄, and at 50 %, it would have been almost exactly 4.5 Tg. The rise between the different reduction scenarios seemed to be linear in Figure 18, although Table B17 showed a slightly non-linear increase. Besides, for the various scenarios in Figure 18, it was obvious that under identical scenarios, by far the largest amount of CH₄ could be saved in the United States, and that DEU and NZL made roughly equal contributions over the ten-year period examined, with Germany contributing slightly more.

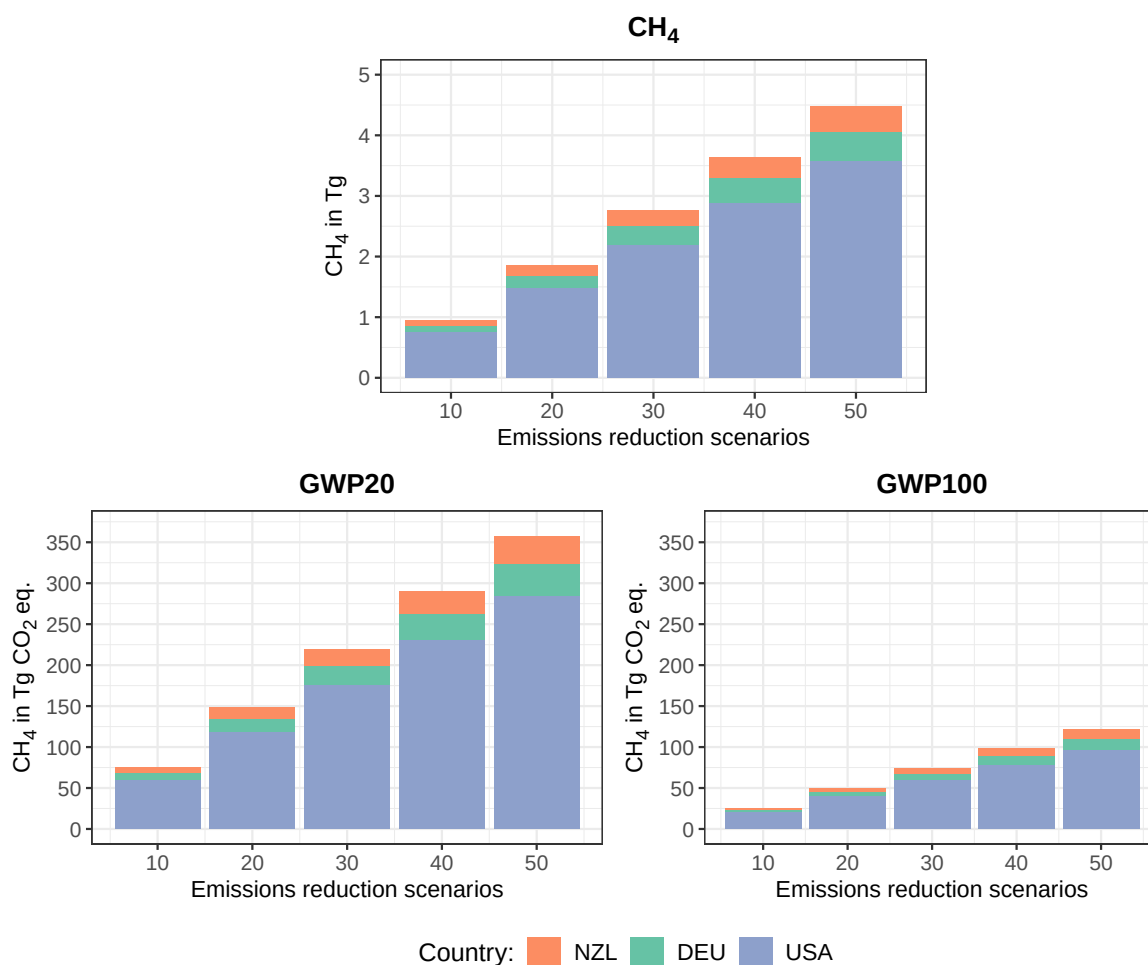


Figure 18: Saved CH₄ through different scenarios (10, 20, 30, 40 and 50 %) of a successful shift of rumen microbiome towards less methanogenic microbes. The saved methane is plotted as Tg CH₄ against the these five "Emission reduction scenarios" (CH₄). The values for the global warming potential (GWP) for 20 years (79.7) and 100 years (27) are plotted as well in Tg CO₂-eq. (IPCC, 2023a). The saved CH₄ is calculated as an average for the decade 2010-2019 for each country (Germany (DEU), New Zealand (NZL) and United States of America (USA))

For global warming potential (GWP), a clear quantitative difference could be observed across the different time horizons. For example, in the 30 % microbiome shift scenario, approximately 220 Tg CO₂ eq. would have been emitted less across all countries for a 20-year time horizon. Over a 100-year time horizon, this figure would be only 75 Tg. The relative dominance of the United States remained evident in these figures.

4.2.4 Prediction for 2030

The total methane emissions for DEU (Figure 19), NZL (Figure 20), and the USA (Figure 21) are shown as a time series from 1990 to 2030. For this purpose, a forecast was performed using three different linear models with the last 5, 10, or 15 years of the data set as reference period for the fit. This forecast includes a scenario in which no further intervention

was made, as well as three scenarios with varying degrees of success in cattle immunization (10, 30, and 50 % shifts in the rumen microbiome) (Section 3.2.5). In addition, the two global methane pledge emission targets for each country are shown in red, with the target of 30 % fewer total methane emissions than in 2020 represented as a triangle and the target of 40 % fewer emissions than in 2010 represented as a circle. Two things stand out when comparing all prediction models across all three countries for total methane emissions. First, all models showed a negative development, although the maximum confidence interval (CI) of the 5-year model for DEU showed a positive trend. Besides, the forecast for the fit with the last 10 or 15 years as time reference showed very similar prediction while the 5-year linear model differed a bit from them. In DEU the 5-year fit predicted higher values for CH₄ production for 2030 than the 10- and 15-year model, whereas for NZL and USA the 5-year model predicted lower values than the other two.

Figure 19 visualizes the results of the prediction for DEU until 2030. The GMP emission goals for DEU were 1.5 Tg and 1.4 Tg CH₄, with the higher emission value corresponding to the 40 % reduction target. The combined CI of the three linear models ranged from 1.4 Tg to 2.25 Tg CH₄, with the CI of the 5-year model (1.4 Tg to 2.25 Tg CH₄) notably wider than the combined interval of the other two models (1.4 Tg to 1.75 Tg CH₄). A closer look at the models in Figure 19 shows that the five-year model predicts a nearly constant emissions trend with only a slight decline (from 1.9 Tg CH₄ to 1.8 Tg), although the confidence intervals are quite wide and even indicate a possible increase in methane emissions. Thus, the 5-year model predicts that DEU would not reach its emission goals. The 10- and 15-year models, on the other hand, show a clear downward trend, with a value of approximately 1.5 Tg CH₄ in 2030, which corresponds almost exactly to the GMP target of a 60 % reduction.

The projected values for reduction scenario 10 hardly differ from the scenario without mitigation measures, deviating by only about 0.1 Tg (Figure 19). A greater difference can be observed for scenario 30. Here, the projected emissions for the 5-year fit would be roughly at the level of the GMP target of 40 % fewer emissions than in 2010, and thus comparable to the values of the 10- and 15-year models under "projection without vaccination." For this scenario, the values for these two models under "with immunization" fall just below the target of 70 % of 2020 emissions, at approximately 1.3 Tg CH₄. The range of the cumulative CI for 2030 extends from 1.1 Tg to just under 2 Tg of emitted CH₄. This means that, for these models, assuming a 30 % shift in the rumen microbiome, no further increase in methane levels would be expected, at least until 2030. Scenario 50, involving immunization of cattle leading to a 50 % shift of the microbiome towards reduced methane production, showed a combined CI

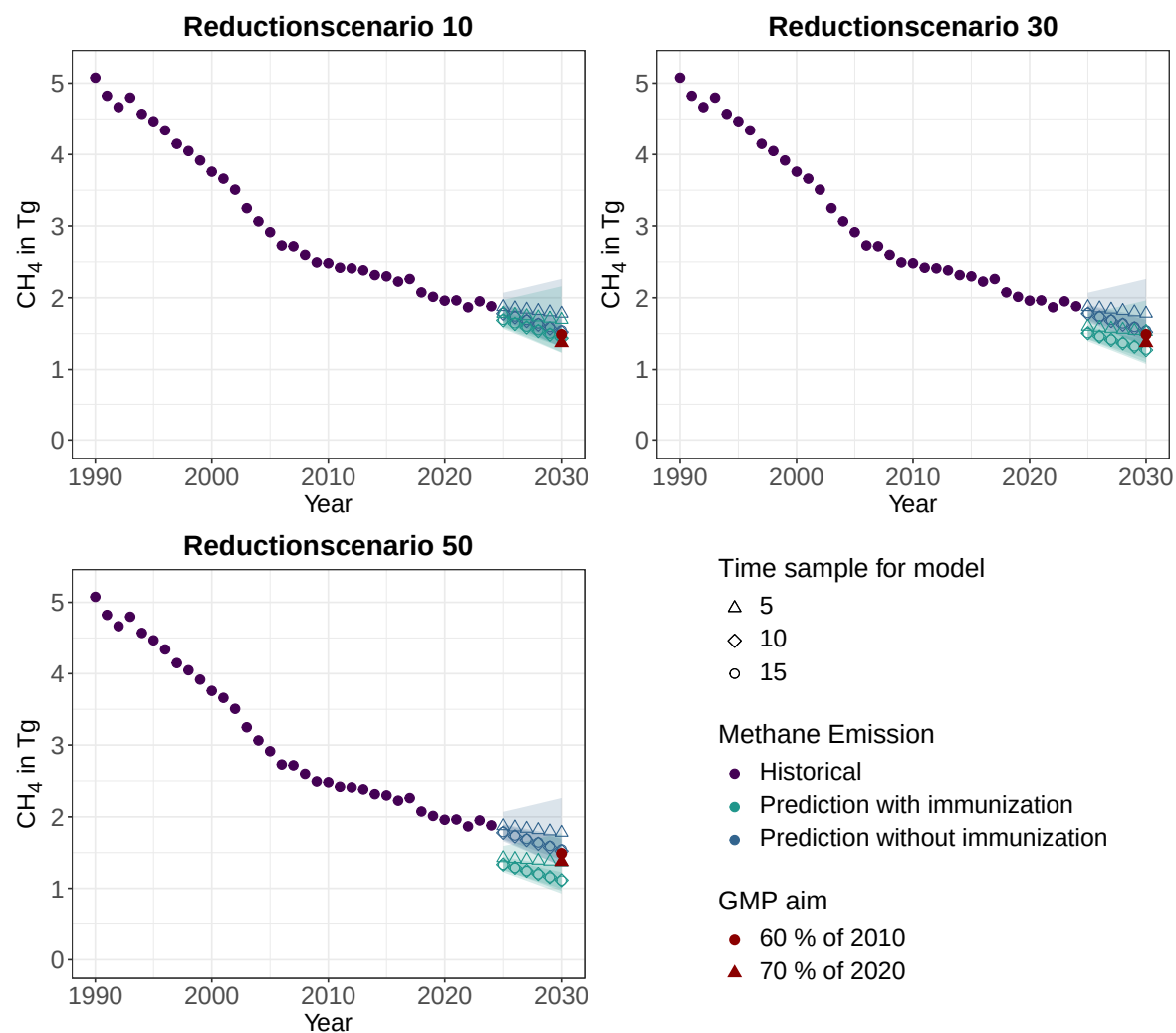


Figure 19: Historical values for total methane emissions in Tg for **DEU** from 1990 to 2024 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in Table B18 along with the corresponding equation and R^2 . Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x , and population size), to calculate the missing emissions from cattle production. In addition to predicting total CH_4 emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50 % shift in the rumen microbiome) (green). For each linear extrapolation, the confidence interval are displayed as a colored area in the corresponding color (confidence level of 95 % was used). The two global methane pledge (GMP) CH_4 emission targets are also shown in red: 60 % of 2010 methane emissions as a circle and 70 % of 2020 methane emissions as a triangle.

ranging from slightly below 1 Tg to 1.75 Tg CH_4 emitted in 2030. Under this scenario, all three models would achieve both GMP emission targets for 2030 (Figure 19), with the 5-year model predicting emissions of 1.4 Tg CH_4 , although the CI still partially exceeds the emission targets.

Figure 20 shows the prediction for NZL. The prediction without immunization had a combined CI ranging from slightly above 1 Tg to 1.35 Tg emitted methane for 2030. Here, the CI was, as before for DEU, notably wider for the 5-year model as for the other two. Neither

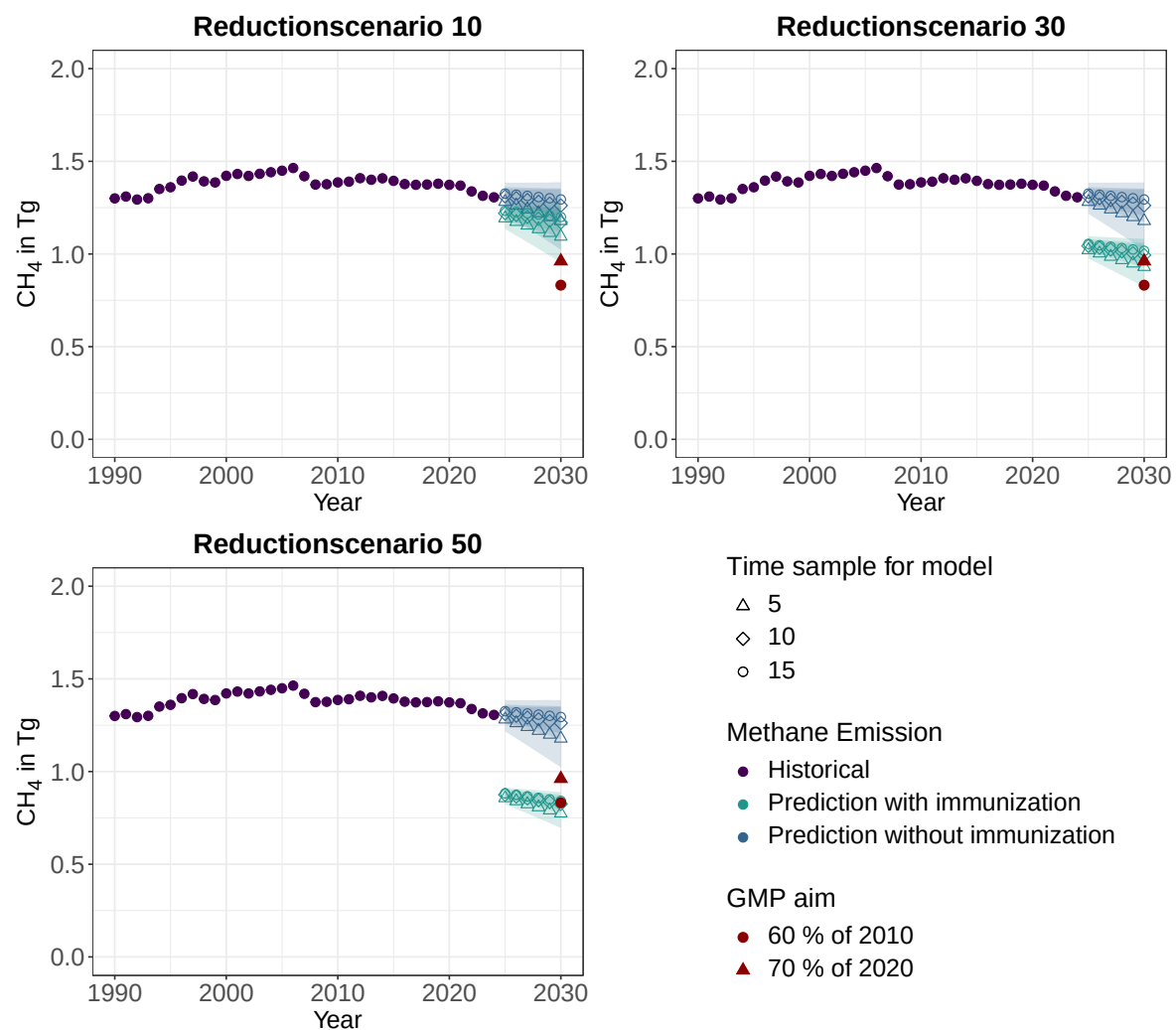


Figure 20: Historical values for total methane emissions in Tg for NZL from 1990 to 2024 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in Table B19 along with the corresponding equation and R^2 . Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and population size), to calculate the missing emissions from cattle production. In addition to predicting total CH₄ emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50 % shift in the rumen microbiome) (green). For each linear extrapolation, the confidence interval are displayed as a colored area in the corresponding color (confidence level of 95 % was used). The two global methane pledge (GMP) CH₄ emission targets are also shown in red: 60 % of 2010 methane emissions as a circle and 70 % of 2020 methane emissions as a triangle.

the values directly predicted through the models (1.2 Tg CH₄ for the 5 year fit and 1.25 to 1.3 Tg for the 10- and 15-year model) nor the CI range achieved one of the two GMP emission targets. The 60 % target relative to 2010 emissions corresponds to 0.8 Tg and 70 % to 2020 emissions at slightly below 1 Tg CH₄.

The difference between the scenarios without and with immunization for reduction scenario 10 was, similar to DEU (Figure 19), around 0.1 Tg. Therefore, it did not help NZL ensure that it would meet its emission goals, although the minimum of the combined CI was

right at the 70 % target (Figure 20). For the reduction scenario 30, this target fell between the direct prediction of the 5-year (approx. 0.9 Tg CH₄) and the extrapolation with the 10- and 15-year model (1 Tg). Besides, the lower limit for the confidence interval met the other GMP aim. For the last assumed scenario (50 % shift of microbiome), the 10- and 15-year linear model predicted methane emission values that were nearly identical to the value of the 60 % target. Therefore, most of the predicted values for the methane emission achieved this GMP aim with values ranging from 0.7 Tg to 0.85 Tg emitted methane for the year 2030.

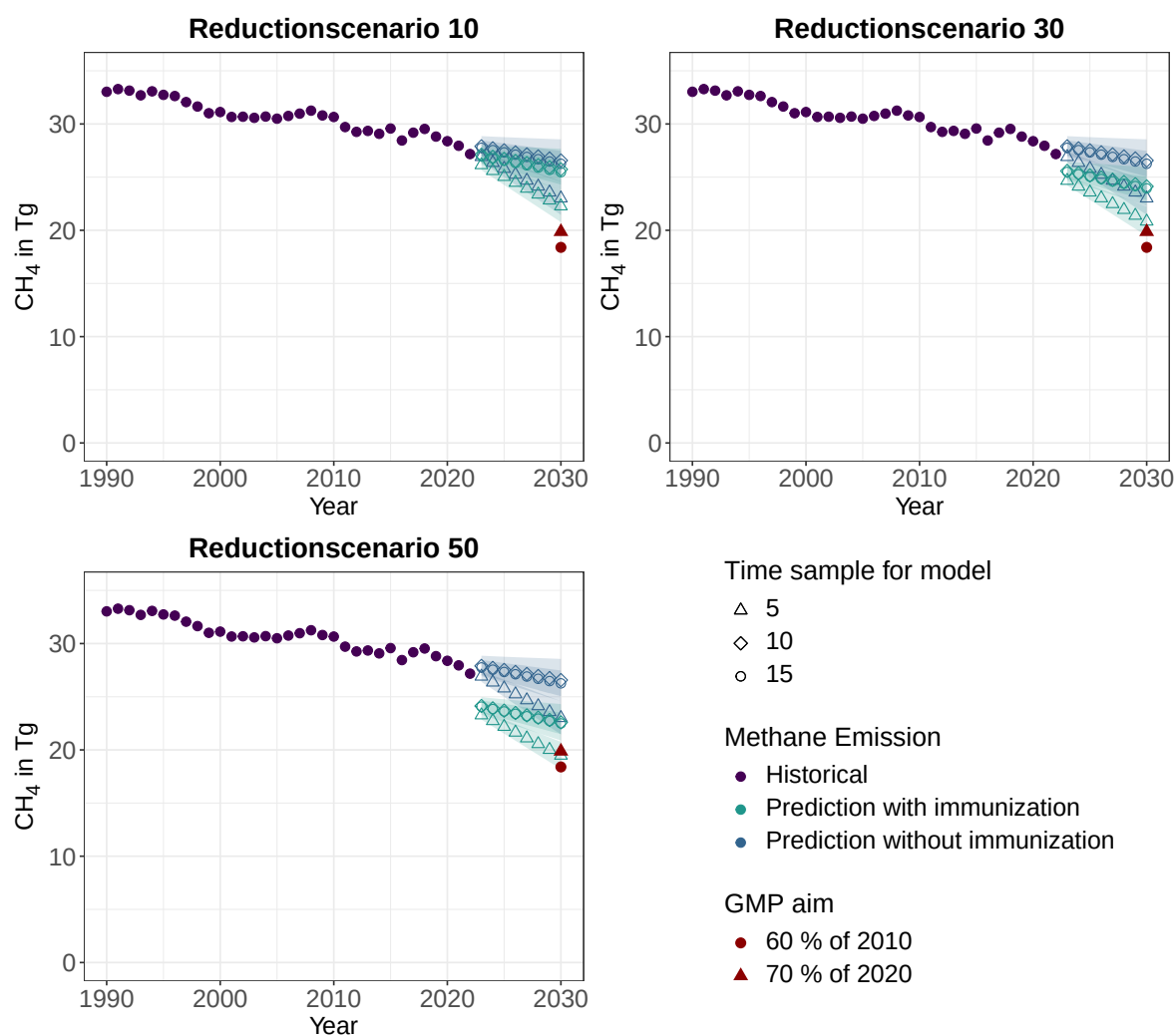


Figure 21: Historical values for total methane emissions in Tg for the **USA** from 1990 to 2022 and the forecast of methane emissions through 2030. The projection was generated using three different linear models with varying time reference periods (the last 5, 10, or 15 years), which are listed in Table B20 along with the corresponding equation and R². Total emissions were not extrapolated directly. Instead, they were determined using a prediction of total emissions excluding enteric fermentation in cattle, as well as the four indicators (DE, GE, NE_x, and population size), to calculate the missing emissions from cattle production. In addition to predicting total CH₄ emissions without additional mitigation measures (blue), the calculation was also performed for three reduction scenarios (10, 30, and 50 % shift in the rumen microbiome) (green). For each linear extrapolation, the confidence interval are displayed as a colored area in the corresponding color (confidence level of 95 % was used). The two global methane pledge (GMP) CH₄ emission targets are also shown in red: 60 % of 2010 methane emissions as a circle and 70 % of 2020 methane emissions as a triangle.

For the USA, the 10- and 15-year models for "prediction without immunization" in Figure 21 predicted an emissions value for 2030 that corresponded almost exactly to the 2022 value (approximately 27 Tg CH₄). Here, the maximum of the CI even showed an increase in emission values, although the models themselves did not predict a positive slope. The 5-year fit, on the other hand, showed a clear decrease by 2030, with a reduction of approximately 4 Tg CH₄, from 27 Tg in 2022 to 23 Tg of methane emitted in 2030. Even considering the combined CI range (22 to 28 Tg CH₄), the lowest value was still notably higher than the nearest GMP emission target. The 70 % target relative to 2020 total methane emissions corresponds to 20 Tg CH₄, and the 40 % target relative to 2010 emissions to approx. 18.5 Tg. The confidence interval range for reduction scenario 10, which differed by approximately 1 Tg from the scenario without additional mitigation efforts, did not meet either target, with its lowest value at approximately 21 Tg of emitted CH₄ in 2030. With reduction scenario 30, the predicted value of 21 Tg CH₄ for the 5-year model was still slightly above the 70 % target, but the minimum value of the CI range at least met this target. The maximum value of the CI range corresponded to approximately 26 Tg, meaning that, under this scenario, none of the values predicted for 2030 exceeded the 2022 value (Figure 21). In the 50 % reduction scenario, the 5-year model prediction, with a value of approximately 19.5 Tg CH₄, was slightly below the 70 % target. However, the predicted value was still above the GMP's 60 % target. The CI range of the 5-year fit, from 18 to 21 Tg CH₄, met the 70 % target. The other fits, as well as their CI range (21 to 24 Tg CH₄), did not meet either of the two GMP targets, even under reduction scenario 50 (Figure 21).

Two points stand out when comparing DEU with the other two countries (Figure 19, Figure 20, and Figure 21). First, in DEU, the predicted range of emission values at least partially met the two emission targets set by the global methane pledge (GMP), whereas in the other two countries, the emission values exceeded the targets. Second, the value for the 60 % target relative to 2010 emissions was higher than the 70 % target relative to 2020 emissions for DEU. For the other two countries, it was the other way around, although the two values were in a similar range for all countries.

5 DISCUSSION

5.1 Biochemistry

5.1.1 General characterization of *Methanobrevibacter*

General microbiological characterization experiments were performed with *M. thaueri*, *M. boviskoreani*, *M. millerae*, and *M. wolinii* to improve our understanding of the poorly understood *Methanobrevibacter*, which is the dominant genus of methanogenic archaea in the rumen of cattle (Baca-González et al., 2020).

Cultivation

In this bachelor thesis, the *Methanobrevibacter* species were successfully cultivated in liquid and solid DSM 1523 and 119. Unlike typical cultivation media described in the literature, these media do not contain rumenfluid (Lee et al., 2013; Li et al., 2023; Miller & Lin, 2002). Most of the experiments were performed in DSM 1523, because there was no visible difference in cultivation between the two media, and DSM 1523 is easier to handle since it does not contain the toxic fatty acid mixture and the sludge fluid. In general, *Methanobrevibacter* reach their growth maximum within two to four days, with a maximum OD600 of 0.3 that does not exceed 0.4. Further growth experiments with a higher time resolution are needed to calculate general growth parameters such as generation time or growth rate, particularly during the exponential growth phase. The highest resolution achieved here was once per day. The detected growth is as fast as that described in the literature for other archaea. However, in this bachelor thesis, the *Methanobrevibacter* did not grow as densely, in some cases reaching only about one-third of the optical density reported by Hoedt et al. (2016) and Li et al. (2023).

A closer look at *M. boviskoreani* reveals that it grows reliably in the described manner and can also be easily spread plated. The only unusual aspect is the formation of precipitates (Figure A1), which makes photometric measurements somewhat more difficult. In this case, it is important to homogenize the culture as thoroughly as possible beforehand, even if the precipitates cannot be completely dissolved. Interestingly though, this behavior is not described for the type strain (JH1^T) of this species and also the other investigated *Methanob.* species did not show this. Why *Methanobrevibacter boviskoreani* AbM4 precipitates remains unclear.

M. thaueri, on the other hand, grew much more unreliably, as greater heterogeneity in the timing of growth onset and growth density was observed both within triplicates and between experiments. In addition, *M. thaueri* lysed quickly after reaching maximal growth, such that

within 4 to 6 days the culture density was low enough to appear visually clear of cells. Furthermore, spread plating on DSM 1523 (1.5 % agar) and pour Plating with DSM 119 (1 % agar) did not work (Figure 5 and Figure A3). Roll plating was successful, but with a notably lower Plating efficiency than for *M. boviskoreani* and *M. wolinii* (Table 5). This growth behavior of *M. thaueri* makes it more difficult to analyze. Thus, *M. thaueri* does not appear to be suitable for basic research due to its unreliable growth.

Future research should further investigate whether *M. boviskoreani* and *M. thaueri* can use a liquid substrate as an alternative to H₂, potentially allowing the upscaling of growth experiments through the use of a plate reader. In contrast to the literature – where Li et al. (2023) reported that *M. boviskoreani* JH1^T could use EtOH as a substrate – neither *M. thaueri* nor *M. boviskoreani* was able to use EtOH as an alternative substrate here. Another substrate that some *Methanobrevibacter* species can use, according to the literature, is formate, which should therefore be tested next (Hoedt et al., 2016; Li et al., 2023).

Cellnumber per mL

For the first time cellnumber per mL was determined for *M. thaueri*, *M. millerae*, *M. wolinii*, and *M. boviskoreani*. Compared to the cellnumber per mL for *E. coli* ($8 \cdot 10^8$ cells/mL) at a OD600 of 1 given by Merck KGaA (2026) the counted cells/mL for the *Methanob.* cultures are quite high (e.g. *M. wolinii*: $10.819 \cdot 10^8$ cells/mL at a OD600 of 0.387; Table B14). Since only one or two cultures have been counted for these organisms so far, further investigation are needed to verify these and investigate the relationship between cell count and OD600.

Besides the small data sample, two further factors contribute to the increased uncertainty. First, the chamber depth of 0.1 mm (LO - Laboroptik GmbH, 2026) caused cells to appear in multiple focal planes, increasing the risk of overlooked or double-counted cells. Second, the still small size of the cells at a 400x magnification plus the tendency to form chains or clumps, increase the risk of overlooked or double-counted cells further due to the problem to distinguish the cells from one another (Figure A2).

One approach to mitigate this risk was to count 150 to 250 cells per large square, rather than less than 150. Thereby, reducing the relative impact of individual miscounts. This, of course, led to an increased workload. Further approaches could utilize the intrinsic fluorescence of methanogenic archaea (Lambrecht et al., 2017) and perform the counting using fluorescence microscopy. This would make it easier to identify the small *Methanobrevibacter* cells at 400x magnification.

Roll plating

In general, the technique works well for *M. boviskoreani*, *M. thaueri* and *M. wolinii*. For *M. boviskoreani* and *M. wolinii* less starting material should be used next time, since there were too many colonies, which are consequently too small for selective picking (see Figure 6 and Table 5). Therefore, it is not proven through sequencing that the right *Methanobrevibacter* grew, but the constant consumption of the H₂/CO₂ atmosphere proves there was definitely anaerobic growth. Still this plating technic has the potential to be a time saving alternative to the spread plating due to less work inside the glove-box. However, it needs improvement for a fitting quantity, so that less but bigger colonies grow.

The plating efficiencies shown in Table 5 were very low, with values ranging from below 1 to 20 ppm. By contrast, Apolinario and Sowers (1996) reached an efficiency of over 50 % for other archaea species. However, the calculated plating efficiencies in this bachelor thesis should be interpreted with caution, since the OD600 of the initial solution was not measured. Instead, it was assumed retrospectively that the cultures had a similar OD to those obtained by cell counting. For future plating efficiency determination, it is important to measure the OD600 in advance and to plate only at an OD600 corresponding to those in Table 4.

5.1.2 Sensitivity tests

Puromycin

Since the selection of successfully transformed *Methanobrevibacter* is based on resistance to puromycin (Puro), it was first necessary to determine whether, and at what concentration, *M. boviskoreani*, *M. millerae* and *M. thaueri* are sensitive to this antibiotic. *M. thaueri* is sensitive to Puro at a concentration at 10 µg/mL and showed no spontaneous resistance within 10 days. *M. boviskoreani* is also sensitive at a concentration of 10 µg/mL, but 20 µg/mL enhances the time until spontaneous resistance occurs. However, at 20 µg/mL spontaneous resistance occurred after 8 days. Hence, cells have to be collected within this time span.

Puro can be used for selection of transformed *M. boviskoreani* and *M. thaueri*, although they are somewhat less sensitive than other archaea species. For example, *Methanococcus voltae* has a minimum inhibitory concentration (MIC) of 2 µg/mL, and *Methanosarcina mazei* has an MIC of 5 µg/mL (Mondorf et al., 2012; Possot et al., 1988). For the upcoming conjugation experiments, a concentration of 20 µg/mL Puro was chosen for selection. Should growth occur within 8 days, it can be assumed that this was mediated by the introduced resistance and not due to spontaneous resistance.

8-Aza

The markerless mutagenesis is based on the counterselection with 8-Aza, therefore *M. boviskoreani*, *M. thaueri* and *M. millerae* has to be sensitive towards the base analogue. Although the *hpt* gene on the plasmid already provides a backup mechanism that induces artificial sensitivity to 8-Aza, as suggested in Klein et al. (2025). The sensitive experiment testing concentrations ranging from 100 to 1000 µg/mL 8-Aza have little to no informativeness, due to the discovery of a sensitivity towards the solvent of 8-Aza dimethylsulfoxid (DMSO) (Figure 9 through Figure 11).

However, for *M. boviskoreani* a sensitivity is detectable for concentration of 500 µg/mL based on a two days delayed growth start and around 0.7 lower maximal OD600 values (Figure 9). *M. millerae* growth results do not allow a clear conclusion due to heterogenous growing through the different concentration preparation as well as between the triplicates (Figure 11). Nevertheless, the fact that all samples exhibit similar inhibition at the maximum concentration compared to the positive control may indicate that, even though the 100 µg/mL solution contains ten times less DMSO in the medium than the DMSO preparation, 8-Aza also inhibits growth here. What argues against this, however, is that no stronger inhibition is observed at higher 8-Aza concentrations, even though the DMSO concentration and base analog concentration increase. *M. thaueri* is very sensitive against DMSO and showed no growth at all except in the positive control (Figure 10).

A plausible explanation for the negative effect of DMSO on the growth of *Methanob.* species is that, as reported by Atcher and Alfonso (2013), DMSO acts as a moderate oxidizing agent toward dithiols at room temperature. Therefore, DTT, which is added to the culture as a reducing agent to remove residual oxygen and keep the medium anaerobic, reacts with the DMSO to form its cyclic structure (Sanz et al., 2002). Thereby, a disulfide bridge is formed, and DMSO is reduced to dimethyl sulfide. Consequently, no reducing agent remains present in the medium, and the anaerobic cultures are exposed to greater oxidative stress.

This hypothesis is supported by the following: First, when samples were collected for photometric measurements, the cultures containing DMSO turned red more quickly and also took longer to lose their color than the positive controls (Section 4.1.5). Second, in the *M. thaueri* measurement series, one preparation contained only 1 mL DMSO and was not inoculated. Within a few days, this preparation culture turned distinctly red and did not fade (Figure 12). However, this also means that the *Methanobrevibacter* themselves can partially reduce the solution, since in this case the cultures did not turn red as quickly or as distinctly as in the pure

DMSO control, although they contained DMSO as well. But the *Methanobrevibacter* do not appear to be able to fully compensate for the absence of DTT, resulting in inhibited growth.

In the future, an alternative solvent should be sought or the counterselection can not be performed properly. In the literature Olson and Lynd (2012) state 50 mg/mL 8-Aza is dissolvable in 1 M NaOH or KOH. A possible problem here could be the influence on the pH of the media, the effect of this on the growth of the *Methanobrevibacter* has to be investigated.

5.1.3 Plasmid constructs

An interesting discovery was made while constructing the plasmids for the transformation: the amino acids sequence of the *pac* gene, which was on the parent plasmid and originally planned to be on the new constructed ones for the conjugation, differs in 5 amino acids from the originally amino acids sequence *Streptomyces alboniger*. This resulted in a different structure prediction by AlphaFold (Figure 13), which, according to Kryshchuk et al. (2021), is expected to be as accurate as experimentally determined structures. Interestingly though, this differing sequence was used in the literature as the amino acid-sequence of *Streptomyces alboniger* (Lacalle et al., 1989). If the changed protein-structure really has an influence on the function or the performed conjugation is not clear. To answer this question it needs further investigation. Just to be on the safe side, for the used plasmids in the conjugation *pac* gene was replaced with the codon optimized original *pac* from *Streptomyces alboniger*.

In general the constructed plasmids followed the scheme in Figure 2, where the selection of a successful conjugation works with a resistance against puromycin via *pac*-gene and a counterselection with a sensitivity against 8-Azahypoxanthin (8-Aza). If the whole-genome sequencing proofs the success of the transformation, a next step for the plasmid experiments would to test the other planned plasmids with a different promoter (*PglnA* instead of *PhmtB*, see Table B15).

5.1.4 Transformation via conjugation

Promising initial results have been achieved for implementing the first genetic toolbox for *M. boviskoreani* and *M. millerae*, even though a final assessment by whole genome sequencing is still pending (see Section 4.1.7). The transformation of *M. thaueri* was not successful. Three cultures in two preparation lines for *M. millerae* and three cultures in one preparation line for *M. boviskoreani* (Table B16) were tested positive by PCR analysis to have taken up plasmid-DNA (Figure 14), although the fragment appears smaller than expected.

This result has to be taken with caution, because the positive PCR signal could result from four other reasons. First, the plasmid could still be present in the media and be extracted along with the gDNA. This scenario is maybe likely for the recovery culture but for the 1st or especially the 2nd transfer, where only 1 % is left of the recovery culture, this seems to be unlikely. Second, the plasmid is amplified through *E. coli* through the corresponding ORI on the plasmid, but the bacterial 16S-Sequencing revealed no contamination with *E. coli* although *Staphylococcus epidermidis* was detected (Figure 14). For this species the plasmid contains no ORI. Third, the primers have an off-target with a similar sized product in the gDNA of present microbes in the culture, which was disproved by a search using “Primer Blast” on NCBI. Fourth, it could be due to an unlikely contamination of the primer set or of a used solution with the plasmids, but this can not be ruled out due to a missing negative control.

To have final clearance a whole genome sequencing is needed. However, whole genome sequencing was not performed, because of the persisting contamination with. Because of that, the next step would be to purify these cultures through plating on selective media and picking a single colonie for the following analyses. If whole-genome sequencing confirms that the plasmid has been successfully integrated, the next step would be to test the counter-selection implanted in the plasmid design using 8-Aza. Also, it would be interesting to test if other transformation methods such as electroporation could work.

Based on the comparison of the six successfully growing cultures, some factors influencing the conjugation can be assumed, although a rather small sample was investigated here and, to verify this, further investigation is definitely needed.

First, the growth status of *Methanobrevibacter* seems to be relevant. Growth only occurred among the 16 of overall 28 preparation lines that used freshly grown cultures (Table B16). This aligns with the literature, where cultures from late exponential or early stationary phase were used for successful transformation (Capovilla et al., 2025; Fink et al., 2021). This could also be the reason why the transformation of *M. thaueri* did not work, since only an old culture was used for this strain. The actual growth phase of the initial cultures used cannot be determined here, due to missing OD600 values for the *Methanobrevibacter* cultures. This should be addressed in future investigations.

Second, all three growing preparation lines used *E. coli* WM6026 as the donor strain for the interdomain conjugation. Future transformation attempts with *M. boviskoreani* and *M. millerae* should focus on this. The methylation pattern of *E. coli* WM6026 therefore does not impact the transformation of *M. boviskoreani* and *M. millerae* (Schwalen et al., 2018, SI).

Third, the recovery time does not appear to be as critical for the success of the transformation as described for *Methanothermobacter thermautotrophicus* ΔH in Fink et al. (2021). However, a slight trend towards a shorter recovery time is visible, as 5 out of the 6 successfully grown cultures were among the preparation lines with a shorter recovery time. In addition, no direct trend is detectable for the OD of *E. coli* WM6026 in this preparation set, although an OD range of 1.24 to 1.79 appears to work.

To improve the conjugation protocol further, it would likely be useful to first test whether a specific growth phase of *Methanobrevibacter* has an effect by measuring its OD600 beforehand, because a clear trend towards fresh grown cultures is already visible for this conjugation experiment. To determine whether the OD600 of *E. coli* WM6026 influences success or whether the ratio of *E. coli* cells to archaea cells does, as already described as a critical factor in Kopp et al. (2004) for the conjugation of *Sorangium cellulosum*, further test series must be conducted. Here, OD600 measurement of the used *Methanobrevibacter* cultures is needed as well.

5.2 Geography

5.2.1 Evaluation of immunization of cattle" as a mitigation strategy

As outlined in Section 2.2.3 two reference data are used for the assessment of the potential impact of the mitigation strategy "immunization of cattle". On the global scale the growth rate of methane (20.9 Tg/year) for the decade 2010 to 2019 (Saunois et al., 2025). On the national scale, the emission targets of the global methane pledge (GMP) to reduce the national emission about 30 % compared to 2020 methane emissions and 40 % to 2010 (Appelhans et al., 2025). DEU, NZL and USA signed the GMP.

National scale

For the evaluation of whether mitigation of methane production in the enteric fermentation processes of cattle can be an important missing piece in achieving the signed emission targets, CH₄ emissions had to be predicted for the year 2030. As described in Section 4.2.4, total emissions were predicted without any additional mitigation, as well as for different success scenarios of shifting the microbiome of cattle towards less methanogenic archaea. For this purpose, predictions were performed via three linear models using different time samples as reference: the last 5, 10, and 15 years. Figure 19 (DEU), 20 (NZL), and 21 (USA) show that the results of the 10- and 15-year models are quite similar, while the 5-year model usually differs somewhat more. This is likely due to the smaller data sample of the 5-year model,

making it more susceptible to changes occurring in the most recent five years. By examining the trends in national methane emissions for individual sectors (see Figure 15), the linear models for each country can be evaluated.

For DEU, it can even be concluded that the 5-year model, predicting a lower methane emission reduction than the other two models (23.5 Tg to 26.5 Tg CH₄), is likely the most plausible (Figure 19). This can be explained by the fact that, in DEU, the energy and waste sectors have remained largely unchanged over the past six years of the dataset (2018–2024), and since emissions are already at such a low level, there is little room for further reduction (Figure 15). From 1990 to 2018, CH₄ emissions from these two sectors decreased continuously. This trend is still factored into the 10- and 15-year models, leading to predictions with a stronger decrease. For NZL (Figure 20) and USA (Figure 21), no such clear attribution can be made. However, the 5-year model can be described as optimistic in both cases, as it assumes that the sharper reduction in methane levels observed over the past few years will continue at this rate. The 10- and 15-year models, on the other hand, are somewhat more pessimistic, as they reflect periods of several years with stronger reductions in the past, which were subsequently followed by years of stagnant methane emissions or even slight increases.

For the three reduction scenarios tested in Figure 19, 20, and 21 - a 10 %, 30 %, and 50 % microbiome shift in the rumen towards less methanogenic archaea - it can be said that the 10 % scenario has too little impact in all three countries to help reach the targets, if these have not already been achieved. The latter case only applies to the 10- and 15-year models of DEU. Therefore, further mitigation of methane emissions is needed in DEU, NZL, and USA, either to enable these countries to meet their emission targets at all, or, in the case of DEU, to ensure that they are met. The 30 % reduction scenario ensures that NZL and DEU would meet at least one of the two reduction targets across all models, although their CI might not fully achieve the emission goals. For NZL, this refers to a 30 % reduction in national methane emissions in 2030 relative to 2020, and for DEU, a 40 % reduction relative to 2010. The fact that, for DEU, the 40 % reduction relative to 2010 target corresponds to a higher emission level than the 30 % reduction relative to 2020 target, whereas the opposite holds true for the other two countries, can be explained by a stronger decrease in methane emissions between 2010 and 2020 for DEU, while total emission values for the other two countries remained nearly stable during this period. The 50 % reduction scenario would even be sufficient for these two countries, with all models and their CI falling below both emission targets (Figure 19 and 20).

The situation differs somewhat for the USA. In none of the scenarios do the pessimistic models (10- and 15-year) meet either of the two emission targets of the GMP (see Figure 21). The optimistic model (5-year) would come close to the 70 % target under the 30 % reduction scenario and fall short of it under the 50 % reduction scenario, but would still meet the 60 % target relative to 2010. This difference between countries can be attributed to the differing sectoral composition of national emissions (Figure 15). While agriculture was the main CH₄ source for DEU and NZL in 2024, in the USA, energy and agriculture contribute equally, and waste, though smaller, still has a substantial impact.

Two conclusions can thus be drawn. First, if the agriculture sector, and therefore EnterFe, is not the dominant methane emission source for a country, addressing it alone will not be sufficient to achieve the GMP targets. For example, in the USA, cattle emission reductions could only become the missing piece to meet the GMP emission goals if the reduction observed in the energy and waste sectors over the past five years continues at a similar rate. However, if efforts to reduce methane emissions from the energy and waste sectors were intensified, the need for cattle mitigation measures might initially be limited. Second, if agriculture is the dominant sector, there appears to be a need for new mitigation options to meaningfully reduce methane production, as agricultural emissions did not change substantially in any of the three investigated countries (Figure 15). CH₄ emissions derived from enteric fermentation in cattle appear to be a valid target for this, provided the reduction achieved is around 30 % or more.

Global scale

On the global scale several things gets quit obvious, if you compare the possible yearly averaged saved methane emission for the different reduction scenarios for the decade 2010-2019 (Figure 18) with the yearly growth rate (20.9 Tg CH₄) calculated in Saunio et al. (2025) for the same decade.

First, no reduction scenario (10, 20, 30, 40, or 50 % shift in the rumen microbiome) would have been sufficient to offset the annual growth rate. This is also true when the savings from all three countries examined are combined. Thus, neither changes in a few countries nor emission reductions in individual sectors are sufficient to stop methane accumulation in the atmosphere or even to enable a decrease in global methane levels. Nevertheless, a reduction scenario of 30 in the three countries (approx. 2.75 Tg CH₄) alone could have offset over 10 % of the global growth rate. Under reduction scenario 50, this figure would even be just over 20 % at 4.5 Tg CH₄. Considering that climate change and rising temperatures are causing natural CH₄ sources to become increasingly active, a prominent example being the thawing

of permafrost (Saunois et al., 2025; Schädel et al., 2026; Turetsky et al., 2020). This means the budget for anthropogenic methane gets relatively smaller if CH₄ should not accumulate in the atmosphere. Therefore, efforts to reduce methane emissions from cattle should aim for a reduction of at least around 30 % in order to make a notable contribution to mitigate climate change.

The second part of the Figure 18 underlines again with the calculation of CO₂-eq., that methane reduction really can be a short-term solution to soften climate change impacts and allow so to adapt/ mitigate further. So would the saved methane emission for the reduction scenario 30 (2.75 Tg) on a time horizon of 20 years correspond to 220 Tg CO₂, for a 100 year time horizon 75 Tg CO₂. The CO₂-eq. for the GWP20 would be roughly the same as the total CO₂ emission of Spain (220.08 Tg CO₂) for the year 2024 (European Commission. Joint Research Centre., 2025).

Finally, it is noticeable in Figure 18 that the amount of methane saved increases linearly with the scenarios. In the scenarios, two values are varied: first, methane conversion factor (Y_m), which has a linear relationship with methane emissions (see eq. (1)), and second, digestibility of food (DE), which has a nonlinear mathematical relationship with methane emissions (see Section 3.2.4). Although the theoretically assumed increase in dietary efficiency leads to further reductions in CH₄ emissions, this component is almost negligible because of the non-visual impact, so it has virtually no relevance to the significance of the method as a mitigation strategy.

Comparison with other mitigation strategies

Another mitigation strategy is the use of food additives, which also affect methanogenic archaea. For example, halogenated methane compounds such as bromoform inhibit the key enzyme (methyl-coenzyme M reductase) involved in methanogenesis (Boadi et al., 2004; Garnett & Eckard, 2024; Malyugina et al., 2025). In this context, the addition of red algae such as *Asparagopsis taxiformis* in in-vivo experiments with beef cattle resulted in an up to 80 % reduction in methane production (Malyugina et al., 2025, Table 4). However, even aside from this best-case scenario, reductions of over 30 % were regularly achieved in-vivo. To be competitive, the immunization strategy should also aim to achieve at least a 30 % reduction.

An advantage of immunization over feed additives is that it does not require daily administration to the animals and can be easily used in grazing animals, whereas feed additives are more difficult to implement in this context because the animals are not fed continuously or there is no controlled feed composition (Baca-González et al., 2020; Malyugina et al., 2025).

5.2.2 Characterization of national emission

The composition of the methane emission from the different sectors for the countries DEU and USA fits the described global pattern in the Section 1 and literature (Appelhans et al., 2025; Mundra & Lockley, 2024) well (Figure 15). So they had both three dominant methane emission sources in 1990: the energy, waste and agriculture sector. The time series (Figure 15) shows for DEU clearly, methane emissions from the energy and the waste sector can already be reduced very effectively using current methods. So that in 2024 the emissions from those two sectors can hardly be reduced any further due to the low emission levels. This reduction trend is also evident for the USA, albeit to a lesser extent. Although it must be noted here as a disclaimer for the strong reduction in DEU, the reduction in the energy sector was achieved partly by outsourcing energy production or coal extraction to other countries, which reduces emissions nationally but not globally (Appelhans et al., 2025). At the same time, as shown in Figure 15, the challenge of reducing emissions from the agriculture sector remains particularly striking (Edelenbosch et al., 2024; IPCC, 2022a). NZL also shows only minor changes in the agricultural sector. However, over the entire period under consideration from 1990 to 2024, there is no other relevant CH₄ emitter besides agriculture. This clearly demonstrates that there can be notably national differences, which must be taken into account when assessing the significance and application of methods. Thus, for the United States, it is likely initially easier and more relevant to reduce emissions from energy and waste.

These national differences are also evident in the share of EnterFe CH₄ emissions from cattle relative to total national emissions in Figure 17. Thus, CH₄ from enteric fermentation processes in cattle is a major and relevant single source in all three countries. However, in the USA in 2022, it accounted for only 22 % of total emissions, whereas in DEU it accounted for just under 50 % and in NZL as much as 60 %. It can, nevertheless, be noted that EnterFe from cattle is a good single methane emission target for the three countries examined.

When examining cattle emissions from enteric fermentation (Figure 16) for the three countries, it is striking that the globally described increase in cattle emissions due to rising demand for animal products (Appelhans et al., 2025; FAO, 2009) is only slightly observable. For DEU there is even a decrease in emissions detectable. This is due to the fact that only OECD countries were considered. The described increase in demand and emissions is primarily attributable to non-ANNEX-I countries.

Furthermore, a comparison with the trends in cattle population numbers in Figure A4, A5, and A6 shows that the amount of methane emissions from cattle correlates very strongly with changes in population size. In addition, DEU shows that while increased production

through higher-performing animals has led to an increase in GE, the simultaneous reduction in the number of animals has, nevertheless, resulted in a reduction in methane emissions (Appelhans et al., 2025; Niebuhr et al., 2026). However, this method of reducing population size and methane emissions cannot be continued indefinitely, and there is also the question of to what extent animal welfare is still ensured in these high-performance animals (Appelhans et al., 2025). It can therefore be concluded that population size is the decisive factor for methane production in cattle farming.

5.2.3 Future steps

For the geography site next steps to improve the calculation and rating would be to check if shifting the microbiome of the rumen can have an influence on other emission sources or is cross-linked to other emission sources. One realistic option here is the manure management. From the calculation described in IPCC (2006b) one value already seems to be promising, the B_0 . B_0 is needed for the calculation of the EF for manure management. The value represents thereby maximum methane production from the manure produced by an animal category. This could be influenced by a different ratio of methanogen to non-methanogen anaerobic microbes. But this values has to be measured, so this hypothesis, therefore, can only be tested once the first successful applications of immunization have been made.

In addition, right now, just the gross effect was investigated off methane and its reduction in cattle. So could a counter effect be that the cow and shifted rumen microbiome produce less CH_4 but more CO_2 , so the shown saved carbon dioxide in Figure 18 could be less. If the immunization strategy is in the in-vivo state, this definitely needs to be checked. Furthermore, it would be interesting to extend the calculations to other countries in order to better assess the impact of such a strategy on a global scale. Of particular interest here would be countries that are not ANNEX-I states, in contrast to the three examined so far, although differences in emission patterns were already evident in those countries as well (see Section 5.2.2)

6 CONCLUSION

The interdisciplinary approach of this bachelor thesis to unit the small scaled investigation of the biochemistry with the greater view of the geography allowed a more holistic view towards the problem of hard-to-abate methane emissions from the agriculture sector.

The analysis of national methane emissions for the three countries of interest (DEU, NZL, and the USA) showed that the agriculture sector is a major potential source for reducing methane emissions. Therefore, as the dominant emission source within the agriculture sector, methane emissions from enteric fermentation processes in cattle are a valuable mitigation target. For DEU and NZL, agriculture is the only sector left with potential for relevant methane reduction (see Figure 15).

The general geodata analyses of the national methane emissions from DEU, NZL and USA proofed the assumption derived from global methane emission pattern, there is a need for a fitting mitigation strategy toward CH₄ emission from enteric fermentation processes in cattle (Appelhans et al., 2025; IPCC, 2023a; Saunois et al., 2025). In addition, this single emission source contributes that much to the national emission (from 25 % in the USA 2022 to 45 % in DEU and 60 % in NZL, Figure 17) it will have a relevant impact on mitigate climate change, if international addressed. Especially the calculation of the GWP20 and GWP100 in Figure 18 showed clearly, methane reduction could be really helpful short-term to stay at least below the 2 °C aim of the UNFCCC (2015). For this purpose, as already pointed out in Section 5.2.1 a shift of the rumen microbiome towards non-methanogenic microbes needs to be at least at 30 %. Whether the immunization of cattle can be helpful for this short-term mitigation, depends on the stage of biochemist development and investigation of a potent and specific vaccine against methanogenic archaea.

The biochemistry part of this bachelor thesis, could collect several basic information on the poorly understood genus *Methanobrevibacter* as the dominant methanogenic archaea in the rumen of cattle (Baca-González et al., 2020, p. 3). Here, promising results were achieved to implement for the first time a genetic toolbox for *M. boviskoreani* and *M. millerae*. For this purpose, a transformation strategy was used based on interdomain conjugation with *E. coli*. If this is confirmed by whole-genome sequencing, it would allow a new approach for identifying a specific and effective antigen against *Methanobrevibacter*, which would be a huge step towards immunization of cattle. Besides this, it could be shown that *M. thaueri* and *M. boviskoreani* can be cultured and plated on media (DSM 1523 and 119) without rumenfluid

and the *Methanobrevibacter* are sensitive towards 20 µg/mL puromycin. In addition, first cellcounts per mL were determined (Figure A2) allowing first calculation of plating efficiency.

One problem has been identified and has to be resolved for future experiments, is that the solvent DMSO has a clear inhibitory effect on the growth of *Methanob.* (Section 4.1.5) due to its oxidative effect (Atcher & Alfonso, 2013). For chemicals that are poorly soluble in water, such as 8-Aza, which is needed for the counterselection, an alternative solvent must be used.

As becomes obvious here, research on a possible vaccine for cattle against their methane-producing microbiome is still in the beginning, therefore some time is still needed until such a vaccine could become available on the market. This implies that the mitigation strategy will not help mitigate climate change within the next decade. Therefore, the best approach to notably reduce methane over the next years is to use the reduction potential in the energy and waste sectors, as is clearly evident for the USA (Figure 15). As shown for DEU (Figure 15), these sectors are easier to address with today's technology (Appelhans et al., 2025; IPCC, 2022b).

Nevertheless, this immunization strategy can be highly useful in the long term for achieving the goal of net-zero emissions within the agricultural sector, at least to some extent (UNFCCC, 2015). This will become even more relevant considering that methane emissions from natural sources, such as permafrost, are expected to increase in the future due to rising temperatures (Schädel et al., 2026). However, it must be noted that even with a mitigation strategy reducing emissions per head of cattle, it will still be necessary to reduce the cattle population to a sustainable size in the future (Edelenbosch et al., 2024). This will be challenging given the growing demand for animal products (Appelhans et al., 2025; FAO, 2009). Here, immunization or other strategies to reduce CH₄ production in cattle could allow for a population size that is both socially and politically acceptable.

Future steps for the biochemical part are to further improve the transformation protocol for *Methanobrevibacter* and conduct initial genetic analyses. A highly interesting value to determine is the actual methane emission of *Methanobrevibacter* via gas chromatography. For the geographical part, the geo-data analyses should be extended to other countries and it should also be investigated whether the mitigation strategy affects or interacts with other emission sectors, such as manure management or pasture carbon sinks.

At last, the basic research on methane production by methanogenic archaea conducted here, may be relevant not only for vaccine development, but also for other mitigation strategies, such as feed additives, and for our general understanding of methane production from natural sources.

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BIOCHEMICAL TERMS

β -sheet	A common secondary structure of proteins. It consists of β -strands arranged in parallel or antiparallel orientations. β -strands are illustrated as arrows. Another typical secondary structure is the α -helix.
16S-Sequencing	16S sequencing targets the 16s genes, which code for the 16S ribosomal RNA (rRNA). The 16S rRNA is a part of the prokaryotic ribosome. These 16-s base sequence is specific for the organism.
AlphaFold	Is a KI based software for predicting in-silico the 3-D structure of proteins. For the prediction it needs an amino acid sequence.
archaea	Archaea constitute one of the three major domains of life (bacteria, archaea, eukaryotes), and, like bacteria, they belong to the prokaryotes (Fritsche, 2024 , p. 3).
auxotrophic	Auxotrophic organism can not produce relevant bio-molecules for growth and surviving. Therefore, they must obtain these from their environment (Fritsche, 2024 , p. 65).
C-terminal end	The end of the amino acid sequence of a protein, where the carboxy group of the terminal amino acid has no covalent connection to another amino acid.
colony	In a microbiological context, a colony is defined as a clearly distinct cluster of cells that can be traced back to a single parent cell through cell division.

conjugation	Conjugation is the process of horizontal gen-transfer between a donor-cell and an acceptor-cell. The connection or "mating" between the cells is made by a pilus (also called "sex-pilus"), which allows the transport of the DNA. The conjugation can also happen between different species or domains. Needed for the conjugation is a plasmid that contains the gene for conjugation (the <i>tra</i> -region) and connection with a pilus (Fritsche, 2024 , p. 207).
counterselection	A selection strategy in molecular genetics used to select against cells carrying a specific genetic marker, typically by exposing cells to a compound that becomes toxic only in its presence. This allows for the selective survival of cells in which the marker has been lost or replaced, enabling markerless genetic manipulation, as opposed to conventional selection.
glove-box	A sealed, airtight chamber system with built-in gloves that allow you to work from the outside while the internal atmosphere remains controlled (e.g. a anaerobic atmosphere).
in-silico	Investigations or experiments are performed on a computer/ via computer simulation software.
in-vitro	Experiments are not performed in/ at the living organism, but in an artificial setting.
in-vivo	Experiments are performed in/ at the living organism.

markerless mutagenesis	A genetic engineering approach that introduces targeted mutations into a genome without leaving behind a selectable marker gene (such as an antibiotic resistance gene) in the final strain. This is typically achieved through a combination of positive selection and counterselection, allowing the marker to be removed after successful genomic integration (Klein et al., 2025).
ORI	Origin of replication, is the region of the DNA, where the machinery responsible for replicating the entire genomic DNA first binds and begins replication.
PCR	The polymerase chain reaction is a in-vitro method to amplify a specific sequence of DNA. For this purpose, specific fitting primer has to be designed, that flank this DNA-sequence on both complementary DNA-strings.
phase-contrast microscope	Phase-contrast microscopy is based on the fact, that objects or medium not only changes the amplitude of light but also the phase of it. The phase-contrast microscope makes this phenomena visual for the human eye (Pawlizak, 2009).
plasmid	A small ring of double-stranded DNA. Occurs naturally in archaea and bacteria, but is also used for scientific purposes. Here often as a vector for a transformation.

Plating	Cultivate microorganism on a solid medium (contains a gelling agent like agar). The plating, therefore, is the process how the organism get distributed on the plate/solid media. Different technics are possible like spreading or pouring (Fritsche, 2024 , p. 364).
pour plating	A plating technique in which the solution containing microorganisms is mixed with the agar-containing culture medium while it is still liquid (at about 45 °C). This mixture is then poured into a petri dish, where it solidifies(Fritsche, 2024 , p. 364).
primer	Short nucleotide sequence that serves as a starting point for the polymerase to replicate the complementary DNA strand. For this purpose, the primer hybridizes to a specific sequence on the singlestranded-DNA template (Fritsche, 2024).
promotor	Region on DNA, which allows binding of relevant enzymes for the transcription of genes. So it is a essential region for gene expression.
restricitionsite	Restrictionsites are a specific sequence of approx. 4 to 10 bp that are recognized by a specific restriction-enzyme. This cuts the double-stranded DNA at this site in a specific behavior.
ribosome	Ribosomes are a mix out ribosomal RNA (rRNA) and proteins. They play a central role in protein biosynthesis by catalyzing the translation of mRNA into an amino acid sequence.

selection	A selection strategy in molecular biology used to select only organisms that have been successfully transformed. Usually, an antibiotic resistance gene is used as a selection marker, enabling the successfully transformed organism to survive in the presence of an otherwise toxic antibiotic.
spread plating	A plating technique in which the solution containing microorganisms is put on the solid agar-containing culture medium. The solution is then spread across the agar plate using a Dragalski spatula (Fritsche, 2024 , p. 364).
suicide plasmid	A plasmid that has no ORI and therefore cannot be replicated in the cell except it gets integrated in the gDNA.
transformation	The process of getting an external DNA into the target organism and its expression. If the target organism is an animal eucaryotic cell it is called transfection.

GEOGRAPHICAL TERMS

ANNEX-I	ANNEX-I and Non-ANNEX-I countries is a classification of the UNFCCC. ANNEX-I are the OECD countries or countries with an economy in transition.
compound events	When two weather extremes occur at the same time or shortly after each other, so that they reinforce each other, as example a heat-wave and droughts (IPCC, 2023b).
CRT	Common reporting table is a template for countries to report its emission of GHG fitting IPCC standards. The UNCTF has obliged the ANNEX-I countries to publish this every year since 1990.
gross	In the context of climate change, the gross effect refers to the direct, physical influence of a greenhouse gas (such as methane) on the Earth's energy balance, considered in isolation. The alternative for the gross effect would be the net effect.

GWP	Global warming potential is a concept used to represent the relative impact of various GHGs on a uniform scale over a specific time period. It takes into account their effectiveness in causing radiative forcing as well as their atmospheric lifetime. CO ₂ serves as a reference, so that the effect of 1 g of greenhouse gas is converted to x g of CO ₂ eq.. The relevant time period is indicated as the number of years following the GWP designation (e. g., GWP20 refers to a 20-year time horizon).
net	Contrary to the gross effect accounts the net effect of GHG also for masking effects. For example, caused by co-emitted cooling agents, such as aerosols released during methane extraction, which can partially offset the gross warming contribution.
planetary boundaries	Is a concept that uses the variability of values of the Holocene as a reference to evaluate thresholds for different processes to maintain a stable system on earth, where human mankind can definitely thrive. The nine main processes are "Climate change", "Biosphere integrity", "Land-system change", "Freshwater use", "Biogeochemical flows", "Ocean acidification", "Stratospheric ozone depletion", "Atmospheric aerosol loading" and "Novel entities" (Planetary Boundaries Science (PBScience), 2025).

radiative forcing	Is a concept that quantifies the change of the energy balance of the whole planet. The unit is W/m^2 . It depends on many factors as e.g. aerosol content of the atmosphere, planetary albedo and the concentration of GHG (Irfan et al., 2024).
zone of an increasing risk	"Refers to the stage when human activities push Earth beyond the Planetary Boundaries, entering a "Zone of Increasing Risk." In this zone, the further boundaries are exceeded, the greater the chance of causing serious damage, destabilizing key Earth system processes, and disrupting life-support functions" (Planetary Boundaries Science (PBScience), 2025).

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APPENDIX

A Figures



Figure A1: Phase-contrast microscope picture of *Methanobrevibacter boviskoreani* AbM4 culture precipitate at 1000 times magnification. *M. boviskoreani* was cultivated in DSM 1523.

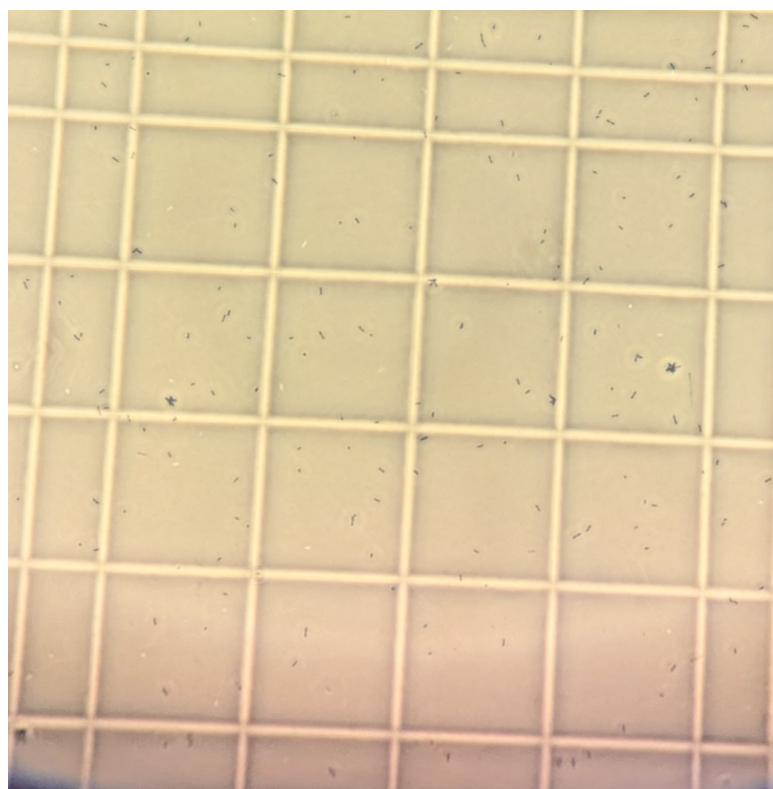


Figure A2: Phase-contrast microscope picture of *Methanobrevibacter boviskoreani* AbM4 at 400 times magnification. One group square with its sixteen small squares of a Neubauer improve counting-chamber is shown.

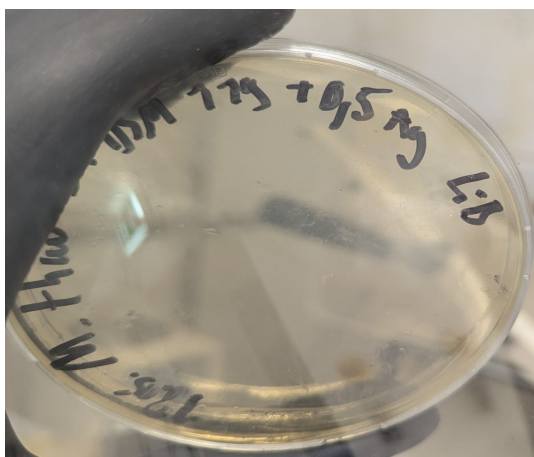


Figure A3: Pour plating result of *Methanobrevibacter thaueri* CW^T DSM 119 (1 % agar). 100 μ L of a full-grown culture have been plated. Cultivated in a jar with H₂/CO₂ gas phase. No growth detected.

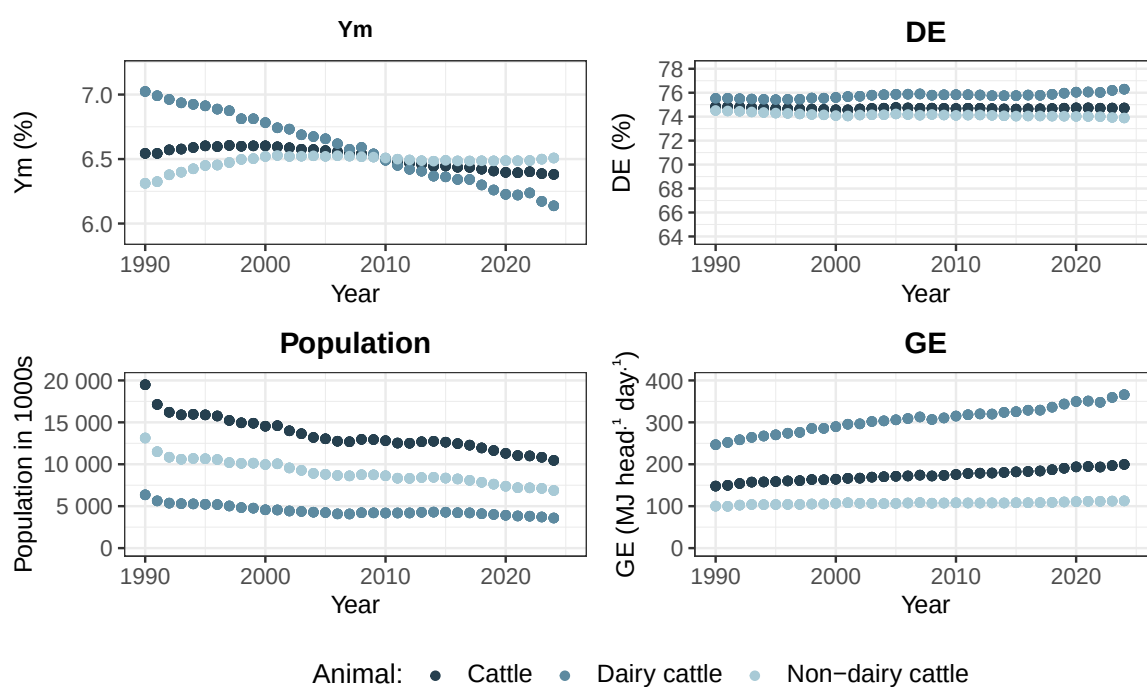


Figure A4: Time series of methane conversion factor (Ym), population-size, digestibility of food (DE) and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for DEU. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

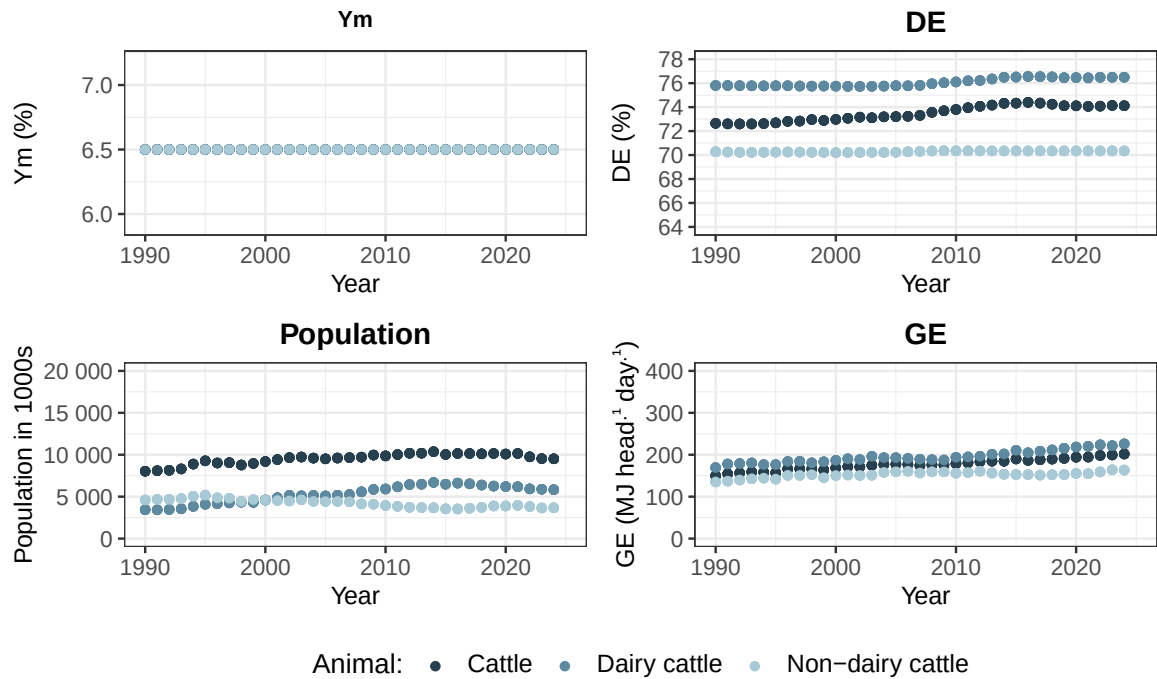


Figure A5: Time series of methane conversion factor (Y_m), population-size, digestibility of food (DE) and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for NZL. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

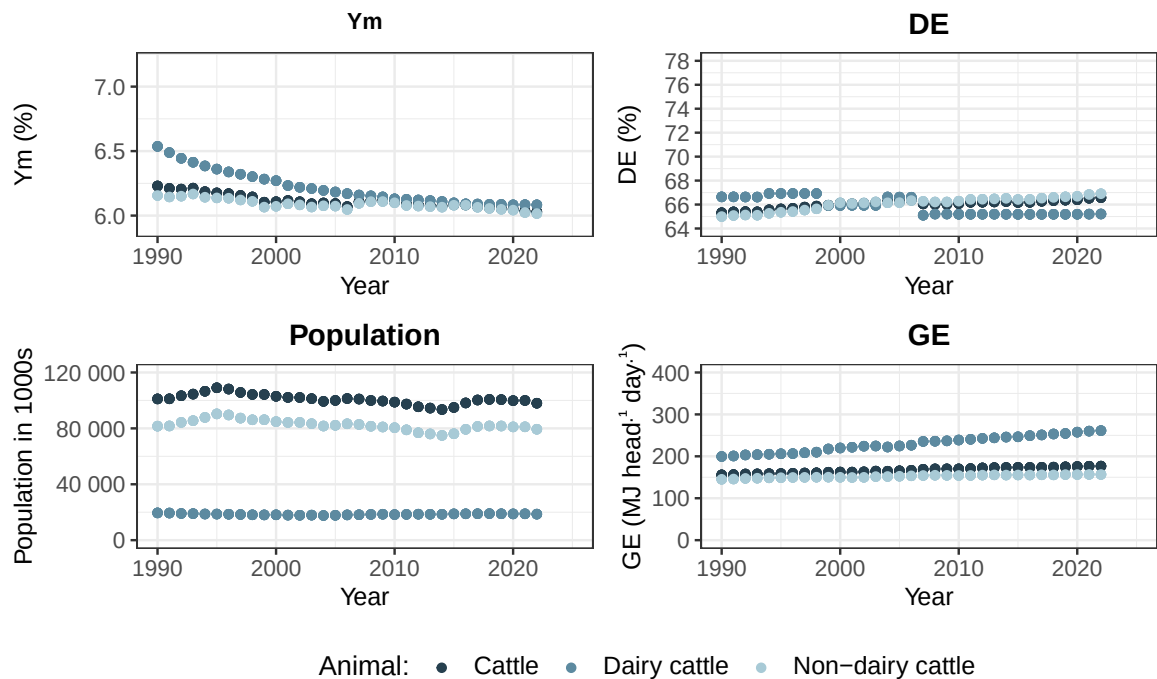


Figure A6: Time series of methane conversion factor (Y_m), population-size, digestibility of food (DE) and gross energy intake (GE) of cattle and its subcategories dairy and non-dairy for USA. The data for dairy and non-dairy cattle is from the published CRT. The values for cattle is calculated as weighted mean of its subcategories, where the population-size is used as weight.

B Tables

Table B1: Composition for the **solution A** as a part of the DSM 1523 and DSM 119. Final concentration of defined chemicals are shown as well. The quantities listed are for 250 mL.

ingredient	quantity	concentration
KH ₂ PO ₄	12.50 g	367.40 mM
MgSO ₄ x 7 H ₂ O	10.00 g	162.28 mM
NaCl	10.00 g	684.46 mM
NH ₄ Cl	10.00 g	747.80 mM
Milli-Q H ₂ O	250.00 mL	–

Table B2: Composition for the **solution A** as a part of the DSM 1523. Final concentration of defined chemicals are shown as well. The quantities listed are for 275 mL. Solutions that have already been premixed are shown in bold. Composition of **Seven vitamins solution** is shown in Table B4.

ingredient	quantity	concentration
CaCl ₂	0.94 g	33.88 mM
Seven vitamins solution	25 mL	10 %
Milli-Q H ₂ O	250.00 mL	–

Table B3: Composition for the **trace element solution SL-10** as a part of the DSM 1523 and DSM 119. Final concentration of defined chemicals are shown as well. The quantities listed are for 1 L.

ingredient	quantity	concentration
HCl (12 %)	10.00 mL	1 %
FeCl ₂ x 4 H ₂ O	1.50 g	7.54 mM
ZnCl ₂	70.00 mg	513.61 µM
MnCl ₂ x 4 H ₂ O	100.00 mg	505.28 µM
H ₃ BO ₃	6.00 mg	97.04 µM
CoCl ₂ x 6 H ₂ O	190.00 mg	798.55 µM
CuCl ₂ x 2 H ₂ O	2.00 mg	11.73 µM
NiCl ₂ x 6 H ₂ O	24.00 mg	100.97 µM
Na ₂ MoO ₄ x 2 H ₂ O	36.00 mg	148.79 µM
Milli-Q H ₂ O	990.00 mL	–

Table B4: Composition for the **seven vitamins solution** as a part of the DSM 1523. Final concentration are shown as well for defined chemicals. The quantities listed are for 1 L.

ingredient	quantity	concentration
Vitamin B ₁₂	100.00 mg	73.78 µM
p-Aminobenzoic acid	80.00 mg	583.35 µM
D-(+)-biotin	20.00 mg	81.86 µM
Nicotinic acid	200.00 mg	1624.56 µM
Calcium pantothenate	100.00 mg	209.85 µM
Pyridoxine hydrochloride	300.00 mg	1458.86 µM
Thiamine-HCl x 2 H ₂ O	200.00 mg	535.76 µM
Milli-Q H ₂ O	1.00 L	–

Table B5: Composition for the **fatty acid mixture** as a part of the DSM 119. The quantities listed are for 1000 mL.

ingredient	quantity	concentration
Isobutyric acid	23.00 mL	2.3 %
DL-2-Methylbutyric acid	27.00 mL	2.7 %
Valeric acid	27.00 mL	2.7 %
Isovaleric acid	27.00 mL	2.7 %
Milli-Q H ₂ O	896.00 mL	–

Table B6: Composition for the **FeSO₄ x 7 H₂O solution (0.1 % w/v)** as a part of the DSM 119. Final concentration was calculated as well. The quantities listed are for 1000 mL 0.1N H₂SO₄ solution.

ingredient	quantity	concentration
FeSO ₄ x 7 H ₂ O	1.00 g	3.60 mM
H ₂ SO ₄ (0.1 N)	1000.00 mL	–

Table B7: Primerlist of the primers used in this bachelor thesis. Besides the name of the primer, also the sequence (5' to 3') is given and its annealing temperature (T_a) with the Q5- and OneTaq-DNA-polymerase.

No.	name	Sequence	T_a Q5 (°C)	T_a OneTaq (°C)
1	27F	AGAGTTTGATCATGGCTCAG	–	45.0
2	1510R	GGTTACCTTGTTACGACTT	–	45.0
17	FW_PhmtB-FseI-NEB	TCAACTTGCCCACTGGCCGGCCCCCATGAACCAACCGATG	61.7	–
22	Met86F	GCTCAGTAACACGTGG	61.0	46.0
23	Met1340R	CGGTGTGTGCAAGGAG	61.0	46.0
38	FW-screen-BBpMr	TTGGA CTCAAGACGATAGTT	46.0	–
39	RV-screen-BBpMr	GGCAGTACACCTTGATAGG	46.0	–
48	FW-Mthau-LHA-hpt_NEB	CATCGGTTGGTTCATGGGGGCGCGCCTTCAGCCTTCTCGATCAATG	58.8	–
≤ 49	RV-Mthau-LHA-hpt_NEB	CCAAATCCATCATAGTACTCACACAAAAATTATTAAAC	58.8	–
50	FW-Mthau-RHA-hpt_NEB	GAGTACTATGATGGATTTGGATAAGGTAATC	57.2	–
51	RV-Mthau-RHA-hpt_NEB	CAGGAAACAGCTATGACCGCGGCCGCTGTTTAAGACTATTTCCAATTCC	57.2	–
52	FW-Mb-LHA-hpt_NEB	CATCGGTTGGTTCATGGGGGCGCGCCGATTTCCAATTCTTG GGG	56.1	–
53	RV-Mb-LHA-hpt_NEB	GAAAATTATGTAATCCTAAAGACTATCTAAATTTAAC	56.1	–
54	FW-Mb-RHA-hpt_NEB	TTTAGGATTACATAATTTTCCTCTTTAATTTAAATTAAAAATATAATTTTAAAAATTG	60.4	–
55	RV-Mb-RHA-hpt_NEB	CAGGAAACAGCTATGACCGCGGCCGCGGAACCAATATCGGTAGC	60.4	–
63	RV_Mthau_pac_opt_Smal_NEB	GGGGTCGGCGGGGCGCCCGGGTTAGGCACCTGGTTTTCTTG	63.4	–
71	RV_Mb_pac_opt_Smal_NEB	GGGGTCGGCGGGGCGCCCGGGTTATGCTCCTGGTTTTCTGG	62.9	–

Table B8: Counted cells for *Methanobrevibacter wolinii* SH^T with an OD600 of 0.387. Cells were counted with a phase-contrast microscope and a Bürker counting chamber at 400x magnification. Per chamber filling four square (Sqr.) with a volume of $4 \cdot 10^{-6}$ mL were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	138	159	226	233	189
2	260	271	293	265	272.25
3	180	197	216	188	195.25
4	207	191	211	227	209

Table B9: Counted cells for *Methanobrevibacter wolinii* SH^T with an OD600 of 0.396. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ mL were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	177	239	263	254	233.25
2	246	241	267	208	240.5
3	175	176	161	221	183.25
4	217	185	205	180	196.75

Table B10: Counted cells for *Methanobrevibacter thaueri* CW^T with an OD600 of 0.235. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ mL were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	189	232	250	265	234
2	237	209	224	263	233.25
3	186	253	230	237	226.5
4	146	187	216	262	202.75

Table B11: Counted cells for *Methanobrevibacter thaueri* CW^T with an OD600 of 0.116. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ mL were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:10 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	255	181	189	171	199
2	127	92	131	193	135.75
3	110	136	153	109	127
4	127	93	109	156	121.25

Table B12: Counted cells for *Methanobrevibacter millerae* SM9 with an OD600 of 0.378. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ mL were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	Sqr. 5	$\bar{x}$ cellnumber
1	186	175	173	156	220	182
2	118	204				161
3	211	178	232	211	184	203.2
4	128	128	201	204	160	164.2

Table B13: Counted cells for *Methanobrevibacter boviskoreani* AbM4 with an OD600 of 0.287. Cells were counted with a phase-contrast microscope and a Neubauer improved counting chamber at 400x magnification. Per chamber filling four large-square (Sqr.) with a volume of $4 \cdot 10^{-6}$ mL were counted. The arithmetic mean ($\bar{x}$ cellnumber) per chamber filling was calculated. The culture was diluted 1:20 in DSM 1523 before counting.

chamber	Sqr. 1	Sqr. 2	Sqr. 3	Sqr. 4	$\bar{x}$ cellnumber
1	123	162	166	170	155.25
2	162	228	205	242	209.25
3	170	128	137	213	162
4	187	174	263	237	215.25

Table B14: The total average of counted cells ($\bar{x}$ cellnumber) for each *Methanobrevibacter* (*Methanob.*) culture with corresponding OD600. Counting was done with a phase-contrast microscope and a Bürker or Neubauer improved counting chamber at 400x magnification. $\bar{x}$ cellnumber was calculated as arithmetic mean of shown cellcounts in tab. B8 through B13. The dilution factor takes into account the dilution of the culture being counted.

culture	OD600	Dilution factor	$\bar{x}$ cellnumber
M. wolinii	0.387	20	216.375
M. wolinii	0.396	20	213.438
M. thaueri	0.235	10	224.125
M. thaueri	0.116	10	145.750
M. millerae	0.378	20	177.600
M. boviskoreani	0.287	20	185.438

Table B15: In-silico produced suicide plasmid constructs for transformation of *Methanobrevibacter thaueri* CW^T and *Methanobrevibacter boviskoreani* AbM4 via conjugation. The listed plasmids follow the general structure shown in Figure 2 and were created in SnapGene. The name of each plasmid is derived from the organism for which it possesses the appropriate homology arms (pMt for *M. thaueri* and pMb for *M. boviskoreani*) and the four named columns, where ORI refers to *Methanob.* in this case and the promoter for the *pac*. *pac_org* is the *pac* with the original sequence derived from *Streptomyces alboniger*, *pac_opt* is the codon optimized version for the affected organism. The size of the constructed is given in bp as well.

name	ORI	promoter	pac	extra feature	size (bp)
pMt_0231_hpt	/ (0)	PhmtB (2)	pac_opt (3)	hpt (1)	6145
pMt_0331_hpt	/ (0)	PglnA (3)	pac_opt (3)	hpt (1)	6254
pMt_0241_hpt	/ (0)	PhmtB (2)	pac_org (4)	hpt (1)	6145
pMt_0341_hpt	/ (0)	PglnA (3)	pac_org (4)	hpt (1)	6254
pMb_0231_hpt	/ (0)	PhmtB (2)	pac_opt (3)	hpt (1)	6214
pMb_0331_hpt	/ (0)	PglnA (3)	pac_opt (3)	hpt (1)	6323

Table B16: Preparation of the conjugation for the transformation of *Methanobrevibacter thaueri* CW^T, *Methanobrevibacter boviskoreani* AbM4, and *Methanobrevibacter millerae* SM9. Two DAP-auxotrophic *Escherichia coli* strains were used for conjugation, WM6026 and WM6029. The final OD600 was measured as a 1:10 dilution with LB-media before the Plating with *Methanobrevibacter*. The *E. coli* for *M. thaueri* contained the plasmid pMt_0231_hpt, for *M. boviskoreani* the plasmid pMb_0231_hpt, and for *M. millerae* the plasmid pMm_0231_hpt. Cultures marked with an asterisk showed visible growth for the first time on the day they were used for conjugation. Preparations from the same *Methanobrevibacter* pre-culture share the same first number (No.). Preparations from the same plate are distinguished by the letters A and B. Before being transferred to a 10 mL DSM 1523 medium containing 200 µg/mL cycloserin (Cyc-Se.) and no DAP, the strains *Methanobrevibacter* and *E. coli* were incubated overnight on agar plates containing 10 mM MgSO₄, 5 mM CaCl₂, and 0.1 mM DAP overnight in a jar (headspace with H₂/CO₂ mix). t_P is the time that elapsed during recovery before Puro (final concentration of 20 µg/mL) was added. The time from this point until the first transfer of the culture to a new DSM 1523 medium with the same concentrations of Cyc-Se. and Puro was measured and denoted as t_{1_t} . t_{2_t} then indicates the number of days that elapsed until the second transfer. The samples marked in red are those that showed visible growth in one of the transferred cultures and were collected for sequencing.

No.	<i>Methanobrevibacter</i>	<i>E. coli</i> (OD600)	t_P [h]	t_{1_t} [h]	t_{2_t} [d]
1.6A	<i>M. thaueri</i>	WM6026 (1.02)	5	17	4
1.6B			19.5	2.5	
1.9A		WM6029 (1.30)	5	17	
1.9B			19.5	2.5	
2.6A	<i>M. boviskoreani</i>	WM6026 (1.22)	5	17	4
2.6B			19.5	2.5	
3.6A			5	17	
3.6B			19.5	2.5	
2.9A		WM6029 (1.15)	5	17	
2.9B			19.5	2.5	
3.9A			5	17	
3.9B			19.5	2.5	
4.6A	<i>M. millerae</i> *	WM6026 (1.24)	5	17	4
4.6B			19.5	2.5	
5.6A			5	17	
5.6B			19.5	2.5	
4.9A		WM6029 (1.59)	5	17	
4.9B			19.5	2.5	
5.9A			5	17	
5.9B			19.5	2.5	
6.6A	<i>M. boviskoreani</i> *	WM6026 (1.79)	4.5	0.5	4
6.6B			21	0.5	3
7.6A			4.5	0.5	4
7.6B			21	0.5	3
6.9A		WM6029 (0.60)	4.5	0.5	4
6.9B			21	0.5	3
7.9A			4.5	0.5	4
7.9B			21	0.5	3

Table B17: Lists saved CH₄ for five scenarios of different success in shifting the microbiom from methanogen towards non-methanogen (10, 20, 30, 40 and 50 %). It was calculated an average for the decade 2010 to 2019 for the three countries Germany (DEU), New Zealand (NZL), United States of America (USA). The CO₂ eq. was calculated for the GWP20 and GWP100 with factors taken from IPCC (2023a, Table 7.15, Chapter 7) (79.7 for GWP20 and 27 for GWP100)

country	red.-scenario	CH ₄ [Tg]	GWP20 [Tg CO ₂ eq.]	GWP100 [Tg CO ₂ eq.]
DEU	10	0.104	8.321	2.819
DEU	20	0.207	16.460	5.576
DEU	30	0.306	24.424	8.274
DEU	40	0.404	32.220	10.915
DEU	50	0.500	39.853	13.501
NZL	10	0.088	7.032	2.382
NZL	20	0.174	13.906	4.711
NZL	30	0.259	20.628	6.988
NZL	40	0.341	27.204	9.216
NZL	50	0.422	33.639	11.396
USA	10	0.750	59.739	20.238
USA	20	1.480	117.953	39.959
USA	30	2.192	174.707	59.186
USA	40	2.887	230.061	77.938
USA	50	3.564	284.071	96.235

Table B18: Listed linear models for predicting relevant values for calculating 2030 CH₄ emission of **DEU**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R² is given as well. R² for the 15-year model of "NDC YM [%]" turns zero because no change is predicted.

time sample	fitted value	equation	R ²
5	total CH ₄ without cattle EnterFe [kt]	$y = -6.25 \cdot x + 13677.282$	0.070
5	DC population size [1000s]	$y = -78.386 \cdot x + 162269.508$	0.956
5	DC Ym [%]	$y = -0.023 \cdot x + 52.062$	0.712
5	DC DE [%]	$y = 0.063 \cdot x - 51.879$	0.735
5	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 1.868 \cdot x - 3630.669$	0.720
5	NDC population size [1000s]	$y = -110.076 \cdot x + 229727.445$	0.893
5	NDC Ym [%]	$y = 0.005 \cdot x - 4.499$	0.821
5	NDC DE [%]	$y = -0.035 \cdot x + 144.304$	0.891
5	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.129 \cdot x - 215.365$	0.888
10	total CH ₄ without cattle EnterFe [kt]	$y = -36.946 \cdot x + 75747.569$	0.792
10	DC population size [1000s]	$y = -76.559 \cdot x + 158578.922$	0.988
10	DC Ym [%]	$y = -0.024 \cdot x + 55.219$	0.946
10	DC DE [%]	$y = 0.057 \cdot x - 40.026$	0.932
10	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 1.941 \cdot x - 3778.256$	0.946
10	NDC population size [1000s]	$y = -168.698 \cdot x + 348278.384$	0.974
10	NDC Ym [%]	$y = 0.002 \cdot x + 2.939$	0.483
10	NDC DE [%]	$y = -0.016 \cdot x + 106.264$	0.749
10	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.209 \cdot x - 377.797$	0.957
15	total CH ₄ without cattle EnterFe [kt]	$y = -40.092 \cdot x + 82103.312$	0.934
15	DC population size [1000s]	$y = -44.459 \cdot x + 93727.349$	0.767
15	DC Ym [%]	$y = -0.023 \cdot x + 52.942$	0.977
15	DC DE [%]	$y = 0.032 \cdot x + 10.375$	0.716
15	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 1.532 \cdot x - 2951.872$	0.940
15	NDC population size [1000s]	$y = -125.537 \cdot x + 261079.881$	0.917
15	NDC Ym [%]	$y = 0 \cdot x + 6.57$	0.000
15	NDC DE [%]	$y = -0.014 \cdot x + 102.431$	0.852
15	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.144 \cdot x - 246.555$	0.863

Table B19: Listed linear models for predicting relevant values for calculating 2030 CH₄ emission of **NZL**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R² is given as well. R² is NA for indicators where the variance of the values is zero. In addition there are two zero values because of rounding, first the slope of the 15-year model for "NDC DE [%]" and the R² for the 15-year model of "NDC population size [1000s]".

time sample	fitted value	equation	R ²
5	total CH ₄ without cattle EnterFe [kt]	$y = -12.753 \cdot x + 26300.235$	0.997
5	DC population size [1000s]	$y = -102.756 \cdot x + 213780.671$	0.887
5	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
5	DC DE [%]	$y = 0.011 \cdot x + 54.263$	0.570
5	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.715 \cdot x - 1352.955$	0.797
5	NDC population size [1000s]	$y = -71.703 \cdot x + 148784.07$	0.732
5	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
5	NDC DE [%]	$y = 0 \cdot x + 70.346$	NA
5	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.937 \cdot x - 1835.457$	0.852
10	total CH ₄ without cattle EnterFe [kt]	$y = -10.058 \cdot x + 20851.018$	0.978
10	DC population size [1000s]	$y = -88.716 \cdot x + 185394.664$	0.930
10	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
10	DC DE [%]	$y = -0.007 \cdot x + 90.206$	0.305
10	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.906 \cdot x - 1740.452$	0.895
10	NDC population size [1000s]	$y = 22.947 \cdot x - 42611.147$	0.211
10	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
10	NDC DE [%]	$y = 0 \cdot x + 70.346$	NA
10	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.508 \cdot x - 968.768$	0.773
15	total CH ₄ without cattle EnterFe [kt]	$y = -9.839 \cdot x + 20408.364$	0.992
15	DC population size [1000s]	$y = -29.407 \cdot x + 65583.652$	0.219
15	DC Ym [%]	$y = 0 \cdot x + 6.5$	NA
15	DC DE [%]	$y = 0.021 \cdot x + 33.254$	0.469
15	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 1.019 \cdot x - 1969.163$	0.966
15	NDC population size [1000s]	$y = -0.097 \cdot x + 3942.39$	0.000
15	NDC Ym [%]	$y = 0 \cdot x + 6.5$	NA
15	NDC DE [%]	$y = 0 \cdot x + 71.161$	0.481
15	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.105 \cdot x - 153.685$	0.096

Table B20: Listed linear models for predicting relevant values for calculating 2030 CH₄ emission of the **USA**. The relevant indicators are total methane emission without cattle enteric fermentation (EnterFe) and Ym (methane conversion factor), DE (digestibility), NE_x (net energy intake), population size for dairy cattle (DC) and non-dairy cattle (NDC). For fitting the linear models three different time periods were used (the last 5, 10 or 15 years of the data set). The fitted value is thereby depended on the time. Besides the equation, the R² is given as well. Slope of the 15-year model for "DC DE [%]" is zero due to rounding issues.

time sample	fitted value	equation	R ²
5	total CH ₄ without cattle EnterFe [kt]	$y = -529.671 \cdot x + 1091556.297$	0.989
5	DC population size [1000s]	$y = -78.221 \cdot x + 176838.967$	0.789
5	DC Ym [%]	$y = -0.001 \cdot x + 8.017$	0.704
5	DC DE [%]	$y = 0.006 \cdot x + 53.297$	0.845
5	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.736 \cdot x - 1400.748$	0.977
5	NDC population size [1000s]	$y = -526.204 \cdot x + 1143925.596$	0.765
5	NDC Ym [%]	$y = -0.01 \cdot x + 27.096$	0.960
5	NDC DE [%]	$y = 0.081 \cdot x - 96.049$	0.952
5	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.173 \cdot x - 295.476$	0.921
10	total CH ₄ without cattle EnterFe [kt]	$y = -245.886 \cdot x + 518212.392$	0.753
10	DC population size [1000s]	$y = 21.479 \cdot x - 24555.96$	0.144
10	DC Ym [%]	$y = -0.003 \cdot x + 12.927$	0.802
10	DC DE [%]	$y = 0.002 \cdot x + 62.028$	0.293
10	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.671 \cdot x - 1269.793$	0.990
10	NDC population size [1000s]	$y = 638.685 \cdot x - 1209266.975$	0.544
10	NDC Ym [%]	$y = -0.006 \cdot x + 19.106$	0.787
10	NDC DE [%]	$y = 0.049 \cdot x - 32.878$	0.820
10	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.113 \cdot x - 173.444$	0.838
15	total CH ₄ without cattle EnterFe [kt]	$y = -236.704 \cdot x + 499649.489$	0.876
15	DC population size [1000s]	$y = 35.307 \cdot x - 52471.673$	0.536
15	DC Ym [%]	$y = -0.005 \cdot x + 16.261$	0.916
15	DC DE [%]	$y = 0 \cdot x + 65.618$	0.011
15	DC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.617 \cdot x - 1159.782$	0.993
15	NDC population size [1000s]	$y = 104.563 \cdot x - 131253.734$	0.040
15	NDC Ym [%]	$y = -0.006 \cdot x + 17.665$	0.876
15	NDC DE [%]	$y = 0.043 \cdot x - 19.257$	0.890
15	NDC NE _x [MJ head ⁻¹ day ⁻¹]	$y = 0.106 \cdot x - 158.969$	0.915

DECLARATION

I hereby declare that I have written the present thesis independently and without the use of sources other than those named. Artificial intelligence (AI) tools were used exclusively for language editing and programming assistance in R and $\text{\LaTeX}$, and not for the generation of scientific content or analysis. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The thesis in the same or similar form has not been submitted to any examination body and has not been published. This thesis was not yet, even in part, used in another examination or as a course performance. Furthermore, I declare that the submitted written (bound) copies of the present thesis and the version submitted on a data carrier are consistent with each other in contents.

Place, date

Signature